

Memory, Expectations, and Climate Adaptation

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Abstract. Smallholder farmers face large and growing weather risk, yet take-up of insurance and other protection remains low. The value farmers place on protection depends on the weather they expect. Where records and forecasts are limited, those expectations are built from memories of past weather, and distorted recall can leave demand for protection poorly aligned with the risks farmers actually face. We measure how much experienced weather survives into perceived risk. In a survey experiment with 686 farmers in eastern Uganda, we elicit remembered and expected weather through visual token allocations and map geospatial rainfall and vegetation records onto the same scale. We find that memory is strongly compressed, with about 90 percent of seasonal weather variation lost in the baseline estimate. Yet remembered weather strongly predicts expectations, so this compression carries into beliefs about future risk. By contrast, salient extremes remain vivid. Farmers who recall an extreme are 19 percentage points more likely to expect another and also plan substantially more adaptation. In a randomized experiment, information on past local weather moves expectations toward the record and raises confidence. Accessible historical information can therefore help farmers assess climate risk and inform adaptation decisions in the face of climate extremes.

Keywords: climate change; drought; climate adaptation; weather risk; agricultural risk; expectations; memory; smallholder agriculture.

JEL codes: D81, D83, D91, O13, Q12, Q54.

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1 Introduction

Smallholder farms produce about a third of the world’s food supply and roughly seventy percent of food calories in the smallholder-dominated regions of Africa, Asia, and Latin America (Ricciardi et al., 2018; Samberg et al., 2016). For the households that run these farms, weather is a central source of income risk, yet take-up of insurance and investment in other forms of protection remain low (Carter et al., 2017; Cole et al., 2013). As dry spells, heavy rainfall, and floods become more frequent and intense, this lack of protection becomes an increasingly important economic concern (IPCC, 2021). This demand problem is partly a problem of perceived risk. The value farmers place on protection depends on the weather they expect. Where historical records are scarce and formal forecasts are limited, those expectations are largely built from remembered experience. Memory therefore shapes both the risk farmers perceive and the protection they seek.

This paper traces and quantifies the mechanism through which experienced weather becomes perceived risk and demand for protection. We measure how objective local weather history enters memory, how remembered weather shapes expectations, and how remembered extremes relate to planned protection, and then experimentally test whether providing the historical record can restore part of the signal lost through memory. The core finding is that memory can distort risk without erasing experience. Ordinary seasonal differences are compressed even as salient extremes remain vivid, so remembered weather can shape expectations while carrying only a fraction of the climate variation farmers actually experienced.

We study this mechanism among 686 smallholder farmers in 60 villages of eastern Uganda, where farming is rainfed and local historical weather records are limited. We first collect detailed information on farmers’ characteristics, farming practices, and adaptation plans. We then elicit remembered and expected weather using a visual probability task in which farmers allocate ten tokens across nine images of the same maize field, ranging from extreme drought to severe flooding. We classify those same past seasons on the nine-picture scale using satellite rainfall and vegetation data, allowing us to compare remembered weather directly with the local record. In a randomized information treatment, farmers see their local weather history in the same visual format between two elicitations of their expectations, while control farmers repeat the elicitation without receiving information.

The first result is that memory preserves only a small share of seasonal weather variation, even though remembered weather strongly predicts expectations. A one-point difference between seasons in the local weather record leaves only about 0.1 points in recall, so roughly 90 percent of the signal is lost. Even under our most conservative correction for noise in

the record, only about a quarter to a third of the signal survives. Yet a one-point wetter remembered history is associated with about 0.44 points wetter expectations for the coming season. Farmers therefore use remembered experience to form expectations, but memory transmits only a heavily compressed version of the weather variation they experienced.

The second result is that salient extremes behave differently. Farmers who freely recall a past drought, flood, or other extreme event are about 19 percentage points more likely to expect another extreme in the coming season. Among farmers planning some adaptation, those who recall an extreme plan to spend 65 to 120 percent more on protection across specifications. Compression and salience therefore coexist. Seasonal variation is largely flattened in memory, while vivid extremes remain influential for perceived risk and planned protection.

The third result is that part of the signal lost through memory can be restored with simple historical information. Showing farmers an accessible visual record of the previous five seasons moves their expected rainfall toward the record, shifts expected planting onset about 2.6 days earlier, and raises confidence in their expectations. We detect no weaker response among farmers who cannot read, even though their initial beliefs lie further from the record. The information changes beliefs.

These results establish three contributions.

First, the paper shows that memory is a quantitatively important filter between experienced weather and subjective risk. The economics of memory emphasizes selective retrieval and the role of past experience in beliefs and risk taking (Bordalo et al., 2023, 2020; Malmendier & Nagel, 2011), while work on climate experience infers memory from behavior or studies how the conditions of recall shape remembered events (Gallagher, 2014; Moore et al., 2019; Patel, 2026). We directly compare season-by-season, recalled distributions with an objective local record and show that compression and salience coexist. This turns memory from an unobserved intermediary into a measurable margin of belief formation.

Second, the paper moves the demand side of climate adaptation upstream from protection choices and expectations to the formation of perceived risk. Much of the literature studies how the supply of protection, including product design, price, and delivery, shapes take-up (Cole et al., 2013; Giné et al., 2008; Karlan et al., 2014), while recent work shows that subjective beliefs affect investment and adaptation (Burlig et al., 2024; Patel, 2025; Zappalà, 2024). We show that those beliefs are themselves shaped by what farmers remember of past weather. This links low demand for protection to a core behavioral mechanism through which past weather is selectively remembered and translated into expectations.

Third, the paper shows that the historical information lost through memory can be partly restored with an accessible visual record. A large literature studies how agricultural information affects beliefs and behavior, with effects depending on how information is presented and delivered (BenYishay & Mobarak, 2019; Cole & Fernando, 2021; Hanna et al., 2014). We show that simple information about farmers’ own local weather history moves expectations toward the record and raises confidence, with no detectable weaker response among farmers who cannot read. Historical information can therefore reach the same belief-formation margin through which memory shapes perceived risk. This makes accessible historical information a low-cost, practical tool for strengthening the risk signal farmers carry into adaptation decisions and, ultimately, preparedness for climate extremes.

Section 2 describes the setting and sample, Section 3 develops the conceptual framework, and Section 4 presents the survey and experimental design. Section 5 presents the empirical results, and Section 6 concludes.

2 Background, Setting, and Sample

2.1 Adaptation, Expectations, and Memory

Demand for protection

Much of the economic literature on climate adaptation in agriculture has focused on the design and delivery of protective technologies. Research has studied insurance, stress resistant crops, savings, credit, and other forms of protection, while varying prices, contract design, payment timing, marketing, bundling, and access to complementary inputs (Carter et al., 2017; Duflo et al., 2011; Emerick et al., 2016). Rainfall index insurance provides a particularly clear example. By tying payouts to a measured weather index rather than assessed losses, it reduces moral hazard and verification costs, yet take up remains low without substantial subsidies (Cole et al., 2013; Giné et al., 2008). A clear lesson from this work is that improving product design and supply can encourage farmers to protect themselves, but its success ultimately depends on demand. Research has therefore turned to the role of personal experience in shaping that demand. Cole et al. (2014) and Stein (2018) show that recent payouts affect subsequent take up, while experimentally induced experience of disaster risk can increase demand for protection (Cai & Song, 2017). This work has greatly improved our understanding of these margins, yet low take up remains persistent. Explaining that

persistence requires a deeper understanding of how farmers form their beliefs about weather risk that drive demand.

Subjective weather expectations

If beliefs about the weather distribution are part of demand formation, the next question is whether they can be measured reliably. A large literature shows that subjective probability distributions can be elicited meaningfully even where literacy and formal schooling are limited (Attanasio, 2009; Delavande, 2014; Delavande et al., 2011; Manski, 2004), and that these beliefs predict economically important choices. Beliefs about monsoon onset predict planting decisions beyond measured risk preferences (Giné et al., 2015), while pastoralists' rainfall expectations respond to new information (Lybbert et al., 2007). Recent work links environmental beliefs more directly to adaptation. Forecasts can shift expectations and investment, prior beliefs shape irrigation responses, and objective environmental information can change both beliefs and input demand (Burlig et al., 2024; Patel, 2025; Zappalà, 2024). Together, these studies show that subjective beliefs matter for adaptation and respond to information. What remains less clear is how farmers form those beliefs when objective records are scarce.

Memory and belief formation

If expectations are formed from past experience, the remaining question is how faithfully that experience is remembered. Classic work on availability and recent models of memory emphasize that retrieval is imperfect and selective, so remembered experience can differ systematically from what actually occurred (Bordalo et al., 2023, 2020; Tversky & Kahneman, 1973). Evidence across settings supports this point. Past experience shapes risk taking and subjective probabilities (Enke & Graeber, 2023; Malmendier & Nagel, 2011), while climate research finds that weather perceptions overweight recent or salient events (Gallagher, 2014; Moore et al., 2019). Recent work has also begun to measure farmers' weather memories directly, documenting systematic distortions in recall and showing that remembered events can shape expectations and adaptation choices (Naso, 2025; Patel, 2026).

What remains less understood is how experienced weather translates into memory, expectations, and adaptation. Where objective records are scarce, farmers may rely heavily on the seasons they remember when forming expectations about the future. We observe each link in this process, from recorded weather to memory, expectations, and planned adaptation, and experimentally test whether providing the local weather record shifts expectations toward the objective history.

2.2 Study Setting

The study covers the districts of Bugiri, Butaleja, and Mayuge in eastern Uganda, on the lowland plains of the Lake Victoria basin plateau at roughly 1,080 to 1,250 meters, west of the cash crop zones on the slopes of Mount Elgon.¹ Rainfall is bimodal (FEWS NET, 2010), with the first and main growing season running roughly from February to July and a second season from August to December. Farming is predominantly rainfed, small scale, and oriented toward own consumption (Uganda Bureau of Statistics, 2017, 2022).

Three features make the setting well suited to studying the demand side of climate adaptation.

First, production is rainfed, so beliefs about the coming season are an important input into planting, investment, and protection decisions. For households farming close to subsistence, errors in judging future weather have direct consequences for production and consumption.

Second, local historical weather records are limited, so farmers’ memories of past seasons are the natural basis for forming expectations about the upcoming season. Asked which sources they use to form rainfall expectations, respondents most often name personal memory of past seasons (61 percent). Formal forecasts reach the area, mostly by radio, but rarely at the village level. Memory is therefore an important input into expectations, though not the only one. The record shown in our experiment presents local weather history in a simple and accessible format that is easier to process than the information otherwise available to farmers. Appendix F.6 reports how farmers gather and weigh weather information.

Third, formal insurance plays little role in protection decisions. Among the 404 respondents who plan any pre season adaptation, only eight, or 2.0 percent, plan to buy crop insurance. Adaptation is financed mainly through savings groups, own savings, and asset sales (Appendix F.6), in line with national data showing that almost no adults hold crop insurance (Financial Sector Deepening Uganda, 2018). The setting therefore lets us study demand formation before insurance has become a routine part of adaptation.

2.3 Sample Selection and Characteristics

We built the sample in three steps (Appendix A.1). From 594 villages identified with our field partner, we first restricted the sampling frame to 442 rural villages in 11 non town subcounties

¹The 2025 pilot ran at a separate site on the Elgon slopes in Bulambuli district. Online Appendix OA-7 describes the pilot, and no result in this paper uses pilot data.

that the field team could reach. We then used satellite rainfall and vegetation data to select 77 candidate villages with sufficient variation in weather histories across seasons and across locations. Before any treatment was assigned, the field team sought permission from village leaders, leaving 60 villages able to host the survey and experiment. These villages, with 686 farmers and 7 to 14 respondents per village, define the study sample. Because village access was determined before randomization, it affects the population to which our results apply but not treatment comparisons. Seven enumerators interviewed all 686 farmers between 10 and 31 March 2026, before the first 2026 growing season.

Table 1: Sample characteristics, by experimental arm.

	Arm A	Arm B	Arm C	All
Respondents	274	293	119	686
Villages	24	26	10	60
Age (years)	42.5 (12.9)	40.0 (12.5)	42.1 (12.7)	41.4
Female	0.434	0.457	0.429	0.443
Household size	7.98 (4.38)	7.64 (4.86)	7.50 (5.26)	7.75
Reads and writes	0.504	0.539	0.462	0.512
Cultivated area (acres)	2.84 (2.22)	2.68 (2.42)	2.97 (2.18)	2.79
No formal education	0.383	0.375	0.454	0.392
Primary education	0.427	0.430	0.319	0.410
Secondary or above	0.190	0.195	0.227	0.198
Plans to grow maize	0.960	0.939	0.983	0.955
Plans to grow beans	0.745	0.795	0.782	0.773
Plans to grow cassava	0.529	0.488	0.580	0.520
Plans to grow groundnuts	0.449	0.464	0.445	0.455
Plans to grow rice	0.164	0.119	0.076	0.130
Uses fertilizer	0.580	0.556	0.597	0.573
Uses improved seeds	0.536	0.509	0.546	0.526
Uses irrigation	0.245	0.212	0.218	0.226
Plans crop insurance	0.019	0.024	0.013	0.020

Notes: Arm means, with standard deviations of continuous variables in parentheses for continuous outcomes. Villages are the randomization units. Literacy denotes self-reported ability to read and write and is available for 684 respondents. The final education category pools secondary, tertiary, and vocational schooling. Crop shares refer to the coming season and need not sum to one. Planned insurance is measured among respondents planning any adaptation ($N = 404$). Source is the March 2026 survey.

Table 1 describes the sample. Respondents are 41.4 years old on average, and 44.3 percent are women. Human capital is limited: just over half can read and write, and 39.2 percent have no formal education. Farms are small, at 2.79 cultivated acres on average, and all 686 respondents identify farming as their occupation.

3 Conceptual Framework

The framework formalizes the argument that organizes the paper. The weather record is filtered through memory, memory shapes expectations, and expectations govern protection. Each link has a parameter we measure or a relation estimated in the literature, and the final subsection adds information about the record as a signal that can enter this process. We take no stand on why memory is imperfect.

3.1 Remembered Weather

Let $W_{s,t}$ denote the weather realized in season t at location s , measured on the nine-picture scale of Section 4.2. A farmer does not carry $W_{s,t}$ forward; asked about season t , she retrieves

$$r_{i,t} = c_i + \beta (W_{s(i),t} - c_i) + \kappa_{i,t} + u_{i,t}, \quad 0 \leq \beta \leq 1, \quad (1)$$

where c_i is the reference point toward which recalled seasons are drawn, $\kappa_{i,t}$ captures the context in which the memory is retrieved, and $u_{i,t}$ is recall noise with variance σ_u^2 . The parameter β is the *recall sensitivity*, the sensitivity of recall to the record: the share of a true difference between seasons that survives in recall. A season two points wetter than another is remembered as two points wetter with $\beta = 1$, but as one fifth of a point wetter with $\beta = 0.1$. Low recall sensitivity, $\beta < 1$, is what we call compression. With $\sigma_u^2 > 0$, neighbors who lived through the same season remember it differently.

Equation (1) is deliberately reduced form: a linear pull toward a central value is among the oldest regularities in judgment and arises in several models of memory.² The linear mapping describes ordinary seasons; vivid extremes may instead be retrieved as events through a separate availability channel (Bordalo et al., 2023; Tversky & Kahneman, 1973). Compression and salience can therefore coexist: ordinary seasons are pulled toward c_i , while a remembered extreme keeps its own weight in what the farmer carries forward. Section 5.3.1 examines this channel.

Measuring β requires a record of $W_{s,t}$ on the scale of recall, which is why the record is classified on the elicitation’s nine-picture scale (Section 4.3), and several seasons per farmer, so that respondent fixed effects absorb her reference point, her use of the scale and the

²See Hollingworth (1910), Huttenlocher et al. (2000) and Petzschnner et al. (2015) for the regularity, and Woodford (2020), Enke & Graeber (2023) and Azeredo da Silveira et al. (2024) for models of Bayesian decoding of a noisy memory trace, cognitive imprecision and optimally imprecise memory. Appendix F.1 tests their comparative statics.

persistent part of $\kappa_{i,t}$, while season indicators absorb anything common to the order of recall.³ Equation (8) is the empirical counterpart, and Section 5.1 tests whether $\beta < 1$ and $\sigma_u^2 > 0$.

3.2 From Memory to Expectations and Protection

The farmer forms an expectation for the coming season from what she remembers and from whatever else she knows,

$$f_i = (1 - \lambda) a_i + \lambda \sum_{k=1}^K w_k r_{i,t-k}, \quad (2)$$

where a_i summarizes other inputs such as forecasts and conversations, the w_k are non-negative weights that sum to one, and $\lambda \in [0, 1]$ is the weight on remembered experience. The elicited baseline response is

$$\tilde{f}_i = f_i + e_i, \quad (3)$$

where e_i is elicitation noise. Substituting Equation (1) into Equation (2) gives

$$f_i = \text{const}_i + \lambda\beta \sum_{k=1}^K w_k W_{s(i),t-k} + \lambda \sum_{k=1}^K w_k (\kappa_{i,t-k} + u_{i,t-k}). \quad (4)$$

A one-point shift in the weighted weather history moves her expectation by $\lambda\beta$, an accounting of how much objective variation survives both steps rather than a causal mediated effect. Memory can therefore be an important input, with λ large, and still pass on little of the record when β is small, while recall noise enters with weight λ . Because both are elicited on the same scales, we estimate the empirical counterpart of λ , a descriptive association, from each farmer’s own recalled history (Section 5.2.1) and pass the record through the estimated β , with the reference point at the middle picture and the average dry lean as a constant, to show how flat perceived histories are (Section 5.2.2).

Expectations matter because they govern protection. A farmer who can pay a cost P before the season to avoid a loss L if the season falls in an adverse region $\mathcal{A}$ of the scale protects when her perceived probability of an adverse season exceeds the cost–loss ratio,

$$d_i = \mathbf{1}\{\hat{\pi}_i(\mathcal{A}) \geq P/L\}, \quad (5)$$

³Patel (2026) shows that the circumstances of retrieval shape what is recalled. Our respondents recall all five seasons in one sitting, so such influences are largely common to a respondent’s seasons, and season indicators leave our estimate similar; we do not study short-lived context effects.

where $\hat{\pi}_i$ is the subjective distribution of which f_i is the mean. Because the decision turns on tail probability and not only on the mean, the full distributions elicited with tokens matter (Appendix F.3). Equation (4) implies that objective weather history enters expected weather only in proportion to $\lambda\beta$. When β is small, differences in experienced weather therefore translate weakly into differences in perceived risk. Whether these differences change protection depends on how they shift subjective tail risk relative to P/L . A remembered extreme that enters $\hat{\pi}_i$ directly, by contrast, can. Compressed memory thus weakens the link between the risk a farmer faces and the risk she perceives, and leaves protection to follow the extremes she remembers, without by itself pinning down average demand. The chain therefore has four links: the record to memory (β), memory to expectations (λ), remembered extremes to expected recurrence, and expectations to protection (Equation (5)). We measure the first three and relate remembered extremes to planned protection (Sections 5.1–5.3.2); the last link is the one that recent experiments estimate (Burlig et al., 2024; Patel, 2025; Zappalà, 2024).

3.3 Information About the Historical Record

If memory discards the record, showing the record restores the signal. Suppose the farmer is shown a summary S_s of the recent local record, as in our information treatment, and revises toward it,

$$f'_i = f_i + \omega (S_{s(i)} - f_i), \quad 0 \leq \omega \leq 1, \quad (6)$$

where ω reflects how much she trusts the record relative to her prior belief. The post-information elicited response is

$$\tilde{f}'_i = f'_i + e'_i. \quad (7)$$

The revision is proportional to the initial distance $S_s - f_i$, so farmers at the record do not move and those furthest away move most. Because the record requires no reading (Section 4.4), ω need not vary with literacy or schooling even when the initial distance does. This lets us test whether human capital differences arise primarily before the signal rather than in the response to it.

Elicitation noise e_i enters the observed revision $\tilde{f}'_i - \tilde{f}_i$ and the observed initial distance $S_s - \tilde{f}_i$ with the same sign, so a regression of one on the other is positive even when $\omega = 0$. Hence ω is identified only relative to a group that answers twice without seeing S_s , the role of Arm A (Section 4.5; Appendix C.2). Section 5.4 tests these predictions.

Along the chain, the framework organizes four empirical tests, which Section 5 takes to the data in order:

- (i) Recall responds less than one for one to the record, $\beta < 1$, and neighbors who lived through the same season remember it differently, $\sigma_u^2 > 0$ (Section 5.1.1).
- (ii) Remembered weather predicts expectations, $\lambda > 0$, so that the record reaches expectations only in proportion to $\lambda\beta$ (Sections 5.2.1 and 5.2.2).
- (iii) Salient extremes survive compression and carry into expected recurrence and planned protection (Sections 5.3.1 and 5.3.2).
- (iv) A historical record moves beliefs toward it, including for farmers who cannot read (Sections 5.4.1 and 5.4.2).

4 Survey and Experimental Design

To implement this framework in the field we designed a survey experiment that combines three elements: a broad survey of farmers and their production environment, a visual elicitation that measures memories and expectations without relying on literacy or formal probabilities, and an information experiment that tests whether those beliefs move when farmers see, and are able to process the local weather record.

4.1 Survey Instrument

Each respondent completed a structured interview covering demographics, farming and land use, access to agricultural information, weather memories and expectations, planned adaptation and spending, risk perceptions, and the location of the main plot. The survey also recorded free recall of extreme weather events, the sources farmers use when forming expectations, and the role of social learning. Online Appendix OA-2 reproduces the full instrument. This breadth lets us characterize who remembers accurately, relate expectations to planned protection, and test whether farmer characteristics predict updating.

4.2 Belief Elicitation

Standard probability questions are poorly suited to a setting where many respondents cannot read and formal probabilities are unfamiliar. A common solution in the expectations

literature is to represent probabilities physically. Respondents allocate a fixed number of tokens across pictured outcomes, and the allocation represents their subjective probability distribution (Delavande, 2014; Delavande et al., 2011). The method requires no literacy and only basic counting. We use it to measure two dimensions of weather beliefs: the conditions farmers associate with a season and the timing of the planting rains.

Our main instrument is a sheet showing the same representative maize field under nine conditions, from extreme drought in picture one, through optimal growing conditions in picture five, to severe flooding in picture nine (Figure 1). The images were built from a large set of photographs of agricultural fields in the study region and combined into a synthetic representation that looks realistic and familiar to respondents while varying the water conditions depicted.

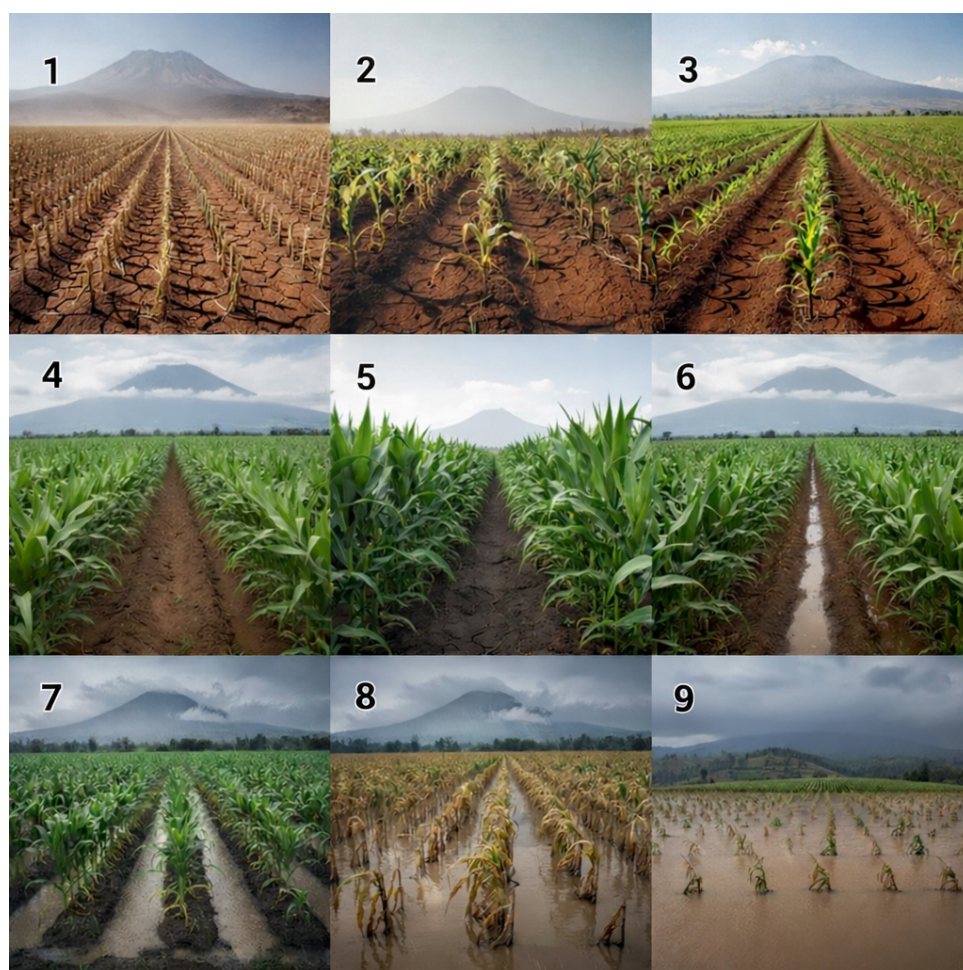


Figure 1: The nine-picture sheet over which farmers allocated ten tokens.

Notes: Source: authors' field materials.

Before the elicitation, the enumerator read a fixed description of all nine images and gave the respondent ten physical tokens. Each token was one vote, to be placed on the image or images that best described the season in question, so the allocation gives a discrete subjective distribution over nine ordered outcomes. We repeated the exercise for each of the five past seasons from 2021 to 2025 and for the coming 2026 season. Respondents who could not remember a past season after two neutral scripted probes were not required to allocate tokens. The share of respondents who placed tokens for a season falls with distance in time, from 99.3 percent for 2025 to 61.7 percent for 2021, yielding 2,637 respondent-seasons. Online Appendix OA-3 reproduces the elicitation script, image descriptions and enumerator instructions.

Rainfall onset

A second instrument elicits when the planting rains begin. Farmers allocate the same ten tokens across ten dated intervals spanning 15 February to 24 April (Appendix Figure 11), which gives a subjective distribution over the timing of onset in the same physical format. We measure onset separately because the timing of the first reliable rains is central to planting. Farmers must decide when there is enough moisture to plant without knowing whether the rains will continue, and the local field team described this timing as increasingly difficult to predict. Planting too early can leave germinating seed exposed if the rains stop, while planting too late shortens the growing period and can reduce yields. We repeat the onset exercise for each past season and for 2026.⁴

Extreme-event recall

Separately from the season-by-season elicitation, respondents were asked to recall any major weather events without being prompted by year, and to report the hazard type, year and perceived severity of each. This free-recall exercise produced 550 first-named events, including droughts, floods, hailstorms and storms, and provides a second measure of weather memory organized around salient events rather than seasons. Respondents were then asked whether they expected a major weather event in the coming season and, if so, of what type and how severe.

⁴Elicited onset is consistently 26 to 32 days later than our rainfall-based benchmark, while the benchmark varies little across the study area. We therefore use onset only for within-person and experimental comparisons, not for cross-sectional measures of accuracy. Appendix B.3 provides details.

Expectations

We measure expectations for the coming season with the same two instruments. Farmers allocate tokens across the nine-picture scale for expected 2026 conditions and across the onset calendar for the expected timing of the rains, and report their confidence. Because memories and expectations are measured on the same scales, we can trace remembered experience directly into beliefs about the future.

The picture scale measures beliefs as well for farmers who cannot read as for those who can; the onset calendar, which relies on dated intervals, is somewhat noisier for farmers who cannot read. In Arm A, where farmers stated their expectation for 2026 twice with no information in between, the correlation of the mean picture across rounds is 0.44 for literate and 0.41 for non-literate farmers, with no detectable difference in agreement by literacy. Recall is also internally consistent, with seasons freely named as droughts recalled 0.36 pictures drier on the token scale and no difference by literacy. Enumerator identity explains under 1 percent of the variance in the mean picture of expected and recalled distributions among respondents interviewed in the same village on the same day, compared with 28 percent for the verbal question of whether an extreme event is expected. The spread of the allocations, expected onset, and the number of seasons recalled depend more on the interviewer, so our main estimates rest on the location of the distributions (Appendix I).

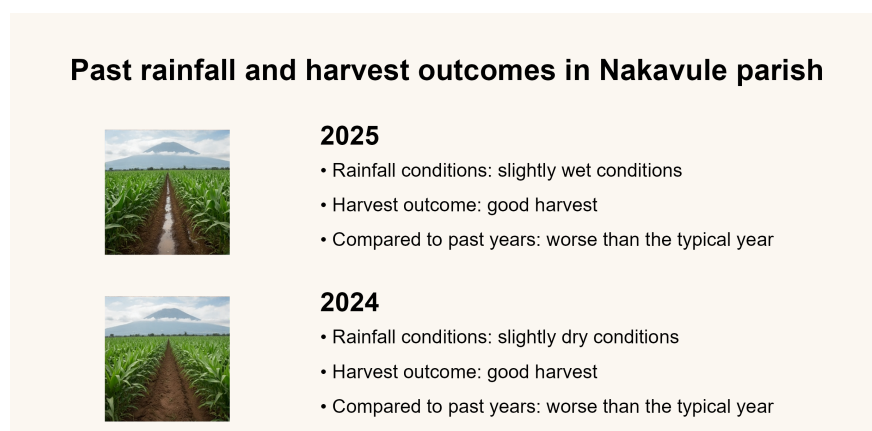
4.3 The Historical Weather Record

Assessing the accuracy of memory requires an objective measure of what occurred on the same scale used in the elicitation. We place each subcounty-season from 2021 to 2025 on one of the nine pictures using a sorting algorithm based on fixed rules that combines satellite rainfall data (Funk et al., 2015) with remotely sensed vegetation data from Sentinel-2. The algorithm combines rainfall over three onset-aligned stages of the growing season with peak vegetation greenness within cropland and maps these conditions onto the nine-picture scale. Appendix B.1 gives the full rule and thresholds.

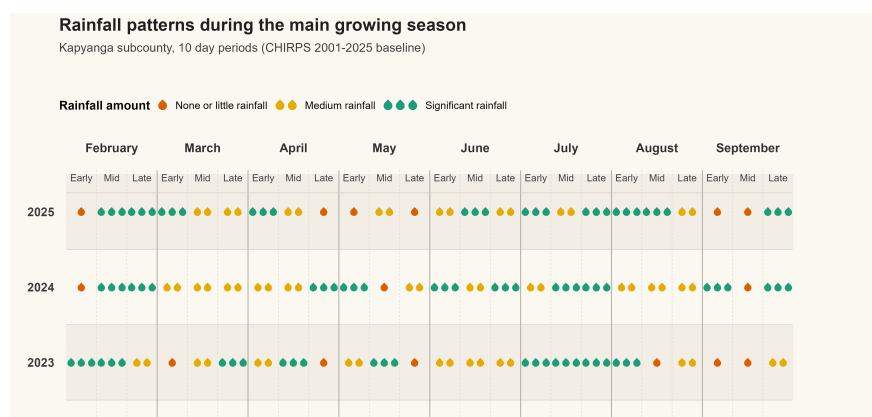
The record varies by subcounty and season. The 60 study villages span 42 parishes and 11 subcounties, giving 55 subcounty-year cells over 2021–2025, while the 686 farms fall within 17 rainfall grid cells. Weather histories are therefore not independent across farms or villages, which the inference described in Section 4.5 and Appendix C.6 takes into account. We benchmark free recall of extreme events separately against the 2001–2025 rainfall record for each farmer’s subcounty.

4.4 The Information Treatment

The information treatment presents recent local weather history in the same visual language as the elicitation. Each treated respondent sees a printed card covering the five seasons from 2021 to 2025. For each season, the card shows the corresponding picture from the nine-picture scale together with short descriptions of rainfall, crop conditions, and how the season compared with the other years shown. A companion calendar summarizes rainfall through the season using one to three water-drop symbols for low, typical, or high rainfall in each ten-day period. Because the record uses the same format as the elicitation, farmers can directly compare it with the seasons they had just recalled. Figure 2 shows the two central sheets, and Appendix B.5 reproduces a complete card.



(a) The overview sheet (excerpt)



(b) The rainfall calendar (excerpt)

Figure 2: The information treatment.

Notes: Excerpts from one parish's card, as shown in Arms B and C. Drops are relative to the subcounty's CHIRPS 2001–2025 baseline. Appendix B.5 reproduces the complete card. Source: authors' field materials.

Three design choices matter for interpretation. First, the cards were generated automatically from the satellite data using the fixed classification rule, so no researcher judgment entered between the record and the information farmers received. Second, the treatment requires no reading. It uses the same pictures and symbols as the elicitation, while the enumerator can communicate the accompanying text orally. Third, the treatment contains no forecast. It reorganizes recent local weather history into an accessible form rather than providing information about the coming season. The intervention is therefore close to Hanna et al. (2014), in that farmers receive a usable summary of experience they have already lived through.

The design also limits what the experiment identifies. Across the 42 parishes, the cards contain only eight distinct weather histories, and the displayed seasons span pictures three to seven, so the experiment does not identify responses to the most extreme conditions. Because the second elicitation follows the treatment within minutes, immediate updating also cannot be cleanly separated from anchoring on the displayed information. Appendix B.2 discusses these limitations and their implications for regressions on card content.

4.5 Randomization and Treatment Arms

We randomized the 60 villages into three arms. Assigning treatment at the village level limits information spillovers between treated and untreated farmers in the same community. We stratified by subcounty and elevation tercile and used elevation and satellite greenness to form 30 matched pairs. Ten pairs served as donor pairs for the smaller Arm C, while the remaining 20 were split between Arms A and B. We rerandomized assignments to improve balance on five geographic covariates using a fixed procedure and seed (Bruhn & McKenzie, 2009; Imai et al., 2009). Table 1 reports arm means, and Appendix A.3 documents the procedure, pairing, and balance tests. Assignment was hard-coded into the survey form five days before fieldwork began, leaving enumerators no discretion. Standard errors are clustered by village. For the main estimates, we also report wild cluster bootstrap p-values (Cameron et al., 2008), while Appendix C.3 reports Romano–Wolf adjustments for families of related hypotheses.

Treatment arms

Figure 3 summarizes the three arms. All farmers complete the same survey and memory elicitation. Arm A then completes the expectation elicitation twice with no information in between, capturing any change from answering twice. Arm B completes the same two

elicitations but receives the card between them. The difference in revisions between Arms B and A therefore identifies the effect of the information and is the primary treatment contrast throughout the paper.

Arm C sees the card before a single expectation elicitation. It serves as a smaller robustness arm. The analysis compares its first-stated beliefs with the pre-information beliefs of Arm A, while a comparison with Arm B after revision tests whether stating an initial belief changes the response to the information. Appendix H.4 reports these comparisons. Recall is elicited before any information in every arm and underlies Section 5.1; the first expectation elicitation in Arms A and B supplies the untreated expectations used in Section 5.2.

Reaction questions

After seeing the card and before stating their second expectation, farmers in Arm B answered six short questions (Online Appendix OA-2): whether the information differed from what they remembered about past rainfall, how surprised they were, whether they wanted to adjust their expectations and, if so, how confident they were in the adjusted expectations, whether historical rainfall information was helpful for planning, and whether they wanted similar information in the future. Arm C answered the same questions before its single elicitation. Arm A answered no comparable questions. The B–A contrast therefore captures the information as delivered, including any effect of these reaction questions, which the data cannot separately identify.

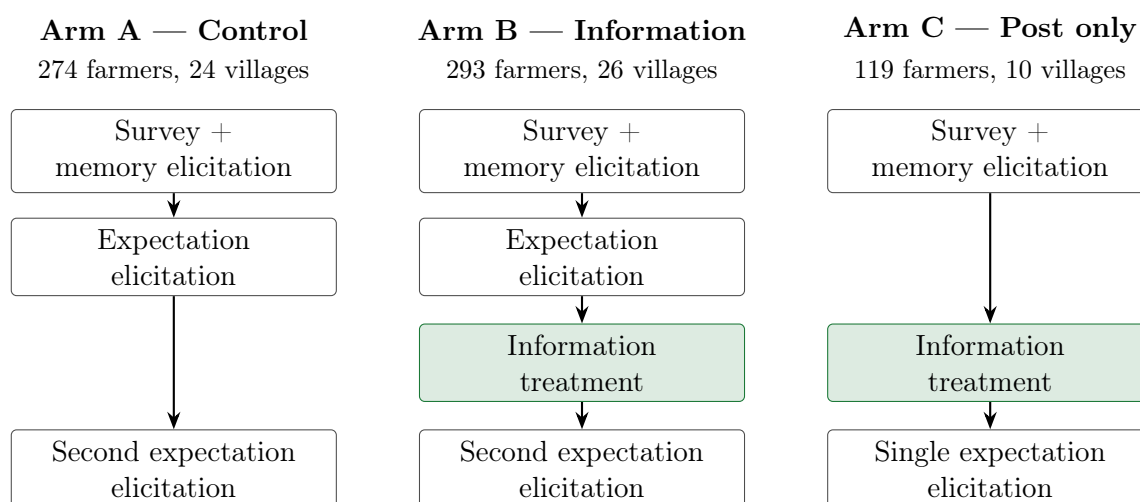


Figure 3: The three experimental arms.

Notes: Sequence of the interview in each arm; green-shaded boxes mark the information treatment. Villages are the randomization units. Source: authors' study design.

5 Results

This section takes the chain of Section 3 to the data, link by link. We first measure what memory keeps of the record (Section 5.1). We then show how remembered weather maps into expectations (Section 5.2), examine how salient extremes carry into expected recurrence and planned adaptation (Section 5.3), and finally test whether providing the historical record changes beliefs (Section 5.4).

5.1 Memory of Past Seasons

Memory preserves only a small share of the variation in past weather. Recalled conditions move far less than one for one with the historical record, and this compression remains substantial after allowing for noise in the benchmark. All estimates in this subsection use memories elicited before any information was shown.

5.1.1 Compression of Recalled Weather

Recall sensitivity

We measure how strongly recall responds to the historical record by regressing each farmer’s recalled conditions for a season on the recorded conditions for that season,

$$\bar{r}_{i,t} = \alpha_i + \beta m_{s(i),t} + \varepsilon_{i,t}, \quad (8)$$

where $\bar{r}_{i,t}$ is the token-weighted mean of respondent i ’s recalled distribution and $m_{s(i),t}$ is the record’s classification of that season in her subcounty. Respondent fixed effects absorb persistent differences in scale use, so β measures recall sensitivity to differences in recorded weather. A value of one implies one-for-one recall, while values below one imply compression.

The estimate is $\beta = 0.0948$ (standard error 0.0342), based on 2,559 respondent-seasons from farmers who recall more than one season. The economically relevant benchmark is one-for-one recall, $\beta = 1$, which is rejected decisively even with inference at the 11 subcounties where the record varies ($p < 0.001$). Against zero, the village-level wild cluster bootstrap gives $p = 0.0091$; Appendix C.6 reports the sensitivity of this test to alternative levels of inference and specifications. A one-point difference between two seasons in the record therefore appears as only about 0.1 points in recall. Figure 4 shows the same pattern visually, with recalled values concentrated near the middle of the nine-picture scale.

The result is not driven by a particular season, selective recall of seasons, or misdating. The slope is similar across all five seasons and among farmers who recall all five, and accounting for misdated seasons leaves the result unchanged (Appendix H.1). Memory therefore preserves little of the differences across ordinary seasons. The weather history farmers remember contains much less variation than the one they experienced.

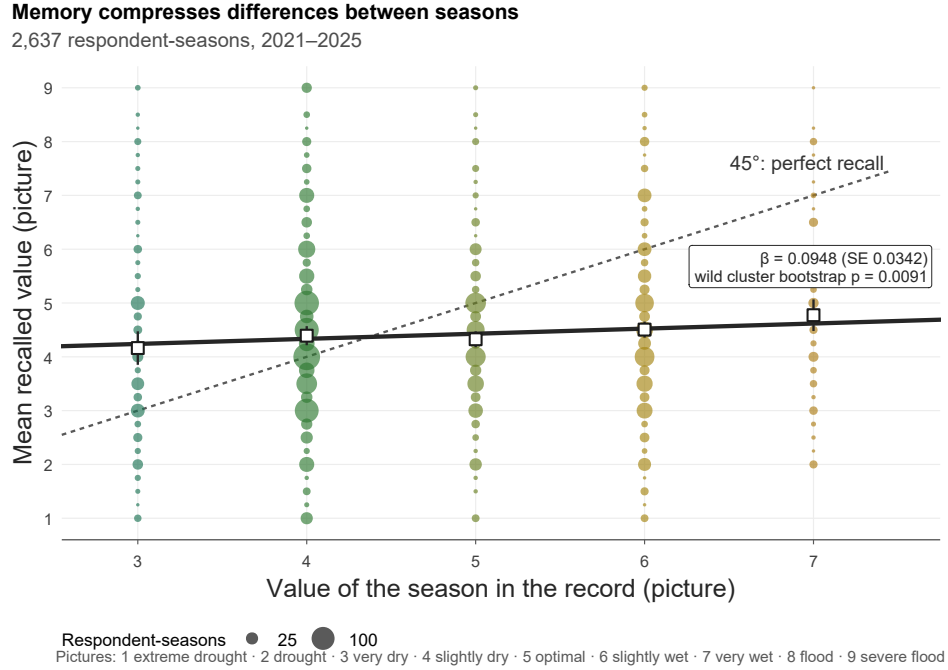


Figure 4: Memory is compressed.

Notes: Discs show recalled against recorded conditions on the nine-picture scale, sized by the number of observations; squares show mean recall at each recorded value with 95% village-clustered confidence intervals. The dashed line denotes one-for-one recall, and the solid line shows the estimated recall slope. The slope is estimated on 2,559 respondent-seasons from farmers who recall more than one season. Source: authors' computation.

Measurement error and mechanisms

A natural concern is that the small recall slope reflects noise in our subcounty-level weather benchmark rather than compression in memory. Alternative corrections for this noise raise the estimated slope but leave it far below one. Instrumenting the card classification with a rainfall-only classification gives a slope of 0.27, while using continuous seasonal rainfall gives 0.28 (95 percent confidence interval [0.10, 0.63]). One-for-one recall is still rejected with inference at the 11 subcounties where weather varies ($p = 0.005$). Even allowing for substantial measurement error, recall therefore preserves only about a quarter to a third of the variation in past weather (Appendix F.2).

The data identify compression more clearly than its underlying mechanism. Appendix F.1 shows that recall is drawn toward a common reference point rather than each subcounty’s own average, and that compression is stronger when recalled distributions are more diffuse. These patterns are consistent with models of imperfect memory discussed in Section 3.1, but they do not distinguish sharply among them.

5.1.2 Bias, Recall Age, and Heterogeneity

Compression is accompanied by a modest dry bias. On average, recalled conditions are 0.324 points drier than the record. This average masks much larger errors at the ends of the scale. Seasons that were dry are recalled 1.163 points wetter, while wet seasons are recalled 2.226 points drier; middle seasons are recalled 0.379 points drier. Both dry and wet seasons are therefore pulled toward the common point identified in Section 5.1.1, with recall settling slightly on the dry side of normal.

Recall age and heterogeneity

Recall accuracy does not decline detectably with age over the five-year window. We measure accuracy using the ranked probability score, with lower values indicating better recall. Older seasons initially appear less accurately remembered, but they were also more extreme, and extreme seasons are harder to recall due to compression. Once we account for these differences in recorded weather, the relationship between age and accuracy disappears. The estimated age gradient is -0.0091 and statistically indistinguishable from zero.⁵ Older memories do, however, become more concentrated. Farmers report narrower distributions for older seasons without recalling them more accurately. Memory therefore becomes more definite with age, but not more accurate.

Literacy strongly predicts recall accuracy. Respondents who cannot read recall past seasons 0.33 standard deviations less accurately than literate respondents (wild cluster bootstrap $p = 0.0001$). Schooling beyond its association with literacy adds little explanatory power, with essentially no relationship between schooling level and accuracy once literacy is held fixed ($p = 0.75$). Older respondents and those in larger households also recall somewhat less accurately. Literacy therefore predicts how accurately farmers remember past weather, but not the direction of their errors.

⁵Recalled planting onset does drift with age: remembered onset moves 1.45 days later for each additional year of recall age, while recalled seasonal conditions show no corresponding drift (Appendix B.3).

5.2 From Memory to Expectations

Memory shapes expectations, but compression limits how much variation in past weather reaches beliefs about the future. We first quantify how remembered history maps into expectations. We then show how the estimated compression maps objective weather histories into much flatter perceived histories and that expectations themselves vary little across places.

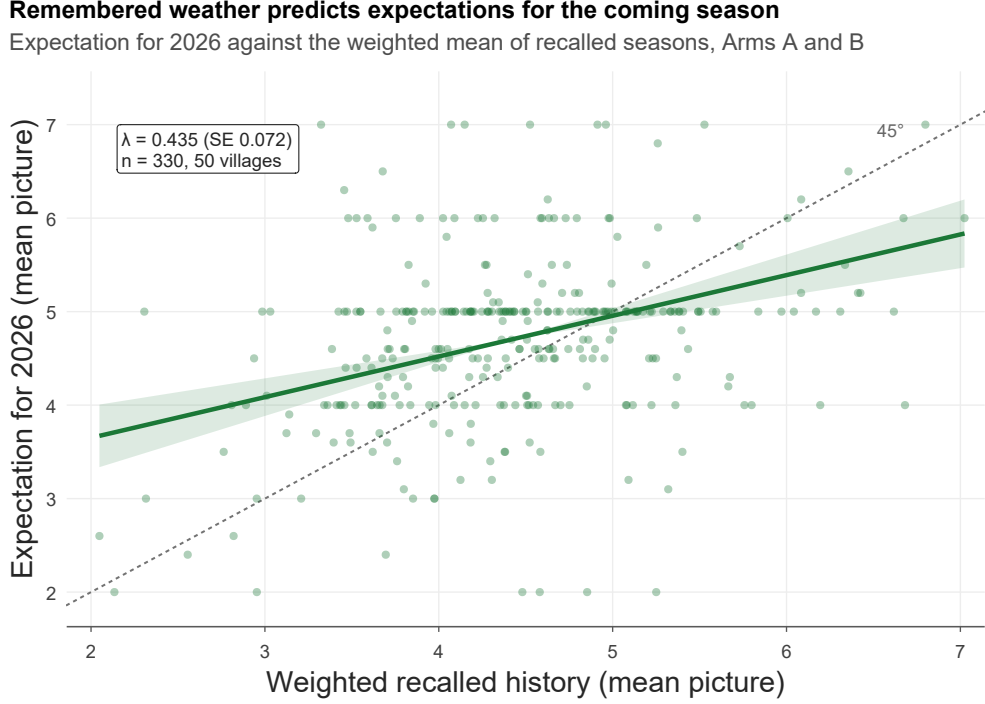


Figure 5: Remembered weather predicts expectations for the coming season.

Notes: Mean pre-information expectations for 2026 plotted against the weighted mean of recalled conditions for the 330 respondents in Arms A and B with complete five-season histories. Both variables use the nine-picture scale. The line shows the fitted relationship from Equation (9). Source: authors' computation.

5.2.1 Memory and Expectations for the Coming Season

Farmers who remember a wetter recent history expect a wetter coming season. Using expectations elicited before any information was shown, we relate each farmer's expected conditions for 2026 to a weighted average of the previous five seasons as she remembers them,

$$\bar{f}_i = \delta_{s(i)} + \lambda \sum_{k=1}^5 w_k \bar{r}_{i,t-k} + \mathbf{x}'_i \boldsymbol{\theta} + u_i, \quad (9)$$

where $\bar{f}_i$ is her mean expectation on the nine-picture scale, $\bar{r}_{i,t-k}$ is her recalled value for an earlier season, and the weights w_k sum to one. Subcounty fixed effects absorb common local differences, while $\mathbf{x}_i$ contains respondent controls. In the framework, λ is the weight placed on remembered experience; empirically, it measures the descriptive association between remembered history and expectations.

Among the 330 respondents in Arms A and B with complete five-season histories, λ is 0.4351 (standard error 0.0717). A farmer whose remembered history is one point wetter therefore expects the coming season to be about 0.44 points wetter.⁶ The estimate remains between 0.40 and 0.45 across specifications with alternative fixed effects, respondent controls, controls for response style, and reweighting for complete histories. A cross-instrument placebo also shows no corresponding relationship from generic scale use (Appendix H.2). Remembered weather is therefore an important input into expectations. Combined with $\beta = 0.0948$, however, a one-point difference in objective weather history translates into only about 0.04 points in expected conditions through memory. The bottleneck is therefore how little variation memory preserves.

5.2.2 Perceived Weather Histories

Expectations vary much less across places than the weather those places experienced. Across the 11 subcounties, mean weather over 2021–2025 has a standard deviation of 0.98 points on the nine-picture scale, while farmers’ average expectation for 2026 has a standard deviation of only 0.28 points. Farmers in different places therefore experienced quite different recent histories but enter the coming season with much more similar expectations. This pattern is consistent with the compression documented in Section 5.1.

To quantify what compression implies, we apply the estimated recall mapping from Section 5.1 to the weather histories each subcounty actually experienced. Here the objective history uses continuous seasonal CHIRPS rainfall mapped onto the nine-picture scale, rather than the discrete card classification used to estimate β . We scale deviations from the middle of the nine-picture scale by the estimated recall sensitivity, $\beta = 0.0948$, and apply the estimated dry bias of -0.324 points. This produces a constructed perceived history rather than an observed expectation.

The effect is large. Under accurate memory, five-season histories differ by 3.58 points across places and years, from clearly dry to clearly wet. After applying the estimated dis-

⁶Equal weighting of the five seasons gives an estimate of 0.4092. In Arm A alone, where farmers never receive information, the estimate is 0.3243 (standard error 0.1131). Appendix H.2 reports these and additional specifications.

tortion, the same histories differ by only 0.34 points, and their standard deviation falls from 0.764 to 0.072. About nine tenths of the variation that distinguishes one recent weather history from another therefore disappears in memory.

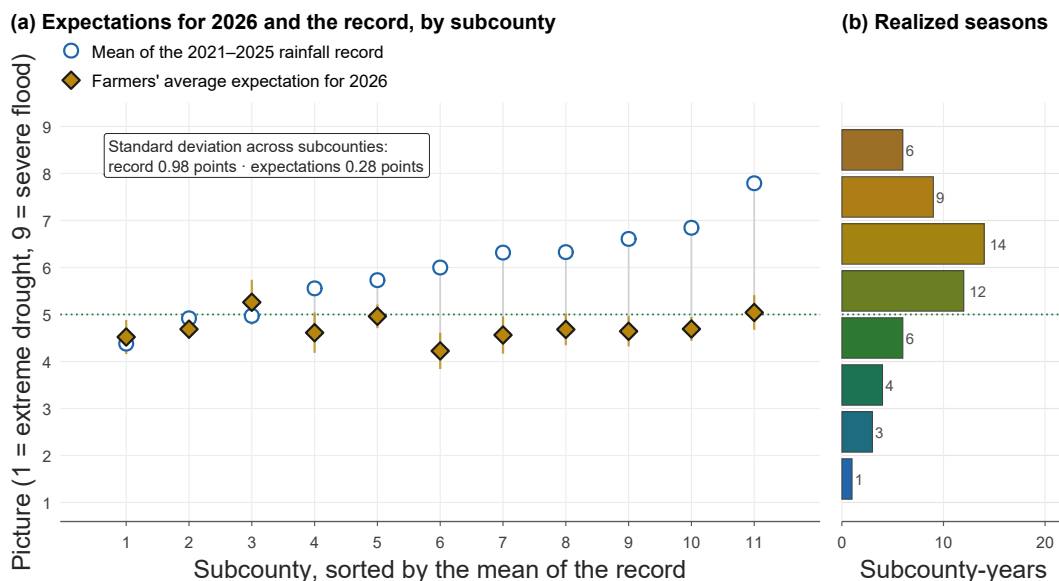


Figure 6: Farmers' expectations vary far less across subcounties than the record.

Notes: Panel (a) compares mean weather over 2021–2025 with average expectations for 2026 across the 11 subcounties. Panel (b) shows the distribution of realized weather across the 55 subcounty-year cells. All outcomes are expressed on the nine-picture scale. Source: authors' computation.

The flatness is not an artifact of the elicitation. Nearly half of respondents place all ten tokens on a single picture, so expectations are not flat because farmers spread probability evenly across the scale. The compression result is also similar when we collapse the scale into dry, normal, and wet categories and when we account for conditions shared within villages (Appendix D.4).

The flatness matters because protection depends on recognizing when extreme conditions are plausible. Before receiving any information, 75 percent of farmers assign no probability to picture three or drier and 90 percent assign none to picture seven or wetter, even though seasons this extreme occur roughly one year in four in the 2001–2025 rainfall record. This is not because extremes disappear from memory. About a third of all recalled tokens lie on picture three or drier, compared with only about a tenth of tokens in expectations for 2026. What appears to be lost is not the memory of extreme seasons themselves, but information about how often they occurred. Appendix F.3 quantifies the implications for protection decisions.

5.3 Extreme Events and Adaptation

Compression characterizes ordinary variation in past weather, but extreme events appear to be remembered differently. We next show that farmers who recall past extremes are more likely to expect another extreme and plan substantially more adaptation.

5.3.1 Extreme Events and Expected Recurrence

Extreme events stand out from ordinary seasonal variation. Farmers who freely recall a past extreme are much more likely to expect another extreme in the coming season. This form of event memory is largely distinct from season-by-season recall, with only weak correlations with its accuracy, concentration, and dry bias.⁷

Among farmers who freely recall a past extreme, 48.1 percent expect an extreme event in the coming season, compared with 27.8 percent among those who recall none. With subcounty fixed effects, recalling an extreme is associated with a 19.4 percentage point higher probability of expecting one in 2026 (standard error 4.5 percentage points). The association remains large when we control for a general tendency to answer yes and when we compare farmers interviewed by the same enumerator, although the latter estimate is less precise (Appendix H.7).

The content of these memories also carries into expectations. Farmers who recall more severe events expect more severe events, and those who remember a particular hazard are more likely to expect the same hazard again. When event recall and season-by-season memory enter together, event recall and the concentration of seasonal memories predict whether a farmer expects an extreme, while recall accuracy does not. Extreme-event memory therefore appears to be a distinct channel through which past experience enters expectations.

Salience does not imply accuracy. We compare each recalled extreme with the 2001–2025 rainfall record for the farmer’s subcounty. Recalled floods line up reasonably well with measured rainfall, but recalled droughts do not. Years named as droughts fall around the 61st percentile of seasonal rainfall, and their longest dry spells around the 53rd percentile. Reported drought severity is also unrelated to the measures of realized dryness available to us. Because our weather record is measured at the subcounty-season level, it may miss highly localized or crop-sensitive dry spells. We therefore interpret this evidence cautiously. Farmers clearly remember drought events, but the years they identify as droughts are not systematically drier in the weather measures we observe.

⁷The correlations are 0.079, 0.126, and -0.107 , respectively.

One possible explanation for this flood–drought asymmetry is that floods are discrete and visible events, whereas judging whether a season was a drought requires comparison with a remembered normal. Compression may make that comparison harder. Consistent with this interpretation, farmers who recall a drought have a stronger dry bias in their season-by-season memories, -0.555 points compared with -0.248 among farmers recalling another hazard. The evidence is suggestive rather than identifying, and Appendix D.6 examines the pattern in more detail.

5.3.2 Remembered Extremes and Planned Adaptation

Remembered extremes strongly predict planned adaptation. Among farmers in Arms A and B, those who recall a past extreme plan to spend 0.77 log points more on adaptation (wild cluster bootstrap $p = 0.0001$). The estimate is 0.78 across all three arms and survives Romano–Wolf adjustment across the full grid of memory, belief, and adaptation outcomes ($q = 0.0002$; Appendix D.5). In our richest specification, farmers who recall an extreme still plan to spend 0.50 log points more ($p = 0.008$) and are 12.1 percentage points more likely to plan any adaptation ($p = 0.042$; Appendix H.7). The association concerns whether and how much farmers plan to adapt rather than any single practice. Crop insurance plays almost no role, with only eight of the 404 farmers planning any adaptation intending to buy it.

Together, the results show two descriptive links around salient events. Farmers who remember an extreme are more likely to expect another and plan substantially more adaptation. We do not interpret our data as identifying the final step from expectations to adaptation. Verbally stated expectations of an extreme also predict planned spending, but that relationship does not survive comparison of farmers interviewed by the same enumerator (Appendix H.7). Existing experimental evidence shows that changes in weather and environmental beliefs can affect investment and adaptation decisions (Burlig et al., 2024; Giné et al., 2015; Patel, 2025; Zappalà, 2024). Our evidence extends this process upstream by showing that remembered extreme events are strongly associated with the beliefs and plans farmers carry forward.

The contrast with ordinary seasons is sharp. With only a small share of recorded seasonal variation surviving in recall, season-by-season memory has little systematic variation left to pass into adaptation plans, and its associations with planned adaptation are mixed across outcomes (Appendix Table 11). Remembered extremes, by contrast, retain a strong association with both expected recurrence and planned protection. Memory therefore appears to transmit salient events much more strongly than ordinary variation in past weather.

5.4 Information and Belief Updating

Providing farmers with their local weather record changes their beliefs. The information moves expectations toward the record and increases confidence, including among farmers who cannot read. Because the card contains historical information rather than a forecast, these results show that making past experience easier to observe can correct part of the information lost through memory.

5.4.1 Effects of the Information Treatment

Showing farmers their local weather history moves expectations toward the record. Farmers in Arms A and B state their expectations twice, but only Arm B sees the card between the two elicitations. Because repeating the elicitation can itself change answers, we estimate the treatment effect from the difference in revisions across the two arms,

$$\Delta y_i = \alpha + \tau B_i + \mathbf{x}_i' \boldsymbol{\theta} + \eta_i, \quad (10)$$

where Δy_i is the change in farmer i 's belief and B_i indicates assignment to Arm B. The coefficient τ therefore measures how much more beliefs change when farmers see the card than when they simply answer the same question twice. Standard errors are clustered by village. Arm C is excluded because its farmers see the card before their only expectation elicitation.

We measure movement toward the record using the distance between the farmer's expected distribution and the distribution shown on the card. Relative to Arm A, the card reduces this distance by 0.134 points (standard error 0.050). A second measure, the ranked probability score against the recorded 2025 season, falls by 0.151 points (standard error 0.051). Lower values on both measures indicate beliefs closer to the record, and the two measures are highly correlated. Both effects remain below 0.10 after Romano–Wolf adjustment across the family of belief-updating outcomes ($q = 0.060$ and 0.063 ; Appendix C.3). The card therefore moves expected rainfall conditions toward the historical record.

Expectations about timing also move. Relative to Arm A, the card shifts expected onset of the planting rains 2.6 days earlier (standard error 0.86; village-clustered $p = 0.004$). This shift appears when farmers revise a belief they have already stated and is weaker with interview-date fixed effects (Appendices H.4 and H.6). Because the rainfall-based onset benchmark often falls at the earliest admissible date on our calendar, we interpret this result as a shift toward earlier expected onset rather than as an improvement in forecast accuracy.

The card also increases confidence in rainfall expectations. Relative to Arm A, stated confidence rises by 0.18 points on the five-point scale (wild cluster bootstrap $p = 0.030$). The same pattern appears in Arm C, where farmers who saw the card before their only elicitation report confidence 0.39 points higher than farmers stating beliefs before seeing the card (randomization-inference $p = 0.004$; Appendix H.4). The information therefore changes not only where farmers place their expectations, but also how certain they feel about them.

The card changes how far farmers revise, not how many revise. The share whose beliefs move closer to the record is 27.7 percent in Arm A and 31.4 percent in Arm B. The difference is therefore not in the frequency of revision, but in its size and direction. Among farmers who revise, moves toward the record are larger in Arm B and large moves away from it are less common (Appendix C.2). Part of the treatment effect therefore comes from preventing beliefs from drifting further from the record.

5.4.2 Heterogeneity by Prior Beliefs and Literacy

Updating depends more on where farmers start than on who they are. For expected onset, the treatment effect is concentrated among farmers whose initial beliefs were not already early. Relative to Arm A, the card shifts expected onset by about four days in the middle and late terciles of initial beliefs, but has essentially no effect in the early tercile. Because each tercile in Arm B is compared with the same tercile in Arm A, this comparison nets out the mechanical revision caused by answering twice. The pattern is consistent with larger revisions among farmers who begin further from the information shown on the card, although the calendar floor prevents us from distinguishing movement toward the record from movement toward the earliest date the elicitation allows.

We find little evidence that updating varies with farmer characteristics. Across 22 pre-treatment characteristics, including education, literacy, age, gender, farm size, information access, and risk attitudes, none predicts a differential response to the card after adjustment for multiple testing. Nor do farmers with better access to forecasts, extension, or past weather information respond differently. These farmers recall past seasons more accurately, but we cannot reject that they compress seasonal differences and update in response to the card as much as others do (Appendices C.3, D.3, H.7, and F.5).

Literacy predicts where farmers start, but not how far they move. Before seeing the card, farmers who cannot read begin about 0.28 standard deviations further from the record than literate farmers, while their expected onset is similar. Schooling adds little once literacy is held fixed. Yet we detect no difference by literacy in how strongly the card shifts expectations.

The visual format therefore reaches the farmers who start furthest from the record, with no detectable weaker response among those who cannot read (Appendix H.7).

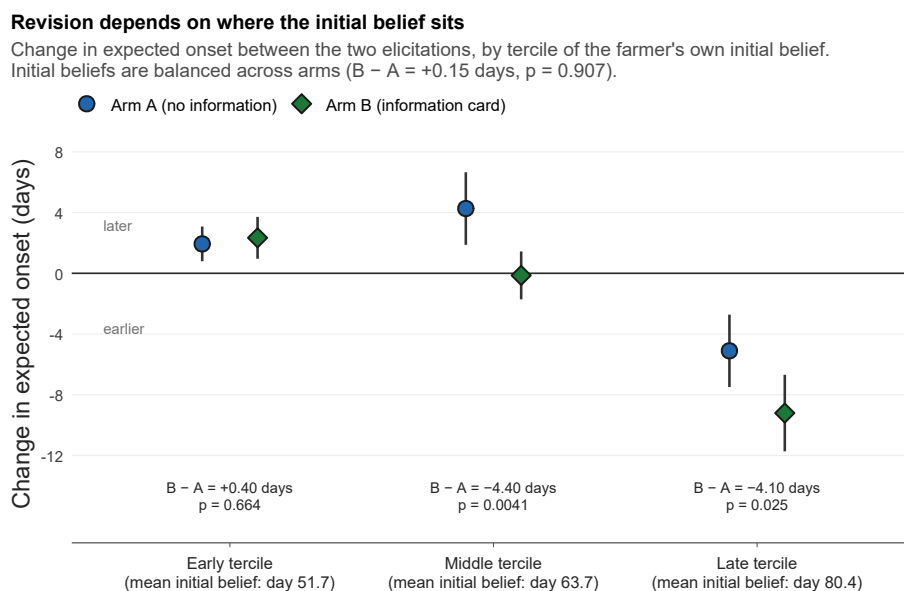


Figure 7: Revision depends on the initial belief.

Notes: Mean change in expected onset between the two elicitations by tercile of initial beliefs in Arms A and B. Negative values indicate revision toward earlier onset; whiskers show 95% village-clustered confidence intervals. Source: authors' computation.

One reason is the format of the information. The card uses the same pictures and symbols farmers had just used to state their own beliefs, so processing it requires neither literacy nor familiarity with formal probabilities. This is consistent with evidence that how information is presented affects whether it is used (BenYishay & Mobarak, 2019; Cole & Fernando, 2021; Hanna et al., 2014). Once the historical record is made accessible in this way, we detect no constraint from literacy or schooling on short-run belief updating.

5.4.3 Planned Adaptation and Farm Outcomes

The card moves beliefs but not, detectably, adaptation plans. Farmers in all arms reported their plans before the card and again at the end of the interview. The share planning any adaptation falls in every arm, by 8.0 percentage points in Arm A and 5.1 points in Arm B, giving a treatment effect of 2.9 percentage points (standard error 4.2; wild cluster bootstrap $p = 0.49$). Effects on planned spending, individual adaptation practices, and financing choices are similarly small and do not survive multiple-testing adjustment. With a minimum detectable effect of about 12 percentage points for the extensive-margin outcome, the experiment is not powered to detect modest changes in plans.

A more targeted test asks whether plans move in the direction implied by the information. Among farmers for whom the card shifts beliefs toward a drier outlook, a one standard deviation larger shift is associated with an 8.2 percentage point increase in planned adaptation relative to Arm A (standard error 4.0; wild cluster bootstrap $p = 0.05$). The sign is consistent with the framework, but the result is not robust to multiple-testing adjustment or alternative specifications, and much of the relationship is already present in plans stated before the card was shown. Conditioning on those baseline plans reduces the estimate to 5.8 percentage points ($p = 0.13$). We therefore treat this as suggestive rather than causal evidence (Appendix F.4).

Three features of the design help explain why beliefs move more clearly than plans. First, the weather history shown on the card is relatively benign, so for many farmers it changes beliefs without clearly increasing perceived risk. Second, plans are re-elicited within minutes, and Arm A shows that repetition alone changes reported plans substantially. Third, crop insurance, the margin most directly linked to weather beliefs, is almost absent in this setting. We therefore do not interpret the experiment as identifying a causal effect of revised beliefs on adaptation. Its stronger result is that beliefs feeding into adaptation decisions can be shifted with simple historical information.

The 2026 season provided little scope to test effects on farm outcomes. Rainfall from February through June was among the wettest in the 25-year record across all study subcounties, no location was classified as dry, and vegetation showed no detectable damage. Farm productivity also remained close to its historical average. We find no detectable treatment effects on satellite-measured farm outcomes (Appendix D.8). Because recorded farm coordinates cluster at interview sites rather than separate plots, these outcomes vary little within villages and the test is effectively at the village level. The experiment therefore shows that beliefs can be moved toward the record, but not whether doing so improves farm outcomes when adverse weather occurs.

5.4.4 Interpreting the Magnitudes

The estimates also help gauge the economic scale of compression and of the information treatment. This exercise is illustrative rather than causal. It combines our estimates with external evidence on how belief changes affect investment and welfare, and we do not treat the onset shift as an established gain in forecast accuracy.

Start with the cost of compression. Under the protection rule in Equation (5), even a farmer who accurately remembers only five seasons loses some of the value of protection

because five seasons provide a noisy picture of risk. With the compression we estimate, perceived risk varies much less with the weather actually experienced, and the forgone value of protection rises further (Appendix F.3). External evidence suggests that such belief errors can matter economically. Among Bangladeshi rice farmers, correcting beliefs about soil salinity changes demand for tolerant seed and can raise profits substantially (Patel, 2025), while in India a forecast that shifted expected monsoon onset changed investment and consumption (Burlig et al., 2024). These studies do not identify the welfare effect of compression in our setting, but they show that the belief distortions we document operate on margins with meaningful economic consequences.

The card is also inexpensive to deliver. Added to an existing visit, it costs about USD 0.80 per farmer (Appendix F.7), below the estimated annual cost of mobile phone advisory services delivered at scale (GiveWell, 2020). The treatment raises confidence and moves expected rainfall toward the historical record, including among farmers who cannot read. These results do not establish a welfare return, since the value of any belief revision depends on its direction and on the actions available to farmers. They do show that local weather history can be made accessible at low cost in a form that farmers use.

6 Conclusion

What farmers remember of past weather is a demand-side margin of climate adaptation. In a survey experiment with 686 smallholder farmers in eastern Uganda, we document the core behavioral mechanism through which remembered weather shapes expectations and planned protection. Memory is highly compressed for ordinary seasons, retaining about ten percent of the recorded differences between them. Yet remembered weather strongly predicts expectations. A remembered history one picture wetter predicts an expectation for the coming season about 0.44 pictures wetter. Extremes behave differently. Farmers freely recall them, expect them to recur, and plan substantially more adaptation. The years they name as droughts, however, are often not dry years in the record. Together, these patterns trace how remembered experience translates into perceived risk and planned protection, while the broader literature establishes the final link from expectations to adaptation. Compression weakens the signal carried by ordinary experience, while salient extremes retain their influence.

The paper makes three contributions. First, it measures memory against an objective weather record, season by season and on the same scale, allowing us to quantify how much experienced variation survives in recall. Second, it moves the demand side of adaptation

upstream from expectations to the memories from which those expectations are formed, showing how compression and salience shape the signal farmers carry forward from past experience. Third, it shows that this signal can be repaired. Presenting the local historical record in a format that requires no reading makes farmers more confident, moves expected rainfall toward the record, and brings expected onset forward, including among farmers who cannot read.

The compression we document is an instance of an old regularity in which remembered magnitudes are drawn toward a central value (Crawford et al., 2000; Hollingworth, 1910; Huttenlocher et al., 2000), and decisions respond too little to fundamentals under cognitive uncertainty (Enke & Graeber, 2023; Enke et al., 2026). Agronomic studies in Uganda and Kenya likewise find that farmers’ accounts of past rainfall diverge from station records (Osahr et al., 2011; Rao et al., 2011), although not everywhere (Salerno et al., 2019). In our sample, farmers with better access to weather information recall past seasons more accurately but compress differences between seasons just as much. We therefore expect the qualitative pattern to hold more widely, while the magnitudes come from one rainfed setting with limited local records and should be replicated elsewhere.

The implication for demand for protection is that low take-up need not reflect only the products on offer. When perceived risk is built from a memory that flattens weather variation, demand responds weakly to much of the weather history farmers have experienced and more strongly to the extremes that remain salient, consistent with insurance take-up that rises after floods and fades in the years that follow (Gallagher, 2014). Information about the past seasons is a low-cost lever on this margin. The information can be generated automatically from public satellite data at extremely low cost per farmer when added to an existing visit. Extension services, local agencies, and providers of protective products could therefore deliver the historical record alongside what they already offer. Such implementation would also create a natural setting to observe whether revised beliefs translate into protective choices under adverse conditions and how long those revisions persist (Cavallo et al., 2017; Coibion et al., 2022).

Climate adaptation depends not only on the protection farmers can access, but also on the signal about future risk they carry forward from past experience. When memory compresses that experience, ordinary variation fades while salient extremes remain. Providing local historical weather information can help inform adaptation decisions and preparedness for climate extremes.

Declarations

Competing interests. The authors declare no competing financial or non-financial interests.

Funding. This study was funded by Erasmus University Rotterdam and the Erasmus Research Institute of Management (ERIM) through departmental research support, as disclosed in the consent form reproduced in Online Appendix OA-5. The funded proposal anticipated satellite-based rather than survey-based outcomes. This difference is documented in Appendix E.

Ethics. The research was reviewed and approved by the Internal Review Board of the Erasmus School of Economics (IRB-E), under approval number ETH2425-0832, granted on 20 May 2025 and valid for three years. The approved protocol covers the single pre-season interview; it does not include a second contact with respondents after May 2026, and no amendment was filed (Appendix E). The approval documentation is reproduced in Online Appendix OA-5.

Pre-registration. The study was registered with the AEA RCT Registry (AEARCTR-0018045) in March 2026, before fieldwork began on 10 March 2026. Appendix E maps every registered hypothesis to where it is reported and documents deviations from the registered analysis.

Data availability. Replication data and code will be made publicly available upon publication. CHIRPS rainfall data are publicly available (Funk et al., 2015).

Generative AI. During the preparation of this work, the authors used Claude (Anthropic) to assist with drafting and editing the manuscript and with preparing and checking analysis code. All AI-assisted output was reviewed and edited by the authors, who take full responsibility for the final content.

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A Sample, randomization, and balance

This appendix documents how the study sample was constructed and how villages were assigned to the three experimental arms. It describes the sampling frame and village selection, access to participating communities, respondent recruitment within villages, the matched-pair randomization and rerandomization procedure, balance across arms, and the fieldwork ultimately completed. Appendix B describes the construction of the objective weather benchmark and information treatment, while the survey and field materials are reproduced in the Online Appendix.

A.1 The sampling frame and village selection

Village selection proceeded in three stages. We first screened candidate districts, then constructed a rural village sampling frame within the selected districts, and finally drew a set of candidate villages using a stratified procedure.

Stage one: district selection. Ten districts were considered before the study area was fixed: Bugiri, Bulambuli, Butaleja, Gomba, Kamuli, Kiboga, Kyankwanzi, Mayuge, Nakasongola, and Namutumba. The screening exercise compared rainfall characteristics across 94 subcounties using CHIRPS pentad data, including seasonal rainfall moments, the frequency of dry pentads, and rainfall-onset timing. Two overlapping data vintages covering 2018–2022 and 2021–2025 were examined. The surviving project documentation does not record the exact criterion by which Bugiri, Butaleja, and Mayuge were selected from the ten candidate districts. We therefore report the screen as documented and do not reconstruct an ex post selection rule.

Stage two: the village sampling frame. The initial village listing contains 594 rows covering 483 distinct village names across 85 parishes and 14 subcounties in Bugiri, Butaleja, and Mayuge. Of these, 511 carry a field-partner assessment, with 327 classified as “Good” and 184 as “Very good.” The remaining 83 have no assessment and are excluded. We then remove settlements identified as town councils, town boards, townships, markets, or towns. This removes three subcounties and leaves an effective rural sampling frame of 442 villages across 11 subcounties and 70 parishes. The population from which the study villages were drawn is therefore rural communities in the 11 non-town subcounties of Bugiri, Butaleja, and Mayuge that the local field partner considered suitable for the study.

Stage three: the candidate-village draw. Villages rated “Very good” received higher sampling priority than those rated “Good.” Within each parish, up to five villages were initially sampled, with the higher-priority villages receiving three times the sampling weight. Where village names differed only by a trailing “A” or “B,” at most one was retained for a given parish. The procedure then capped each subcounty at seven villages and sampled up to 80 villages from the remaining pool using the same three-to-one priority weighting. With a fixed random seed of 321, this procedure selected 77 candidate villages.

From 77 candidates to 60 study villages. The 77 villages were a preselection rather than the final experimental sample. Before any treatment was assigned, the field team visited the candidate communities and sought permission to conduct the survey, meeting local council leadership and community mobilizers who would later help convene respondents. Access was secured in 60 villages. These 60 villages constitute the experimental sample.

This access stage restricts the population to communities whose local leadership agreed to host the study. It therefore matters for external validity. Participating villages may differ from villages that did not participate in local institutional capacity, prior contact with researchers, or other unobserved characteristics. We cannot determine the direction or magnitude of this selection because no survey data were collected in the 17 candidate villages that did not enter the final sample.

The access process does not affect the internal treatment comparison because it was completed before randomization. The 60-village sample was fixed first, and treatment assignment was conducted only afterward, on 5 March 2026, five days before the first interview on 10 March. No village entered or left the sample after treatment assignment, and treatment status did not exist when village access was negotiated. Selection into the study therefore determines the population to which the experiment applies but not the validity of treatment comparisons within that population.

A limitation of the field records is that the identities of the 17 candidate villages that did not enter the final sample, and the reasons for their exclusion, were not recorded systematically. We therefore cannot compare participating and nonparticipating candidate villages on the weather characteristics used in the original sampling frame. The archived sampling documentation also contains one description error. It refers to a final-sample category labeled “Yes,” although no such assessment category was used. The final sample is instead defined by the 60 villages in which access was secured.

The realized sample. Table 2 summarizes the geographic structure of the final sample. Twenty-eight of the 42 participating parishes contribute one village, ten contribute two villages, and four contribute three.

Table 2: The realized village sample, by district.

District	Subcounties	Parishes	Villages	Respondents
Bugiri	4	16	19	215
Butaleja	3	11	18	197
Mayuge	4	15	23	274
Total	11	42	60	686

Notes: The final sample contains 60 villages across 42 parishes and 11 subcounties in Bugiri, Butaleja, and Mayuge. Respondent locations are assigned using the standardized village-to-district mapping used in the survey rather than the free-text district field, which contains multiple spelling variants. Source: March 2026 survey.

Geographic coordinates. Exact village coordinates were not collected. Villages were instead assigned the centroid of their parish polygon using the Uganda Bureau of Statistics 2016

parish boundaries. Coordinates were matched by district and parish, with subcounty used to resolve repeated parish names. Two parish-name spellings were standardized to match the official geographic boundaries.

This means that the 60 study villages are represented by only 42 distinct coordinate pairs because villages in the same parish share a centroid. Geographic variables used in the randomization are consequently measured at parish or subcounty level rather than at the village itself. Maps in this paper therefore show parish-centroid locations and should not be interpreted as exact village coordinates.

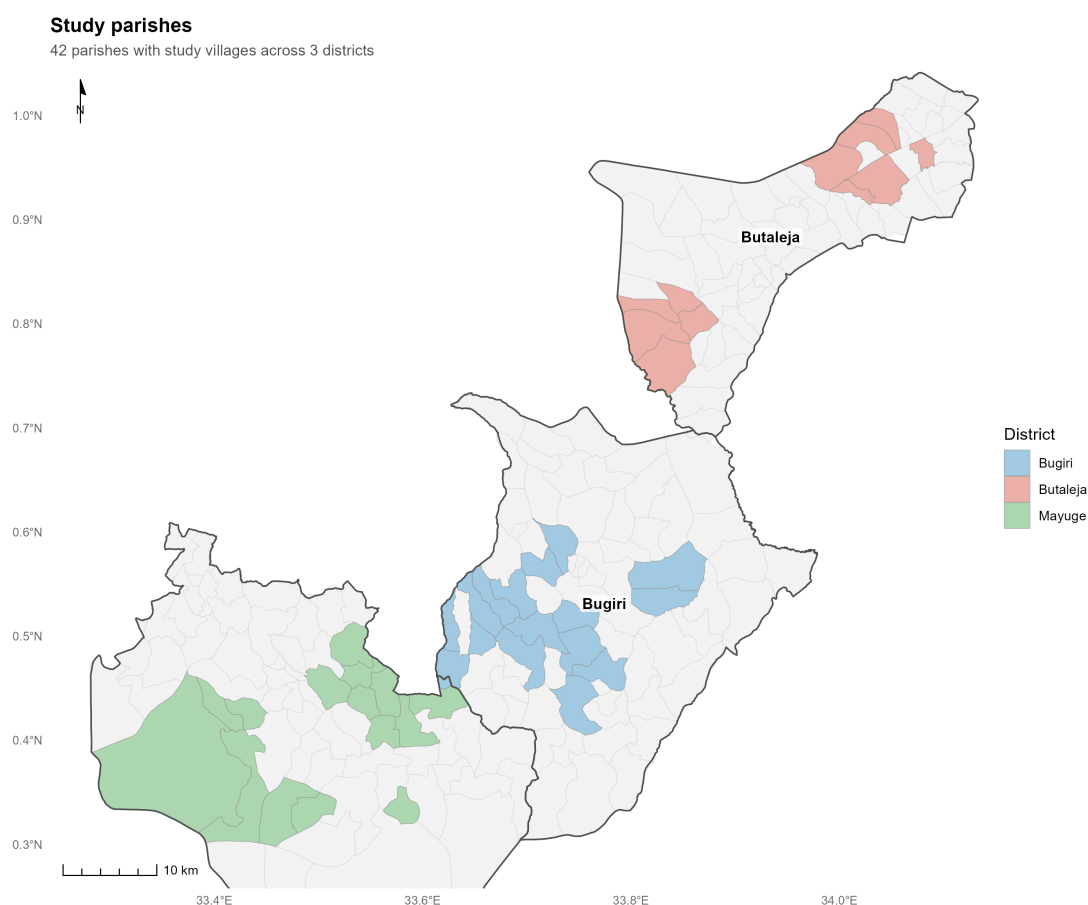


Figure 8: The study area: 42 parishes containing study villages across the three districts.

Notes: Parishes containing at least one of the 60 study villages are shown by district. Unshaded polygons are other parishes within Bugiri, Butaleja, and Mayuge. Villages are represented geographically by parish centroids, so parish is the finest mapped unit. Figure 9 shows the corresponding locations separately by district. Source: authors' construction using Uganda Bureau of Statistics 2016 parish boundaries.

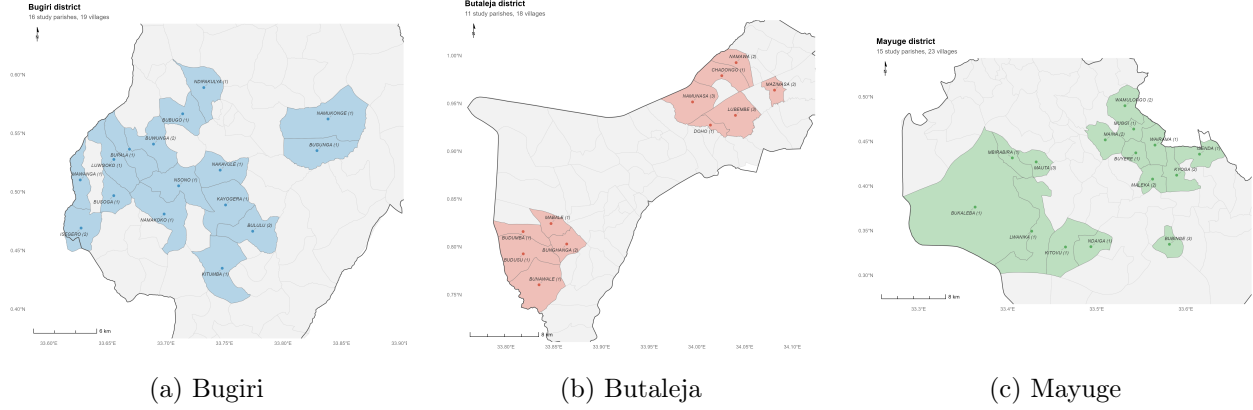


Figure 9: Study villages by district.

Notes: The study contains 19 villages in Bugiri, 18 in Butaleja, and 23 in Mayuge. Locations are plotted at parish centroids, so villages within the same parish share a coordinate and may overlap on the map. Source: authors' construction.

A.2 Respondent selection within villages

The surviving field documentation does not record a formal within-village sampling rule. In particular, it does not establish whether respondents were drawn from a household listing, selected using a spatial rule or quota, or recruited from lists maintained by local leaders. We therefore do not characterize within-village recruitment as random and do not make claims about refusals or replacement that are not documented. The March 2026 registration states the intended rule: no household selection protocol, with enumerators interviewing any eligible farmer, male or female and aged 15 or older, present in the village. Respondents were convened with the help of community mobilizers (Appendix A.1). The records do not show when respondents were convened or whether the mobilizers knew the village's arm, which was fixed on 5 March 2026, after village access had been negotiated.

The realized sample contains 686 respondents across all 60 villages, with between 7 and 14 respondents per village, a median of 12, and a mean of 11.4. Only two villages contain fewer than ten respondents, with 7 in Buhwaya and 9 in Mawanga, while three villages contain 14. This distribution is consistent with an approximate field target of twelve respondents per village, but such a target is not documented in the surviving protocol and should therefore be interpreted only as an inference from the realized sample.

Coverage is similar across experimental arms. Arm A contains 274 respondents in 24 villages, with 7 to 14 respondents per village. Arm B contains 293 respondents in 26 villages, with 9 to 14 per village. Arm C contains 119 respondents in 10 villages, with 11 to 14 per village. Mean village sample sizes are 11.4, 11.3, and 11.9 in Arms A, B, and C, respectively.

Respondent identifiers were designed to incorporate enumerator and village codes, providing an additional check that interviews were associated with the intended field assignment. All 686 respondent identifiers are unique. Of these, 683 follow the intended coding structure and identify one of the seven enumerators. Three contain malformed codes. These appear

to be data-entry errors rather than evidence of additional enumerators and do not affect treatment assignment or respondent uniqueness.

A.3 Randomization

Village-level assignment. Treatment is assigned at village rather than household level because the intervention consists of information that can readily diffuse between nearby farmers. Respondents receive a printed summary of local weather history, while social interactions are themselves an important channel through which agricultural information can spread (BenYishay & Mobarak, 2019; Conley & Udry, 2010). Household-level treatment assignment within the same village would therefore create a substantial risk of information spillovers into the control group. The corresponding cost is statistical power. The experimental treatment varies across 60 villages rather than 686 individuals, and inference is clustered accordingly.

Randomization covariates and strata. Five geographic covariates were used in the randomization. The first is mean elevation within a 500-meter radius of the parish centroid, measured using SRTM. The remaining four are subcounty-level characteristics calculated over the 2001–2025 historical weather record: mean seasonal rainfall, the coefficient of variation of seasonal rainfall, mean peak EVI, and the coefficient of variation of peak EVI. Only elevation varies within subcounty.

Villages were stratified by subcounty and elevation tercile. Elevation terciles are divided at 1,118 and 1,171 meters, placing exactly 20 villages in each tercile. Crossing these groups with the 11 subcounties produces 17 nonempty strata containing between one and six villages.

Pair formation. Within each stratum, villages were paired by nearest-neighbor matching on standardized elevation and mean peak EVI. At each step, the two closest remaining villages were paired until fewer than two villages remained. This produced 25 within-stratum pairs. Ten villages remained unmatched because they belonged to strata with odd numbers of villages. These villages were pooled and matched using the same covariates, with a strong preference for matches within the same subcounty. This produced five additional pairs. Three remain within subcounty, while two necessarily cross subcounty boundaries: Kityerera with Mpungwe and Buwunga with Kigandalo. All 60 villages are therefore contained in 30 matched pairs. Pair-matched cluster randomization follows Imai et al. (2009).

Matching was close for most pairs. Fifteen of the 30 pairs have no distance on the standardized matching variables, nine have distances at or below 0.32 standard-deviation units, and four have distances between 0.62 and 1.65. The remaining two are the forced cross-subcounty pairs and were deliberately penalized in the matching procedure to ensure that within-subcounty matches were selected whenever possible.

Assignment of the three experimental arms. Arm C is smaller than Arms A and B, so treatment assignment uses ten of the 30 matched pairs as donor pairs for Arm C. Within each donor pair, one village is randomly assigned to Arm C and the second is randomly assigned to either Arm A or Arm B. In each of the remaining 20 pairs, one village is assigned

to Arm A and the other to Arm B. Arm A and Arm B therefore each receive 20 villages from the regular matched pairs plus the villages assigned to them from the ten Arm C donor pairs. In the accepted assignment, four donor-pair villages were allocated to Arm A and six to Arm B, yielding 24 villages in Arm A, 26 in Arm B, and 10 in Arm C.

Rerandomization. We used rerandomization to improve balance on the five geographic covariates. A candidate assignment was accepted only if three conditions were satisfied simultaneously. The largest absolute standardized difference between Arms A and B across the five covariates had to be no greater than 0.10. The corresponding maximum difference for either the A–C or B–C comparison had to be no greater than 0.25. Finally, the difference in the number of villages assigned to Arms A and B could not exceed four.

The tighter balance requirement for Arms A and B reflects their role as the primary experimental comparison, while Arm C is a smaller robustness arm containing only ten villages. Candidate assignments were generated sequentially from a fixed randomization procedure. The accepted assignment was the 844th candidate draw, corresponding to random seed 966, and was finalized on 5 March 2026. Balance is assessed following Bruhn & McKenzie (2009).

Balance achieved by the design. Table 3 reports balance on the five randomization covariates. The largest standardized difference between Arms A and B is 0.0847, for mean peak EVI. The largest difference involving Arm C is 0.2331, for elevation between Arms B and C. Mahalanobis distances between the five-dimensional arm means are 0.0164 for A–B, 0.3334 for A–C, and 0.3003 for B–C. Treatment assignment is also distributed across all three districts. Bugiri contributes 7, 9, and 3 villages to Arms A, B, and C; Butaleja contributes 8, 8, and 2; and Mayuge contributes 9, 9, and 5.

Assignment in the field. Treatment assignment was fixed before fieldwork and preloaded for each of the 60 study villages. Enumerators selected the village from the study list, after which the corresponding experimental arm was assigned automatically and displayed for confirmation. Enumerators therefore had no discretion over treatment assignment. This procedure differed from the pilot, in which treatment status could be selected manually, and was adopted specifically to eliminate that source of discretion.

Realized assignment matches the randomized assignment exactly. Arm A contains 274 respondents in 24 villages, Arm B contains 293 respondents in 26 villages, and Arm C contains 119 respondents in 10 villages. No respondent is associated with a treatment status different from the assignment of the reported study village.

Figure 10 shows the geographic distribution of the randomized assignment.

Table 3: Design-stage balance across arms on the five randomization covariates.

	Arm A ($n = 24$)		Arm B ($n = 26$)		Arm C ($n = 10$)	
Village covariate	Mean	SD	Mean	SD	Mean	SD
Elevation (m)	1147.167	51.908	1149.538	47.171	1138.600	35.616
Season rainfall (mm)	709.922	9.962	710.715	9.724	710.673	10.022
Rainfall CV	0.137	0.017	0.137	0.016	0.138	0.021
Peak EVI mean	0.593	0.029	0.591	0.027	0.591	0.027
Peak EVI CV	0.076	0.026	0.075	0.024	0.075	0.027
	A – B		A – C		B – C	
Elevation (m)	−0.0505		0.1826		0.2331	
Season rainfall (mm)	−0.0817		−0.0773		0.0044	
Rainfall CV	−0.0148		−0.0899		−0.0751	
Peak EVI mean	0.0847		0.0801		−0.0047	
Peak EVI CV	0.0342		0.0290		−0.0052	
Mahalanobis D^2	0.0164		0.3334		0.3003	

Notes: The upper panel reports arm means and standard deviations for the five geographic covariates used in the matched-pair rerandomization. The lower panel reports pairwise mean differences divided by the pooled standard deviation across all 60 villages and the Mahalanobis distance between arm centroids over the five covariates jointly. Elevation is measured within a 500-meter buffer around the parish centroid. The rainfall and vegetation measures are subcounty-level quantities constructed from the 2001–2025 CHIRPS and Sentinel-2 historical record and are therefore constant within subcounty. Respondent-level balance is reported in Appendix A.4. Source: authors’ computation from CHIRPS (Funk et al., 2015) and Sentinel-2 data.

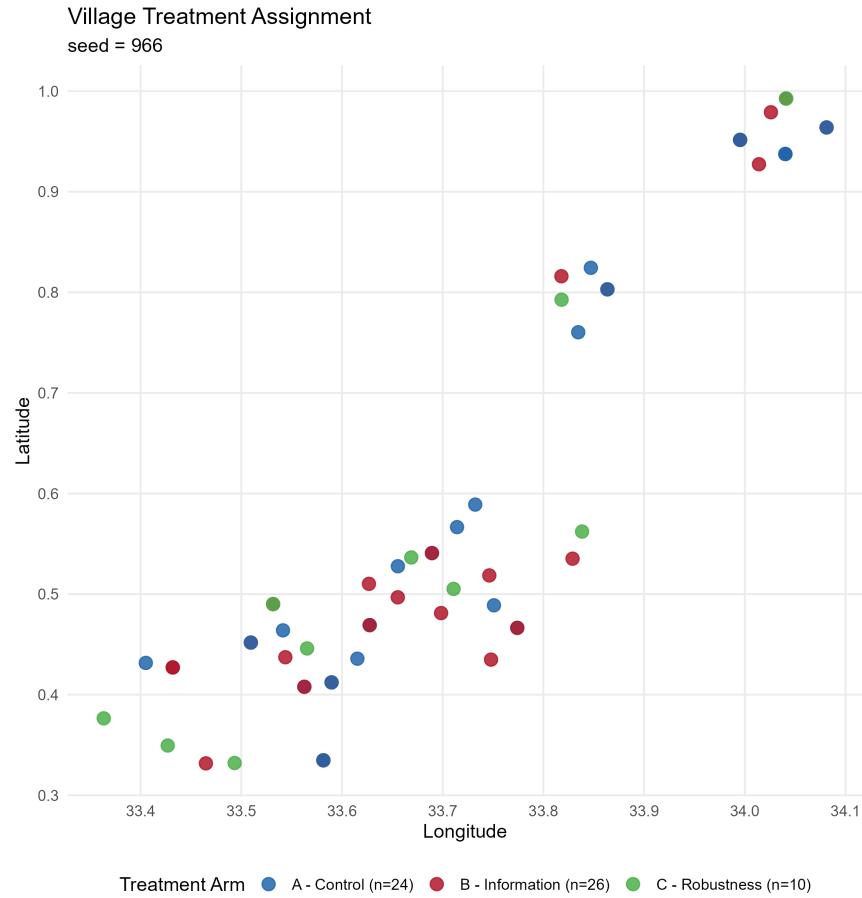


Figure 10: Village treatment assignment.

Notes: The 60 study villages are shown by experimental arm: 24 control villages in Arm A, 26 information villages in Arm B, and 10 villages in Arm C. Treatment assignment is distributed throughout the 11 study subcounties rather than concentrated geographically. Locations are represented by parish centroids, so villages within the same parish overlap and the 60 villages appear at 42 distinct coordinates. The accepted rerandomization draw used seed 966. Source: authors' construction.

A.4 Respondent characteristics and balance

Table 1 in the main text reports respondent characteristics by experimental arm. This section provides additional balance evidence, distinguishing the geographic variables explicitly balanced by the design from respondent characteristics that were not used in the randomization.

Design-stage balance. Table 3 shows that the rerandomization achieved the balance restrictions imposed on the five geographic covariates. The largest standardized A–B difference is 0.0847 and the largest standardized difference involving Arm C is 0.2331, both within the pre-specified acceptance thresholds. Pair-matched cluster randomization follows Imai et al. (2009), balance is assessed following Bruhn & McKenzie (2009), and covariate adjustment where used follows Lin (2013).

Respondent-level balance. Respondent characteristics were not included in the rerandomization because they were measured only after village assignment. We assess ex post balance across 74 pre-treatment respondent characteristics using village-clustered comparisons between experimental arms. Seven of the 74 characteristics differ between Arms A and B at the ten percent level, compared with 7.4 differences expected by chance under independent tests. Eleven differ between Arms A and C, consistent with the smaller size of Arm C and the wider balance tolerance applied to comparisons involving that arm.

Among the respondent characteristics displayed in Table 1, the largest A–B difference is age. Respondents average 42.5 years in Arm A and 40.0 years in Arm B, corresponding to a standardized difference of 0.200 and a village-clustered p -value of 0.066. No other characteristic shown in the main descriptive table differs between Arms A and B at the ten percent level. Age is included in the pre-specified covariate set used in the treatment-effect regressions.

Two initial belief measures are especially relevant for the updating analysis and are balanced across the main experimental arms. Before treatment, expected planting onset differs between Arms A and B by 0.149 days on the fielded calendar ($p = 0.91$), while the mean weather expectation differs by -0.051 bins. Neither difference is statistically distinguishable from zero. The pre-card distance to the card is less well balanced: Arm B starts 0.19 standard deviations further from the card (randomization-inference $p = 0.004$), although an omnibus test over ten pre-card belief levels does not reject balance ($p = 0.31$; Appendix H.3).

A.5 Realized fieldwork

Fieldwork dates. Fieldwork took place from 10 through 31 March 2026 over 20 interviewing days. All 686 interviews were completed during March, with the first recorded submission on 10 March and the final submission on 31 March. The first 2026 growing season had not yet resolved, so all season-ahead beliefs analyzed in the paper were elicited before the

corresponding weather outcome was known. No interviews were recorded on 15, 17, or 22 March.

Daily interviewing volume increased over the field period. The first five interviewing days contained 32, 31, 31, 35, and 34 interviews. The final full week contained 46, 45, 44, 46, and 48 interviews, followed by 32 interviews on the closing day. The smallest daily total was 25 interviews on 19 March.

Field team. Seven enumerators conducted the survey. They completed 94, 92, 101, 97, 97, 108, and 94 interviews, respectively, giving a range of 92 to 108 interviews per enumerator. Three respondent identifiers contain malformed enumerator codes, but the field records provide no evidence of an additional enumerator. Each enumerator worked across 56 to 59 of the 60 villages, indicating that the field team moved through the study area together rather than assigning separate sets of villages to individual enumerators. All survey submissions were made through the same project account.

Completion. All 60 randomized villages were visited and 686 interviews were completed, with 7 to 14 respondents per village. The realized data contain 686 distinct respondent records and no partial or invalid submissions. Every respondent is associated with one of the 60 sampled villages. The analysis sample therefore coincides with the completed fieldwork sample, with no attrition between data collection and the main respondent-level analysis.

Recall and expectation elicitation. The nine-picture weather exercise shown in Figure 1 was administered for each season from 2021 through 2025 and for the upcoming 2026 season, with the forward expectation elicited a second time where required by the experimental arm. The complete instrument is reproduced in Online Appendix OA-2.

For each past season, respondents were first asked whether they could remember the season. Those unable to recall it after two neutral prompts were not required to allocate tokens. Recall coverage falls from 99.3 percent for 2025 to 61.7 percent for 2021, producing the 2,637 respondent-year observations used in Section 5.1. Because the instrument records whether a respondent reports being unable to remember a season, rather than directly observing whether the underlying memory has been lost, this decline should be interpreted as declining elicited recall coverage rather than as a direct measure of forgetting.

Extreme-weather memory was elicited separately without prompting respondents with particular years. Respondents named the hazard, year, and perceived severity of remembered extreme events. The resulting data contain 550 first-mentioned events. Across all reported events, respondents name 210 droughts, 134 floods, 103 hailstorms, and 36 storms.

Interview duration. Actual interview duration cannot be recovered from the available survey records because interview start and end times were not retained. We therefore do not report an empirical measure of interview duration.

Data vintage. All fieldwork counts and descriptive statistics in this appendix are based on the frozen survey export obtained on 1 April 2026 at 08:28:56, immediately after completion of fieldwork.

B The objective benchmark and the information cards

This appendix documents how the objective weather benchmark and information treatment were constructed. It describes the rule used to map observed rainfall and vegetation into the nine-picture weather scale, the geographic resolution of the resulting information, the limits this places on analyses of card content, the construction of the planting-onset measure, and one complete information packet as respondents saw it.

B.1 Construction of the information cards

A central feature of the treatment is its geographic resolution. Weather conditions were constructed at the subcounty level and then presented on cards labeled by parish. The study covers 11 subcounties and 42 parishes, so the 42 parish-labeled cards are generated from 11 underlying subcounty weather histories. Across the five displayed seasons, the historical record therefore contains 55 independent subcounty-year cells rather than 210 independent parish-year observations. This structure follows directly from how the treatment was constructed and determines the effective geographic variation in card content.

Geographic unit and data sources. Each subcounty is represented by a centroid surrounded by a two-kilometer extraction buffer. Rainfall is measured using CHIRPS (Funk et al., 2015), with a historical baseline covering the 25 seasons from 2001 through 2025. The information cards display the five seasons from 2021 through 2025. Vegetation conditions for these display years are measured using Sentinel-2 L2A surface reflectance at 10-meter resolution.

Season and rainfall onset. Rainfall onset is identified from daily CHIRPS observations. Beginning on 10 February, onset is defined as the first date for which rainfall over the preceding 21 days reaches at least 25 millimeters and rainfall over the subsequent seven days reaches at least a further 10 millimeters. The search window ends on 15 May, with 30 April used as a fallback if the threshold is never met. The growing season then runs from the detected onset date until the earlier of 180 days after onset and 30 September. A separate fixed window from 10 February through 30 September is used for the rainfall-calendar component of the information treatment so that calendar coverage remains identical across years.

An important feature of this onset definition is that it produces almost no geographic or interannual variation in the study area. Across the 55 subcounty-year cells, detected onset falls on 10 February in 48 cases and on 11 February in the remaining seven. The accumulation threshold is therefore met at or within one day of the beginning of the search window in every case. By contrast, the onset calendar shown to respondents begins on 15 February and ends on 24 April. The rainfall-based benchmark consequently falls before the first elicitation interval in every subcounty-year. This is why the benchmark maps to the earliest onset interval everywhere and why we do not use the instrument to make cross-sectional statements about onset accuracy. Appendix B.3 discusses this issue in detail.

Vegetation and the cropland mask. Vegetation is measured using EVI2, calculated for each Sentinel-2 observation as

$$\text{EVI2} = 2.5 \frac{\text{NIR} - \text{Red}}{\text{NIR} + 2.4 \text{Red} + 1}.$$

Pixels classified as cloud, shadow, snow, or missing are excluded. For each subcounty-year, the cropland mask is based on four conditions. The 10th percentile of EVI2 between 15 January and 31 March must be below 0.20, the 90th percentile between 1 May and 15 September must exceed 0.55, the difference between these two measures must be at least 0.25, and the seasonal vegetation peak must occur between 1 May and 15 September. The resulting mask is smoothed using a local majority rule.

Daily mean vegetation is retained only when at least 30 percent of cropland pixels contain valid observations. Isolated daily outliers are removed using a rolling median and median absolute deviation criterion, after which the time series is smoothed using a three-day rolling median. Peak EVI is defined as the maximum of this smoothed series between 1 May and 15 September. Across the 55 subcounty-year cells, peak EVI ranges from 0.468 to 0.748, and every seasonal vegetation peak falls within the expected April-to-September growing-period window.

Mapping observed weather into the nine-picture scale. Each subcounty-year is assigned to one of the nine weather pictures by combining vegetation performance with the direction of rainfall conditions.

The vegetation component places peak EVI into six fixed bands. Values of 0.65 or above are classified as optimal, values from 0.62 to below 0.65 as near-optimal, values from 0.55 to below 0.62 as good, values from 0.45 to below 0.55 as average, values from 0.35 to below 0.45 as bad, and values below 0.35 as failed.

The rainfall component is constructed over three stages relative to detected onset. Establishment covers days 0 through 30, flowering covers days 31 through 75, and the pre-harvest stage begins on day 76. Within each stage and subcounty, rainfall amount is classified relative to the 2001–2025 distribution. Stage rainfall below the 25th percentile is classified as dry, rainfall between the 25th and 75th percentiles as balanced, and rainfall above the 75th percentile as wet.

We separately characterize the distribution of rainfall within each stage. Pentad rainfall is classified as even, moderate, or lumpy using both the coefficient of variation across pentads and the share of stage rainfall falling in the wettest pentad, benchmarked against the median and 75th percentile of the corresponding historical distributions. Rainfall amount and within-stage distribution are combined into a stage score ranging from -2 for dry conditions to $+2$ for wet and evenly distributed conditions. Establishment receives a weight of 0.25, flowering a weight of 0.40, and pre-harvest a weight of 0.35. The resulting weighted score is divided at -1.0 , -0.3 , 0.3 , and 1.0 to classify the season as dry, slightly dry, balanced, slightly wet, or wet.

Two additional rules address cases in which this stage-based classification may miss important rainfall patterns. First, a dry-spell rule classifies a season as dry when the longest sequence of days with rainfall below one millimeter reaches 44 days. A balanced season is

reclassified as slightly dry when the longest such sequence reaches 28 days. This rule does not bind in the realized sample and changes none of the 55 subcounty-year classifications.

Second, when the stage-based rainfall classification is balanced but vegetation is below the optimal range, an additional measure determines whether the season is placed on the dry or wet side of the scale. This measure equally weights seasonal rainfall relative to its 25-year mean and the complement of the share of ten critical ten-day periods from early March through early June with little or no rainfall. Values of 1.10 or above are assigned to the wet side, values of 0.90 or below to the dry side, and intermediate values are assigned to the slightly dry category. This rule applies in 17 of the 55 subcounty-year cells, all of which are classified as slightly dry.

The final picture bin is determined by crossing the vegetation band with the five rainfall categories. For optimal and near-optimal vegetation, dry through wet rainfall conditions map respectively to bins 4, 5, 5, 5, and 6. For good vegetation, they map to bins 4, 4, 4, 6, and 6. For average vegetation, they map to bins 3, 3, 3, 7, and 7. For bad vegetation, they map to bins 2, 2, 2, 8, and 8. For failed vegetation, they map to bins 1, 1, 1, 9, and 9. These assignments follow a fixed lookup rule and are not smoothed or estimated from the realized survey outcomes.

Variation in the realized weather classifications. Among the 55 subcounty-year observations, seven are assigned to bin 3, twenty-one to bin 4, eleven to bin 5, thirteen to bin 6, and three to bin 7. No realized subcounty-year falls into bins 1, 2, 8, or 9. Across the 11 subcounties, the five displayed seasons generate eight distinct unordered sets of weather bins and nine distinct ordered sequences. Buwunga, Mutere, and Nankoma each display the set {3, 4, 4, 5, 5}. Kigandalo and Kityerera share the ordered sequence 4, 6, 4, 6, 3, from 2025 back through 2021, while Buwunga and Nankoma share 4, 5, 5, 4, 3. These limited differences in card content determine what can be identified from regressions using characteristics of the displayed weather history, as discussed in Appendix B.2.

The printed information packet. The cards were produced as high-resolution printed sheets at 300 dpi. Each parish-labeled overview carries the heading “Past rainfall and harvest outcomes in [parish] parish” and presents one block for each of the five displayed years, ordered from most recent to least recent. Each block places the assigned weather picture beside three short statements constructed mechanically from the underlying weather measures.

The first statement describes rainfall conditions using the language associated with the nine weather bins, ranging from “extreme drought” and “drought conditions” through “well balanced rainfall” to “flood conditions” and “severe flooding.” The second describes the harvest outcome using peak EVI thresholds at 0.35, 0.43, 0.50, 0.55, 0.60, and 0.65, with descriptions ranging from “crop failure or very little harvest” to “excellent harvest.” The third compares vegetation performance with the other five displayed seasons using the mean and standard deviation of peak EVI within that card. This final statement is therefore relative to the five displayed years rather than to the longer historical distribution.

A separate rainfall calendar divides each month into three approximately ten-day periods. Rainfall in each period is classified relative to the 2001–2025 distribution for the same

subcounty and calendar period, using terciles. When fewer than eight historical observations are available for a particular period, month-level terciles are used instead. The three rainfall categories are displayed using one, two, or three water-drop symbols.

Each parish packet contains seven sheets. Five show rainfall and vegetation for the individual seasons from 2021 to 2025, one summarizes rainfall over ten-day periods across all five seasons, and one provides the parish-labeled overview. Across the 42 parishes, this produced 294 printed sheets. Because the annual panels and rainfall calendars are constructed at subcounty level, only 108 of these sheets contain distinct underlying content. Information packets were assembled by village and shown only in the 36 treated villages, comprising 26 villages in Arm B and 10 in Arm C. Appendix B.5 reproduces one complete packet.

B.2 Scope of the card content

The construction above places clear limits on what can be learned from regressions that use characteristics of the information cards themselves. Three features are particularly important.

There are few independent information histories. Although 42 parish-labeled cards were printed, their weather content is generated from 11 subcounty histories and reduces to only eight distinct card distributions and nine distinct ordered sequences over the five displayed years. Regressions using the distribution of weather shown on the card therefore rest on only eight distinct information profiles across 11 subcounties. Village-level clustering does not create additional independent variation in these card-level regressors.

The experiment contains no extreme card content. Across the 210 parish-year displays, realized weather bins range only from 3 through 7. No respondent is shown a season in bin 1, 2, 8, or 9. An indicator for exposure to an extreme card is therefore constant at zero and cannot be estimated. The nearest available proxy, the distance of the most extreme displayed bin from the center of the scale, varies only between one and two bins and produces no detectable relationship with updating. The complete information packet reproduced in Appendix B.5 illustrates the limited support of the treatment distribution.

Card shape explains little of the variation in revision. Measures of the distribution shown on the information card, including normalized entropy, standard deviation, and range, do not meaningfully predict the magnitude of belief revision. A joint Wald test over six card characteristics is statistically indistinguishable from zero, although the available variation is too limited for this null to be especially informative. A variance decomposition gives a clearer sense of magnitudes. The eight distinct card distributions, using seven degrees of freedom, explain $R^2 = 0.026$ of the variation in revision magnitude. By comparison, the respondent's absolute initial gap explains $R^2 = 0.081$ with one degree of freedom, while village indicators explain $R^2 = 0.090$ with 25 degrees of freedom.

One exploratory decomposition does reveal an association with information content. We separate the respondent's initial gap from the card into two components. The first is the

distance between the card and the leave-one-out consensus initial belief in the respondent’s village. The second is the distance between that village consensus and the respondent’s own initial belief. Because the first component does not contain the respondent’s own elicited belief, it is not mechanically generated by measurement error in that individual’s initial elicitation.

Pooling the 567 respondents in Arms A and B across 50 village clusters, the relationship between this village-consensus component and revision has a slope of 0.141 in Arm A. The Arm B interaction is +0.2282 with a standard error of 0.1095, giving an Arm B slope of 0.3692. We interpret this result cautiously for two reasons. First, the regressor combines variation from eight distinct information cards with variation in 50 village-level consensus beliefs, so the interaction cannot be attributed to card variation alone. Second, the interpretation assumes that the village consensus does not itself contain a common elicitation shock arising from enumerator, session, or other village-level influences. Because this analysis was exploratory and the underlying card variation is limited, we report it here rather than in the main text.

B.3 The onset calendar: measurement caveats

Planting onset is elicited using the same ten-token format as the weather-amount exercise. Respondents allocate tokens across ten dated intervals running from 15 February through 24 April. Nine intervals cover seven days and the third covers six days. Figure 11 reproduces the fielded calendar, and Online Appendix OA-2 transcribes the interval labels. Two features of the instrument constrain how onset magnitudes can be interpreted.

FARMER SURVEY CALENDAR: Start of Rains			
Allocate 10 tokens across the intervals to indicate when you remember / expect the seasonal rains started			
Interval	Date range	General season	Token(s)
1	15–21 Feb	Very dry; land still hard; few early land preparers start ploughing.	
2	22–28 Feb	First scattered showers in some low-lying areas; usually still unreliable.	
3	1–6 Mar	Land preparation intensifies; some early planters risk maize if showers continue.	
4	7–13 Mar	Transitional period; first rains often occur; seedbed preparation peaks.	
5	14–20 Mar	Some areas record 1–2 good rains; not yet sustained in the highlands.	
6	21–27 Mar	Typical onset for lower slopes; first main planting begins.	
7	28 Mar–3 Apr	Onset consolidates; continuous rain spells; widespread maize and bean planting.	
8	4–10 Apr	Average onset period for mid-slopes; moisture reliable.	
9	11–17 Apr	Late planters begin; upper slopes catch up.	
10	18–24 Apr	Late onset years often start here; second wave of planters.	

Figure 11: The ten-interval planting-onset calendar.

Notes: Planting-onset calendar used in the ten-token elicitation, covering ten intervals from 15 February to 24 April. Respondents could select *No Memory* after two neutral probes. Appendix B.4 describes the scale construction. Source: authors’ field materials.

Elicited onset differs systematically from the rainfall benchmark. Reported onset falls substantially later than the CHIRPS-based benchmark in every displayed season. On the fielded day scale, mean recalled onset and the corresponding rainfall benchmark are 73.0 and 41.2 in 2021, 70.5 and 41.1 in 2022, 71.3 and 41.3 in 2023, 69.9 and 41.1 in 2024, and 67.4 and 41.0 in 2025. The resulting gap is approximately 26 to 32 days.

Part of this difference is built into the measurement scales. The earliest interval available to respondents has a midpoint of day 49, whereas the objective benchmark is day 41 or 42 in every subcounty-year. The two measures may also represent different concepts. The rainfall algorithm identifies the first sustained rainfall event satisfying its threshold, whereas farmers may interpret onset as the beginning of sufficiently reliable planting rains. A further concern is that some descriptions on the onset calendar originated in the Elgon pilot and refer to highlands and slopes, as documented in Online Appendix OA-7. The data cannot distinguish among these explanations. We therefore make no cross-sectional claims about the absolute accuracy of elicited onset. The analysis uses the onset measure only for within-person comparisons and randomized treatment contrasts.

The objective benchmark maps to the first interval everywhere. Because CHIRPS-based onset occurs before the first elicitation interval in every subcounty-year, the objective benchmark maps to interval one across all 210 parish-year displays. Cross-sectional measures of onset accuracy therefore contain no independent variation in the benchmark. The rule places onset on the first admissible day of its search window in 48 of the 55 subcounty-seasons, and field staff and local respondents reported that the main planting rains had not yet begun at the time of the interviews, so the benchmark captures early rain rather than the planting rains farmers refer to. Under agronomic definitions that screen out false starts, onset falls inside the calendar and the card does not reduce the distance between beliefs and the record (Appendix H.6). We therefore interpret the treatment effect in Section 5.4.1 as a shift toward earlier expected onset, not as movement toward the recorded timing.

B.4 The day scale of the fielded calendar

Every onset result reported in days is calculated directly from respondents' ten-token allocations using the midpoints of the calendar intervals actually fielded. The calendar covers 15 February through 24 April. Its interval midpoints are days 49, 56, 62.5, 69, 76, 83, 90, 97, 104, and 111 of the year. A respondent's expected or recalled onset is the token-weighted mean of these values and must therefore lie between 49 and 111.

Because the third interval contains six rather than seven days, these midpoints are not exactly affine in the interval number. The day-scale measure is therefore not simply a linear rescaling of the interval-scale measure. We calculate and compare both representations. Results expressed directly on the interval scale, slopes that are invariant to affine transformations, and standardized ratios are substantively unchanged. The main onset treatment effect on the interval scale is -0.3776 intervals (SE 0.1270, $p = 0.0046$). As a support check, every day-scale statistic reported in the paper lies within the feasible range of 49 to 111 days. The initial-belief tercile means of 51.7, 63.7, and 80.4 days all satisfy this restriction.

B.5 One complete information card set

Respondents in Arms B and C were shown a printed information packet corresponding to their parish and subcounty. Figures 12 through 14 reproduce the complete packet used in Nakavule village, Nakavule parish, Kapyanga subcounty, Bugiri district, which was assigned to Arm B. The packet contains seven sheets. One provides an overview of the five previous seasons, one presents rainfall in ten-day intervals for all five seasons, and five present rainfall and vegetation separately for each season. The overview is labeled at parish level, while the rainfall calendar and annual rainfall-and-vegetation panels contain subcounty-level weather information.

Treatment administration. In Arm B, enumerators were instructed to show respondents the local historical rainfall charts after the first expectation elicitation. After viewing the packet and before repeating the expectation exercise, respondents answered six questions about their reaction to the information (Online Appendix OA-2): whether the information

differed from what they remembered about past rainfall, how surprised they were on a five-point scale, whether they would like to adjust their expectations and, if so, how confident they were in the adjusted expectations, whether historical rainfall information is helpful for planning, and whether they would like to receive similar information in the future. Respondents then repeated the same nine-picture weather exercise and ten-interval onset exercise for the upcoming season. Arm C answered the same six questions after seeing the packet and before its single elicitation. The pilot version of this block also asked whether the information changed respondents’ planned adaptations; those items were removed before the main fieldwork (Online Appendix OA-7), and adaptation plans were instead asked again in Arms A and B after the second elicitation.

As discussed in the main text, Arm A answered no comparable questions between its two elicitations. The treatment effect therefore identifies the information intervention as delivered, including any additional effect of these questions, and the two components cannot be separated. The order in which the different sheets were physically presented is not recorded, and neither is how long the packet remained in view or whether respondents kept any sheet; packets were shared among the respondents of a village (Appendix F.7). We therefore make no claim about sequencing effects within the information packet.

Resolution and support of the packet. The overview is the only sheet labeled at parish level, although its weather classification is generated from the same underlying subcounty record. The rainfall calendar and five annual panels are identical for every treated village within Kapyanga subcounty. This illustrates why the 42 parish-labeled packets reduce to a much smaller number of distinct displayed weather distributions.

The Nakavule overview also illustrates the limited support of the treatment. It classifies 2021, 2022, and 2025 as “slightly wet conditions,” 2024 as “slightly dry conditions,” and 2023 as “well balanced rainfall.” These correspond to bins 4 through 6 of the nine-picture scale. No season in this packet is close to either extreme, and the same is true of the treatment as a whole. No respondent was shown a season classified as drought, extreme drought, flood, or severe flooding.

Past rainfall and harvest outcomes in Nakavule parish



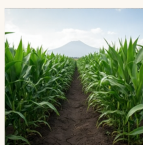
2025

- Rainfall conditions: slightly wet conditions
- Harvest outcome: good harvest
- Compared to past years: worse than the typical year



2024

- Rainfall conditions: slightly dry conditions
- Harvest outcome: good harvest
- Compared to past years: worse than the typical year



2023

- Rainfall conditions: well balanced rainfall
- Harvest outcome: excellent harvest
- Compared to past years: much better than the typical year



2022

- Rainfall conditions: slightly wet conditions
- Harvest outcome: very good harvest
- Compared to past years: better than the typical year



2021

- Rainfall conditions: slightly wet conditions
- Harvest outcome: good harvest
- Compared to past years: much worse than the typical year

Figure 12: Information card, sheet one: overview of the five previous seasons. Nakavule parish, Kapyanga subcounty, Arm B.

Notes: The overview presents the seasons from 2025 back through 2021. For each season, it displays the same picture used in the token elicitation together with short descriptions of rainfall conditions, harvest conditions, and performance relative to the other displayed years. The sheet is labeled at parish level, while the underlying weather classification is constructed at the subcounty level as described in Section B.1. Source: authors' field materials.

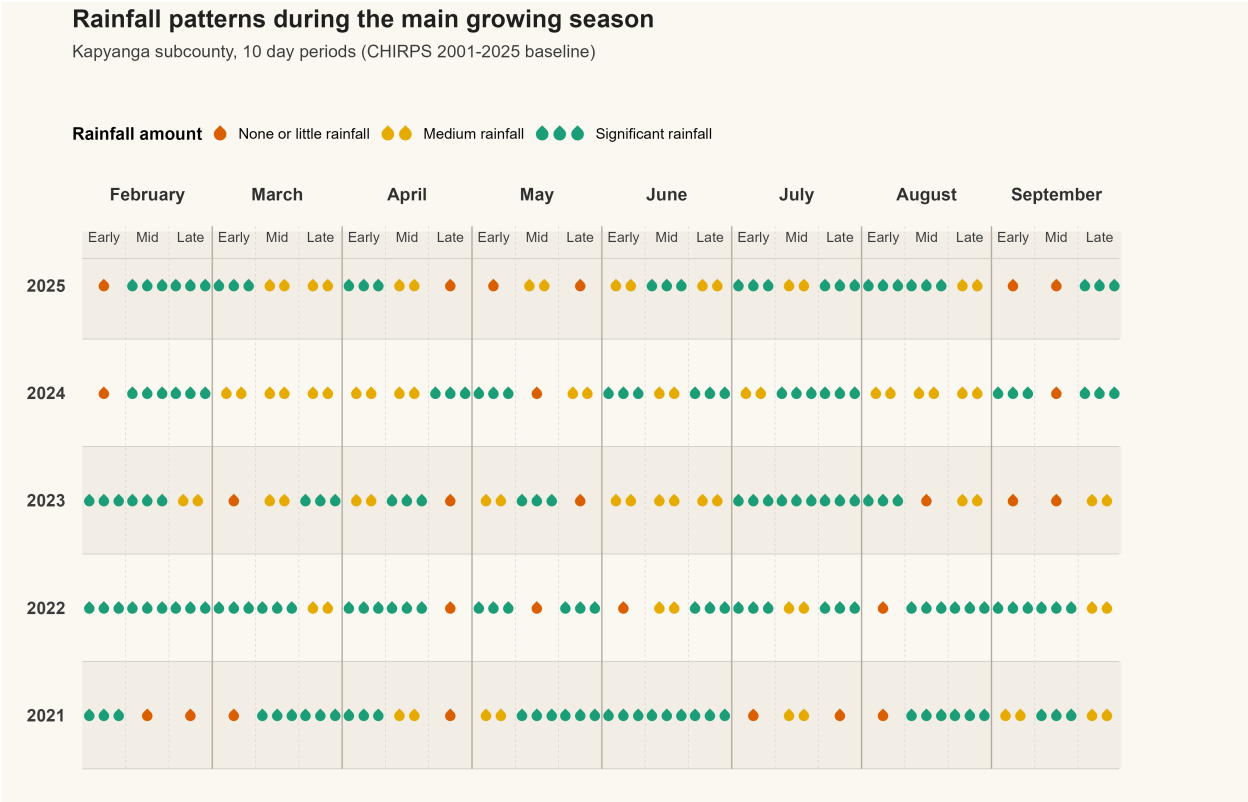


Figure 13: Information card, sheet two: ten-day rainfall calendar for the five previous seasons. Kapyanga subcounty.

Notes: Each row represents one season and each column an approximately ten-day period from February through September. Cells display one, two, or three water-drop symbols according to rainfall relative to the 2001–2025 historical distribution for the same subcounty and calendar period (Funk et al., 2015). This sheet is constructed at subcounty level and is identical for all treated villages within Kapyanga. Source: authors’ field materials.

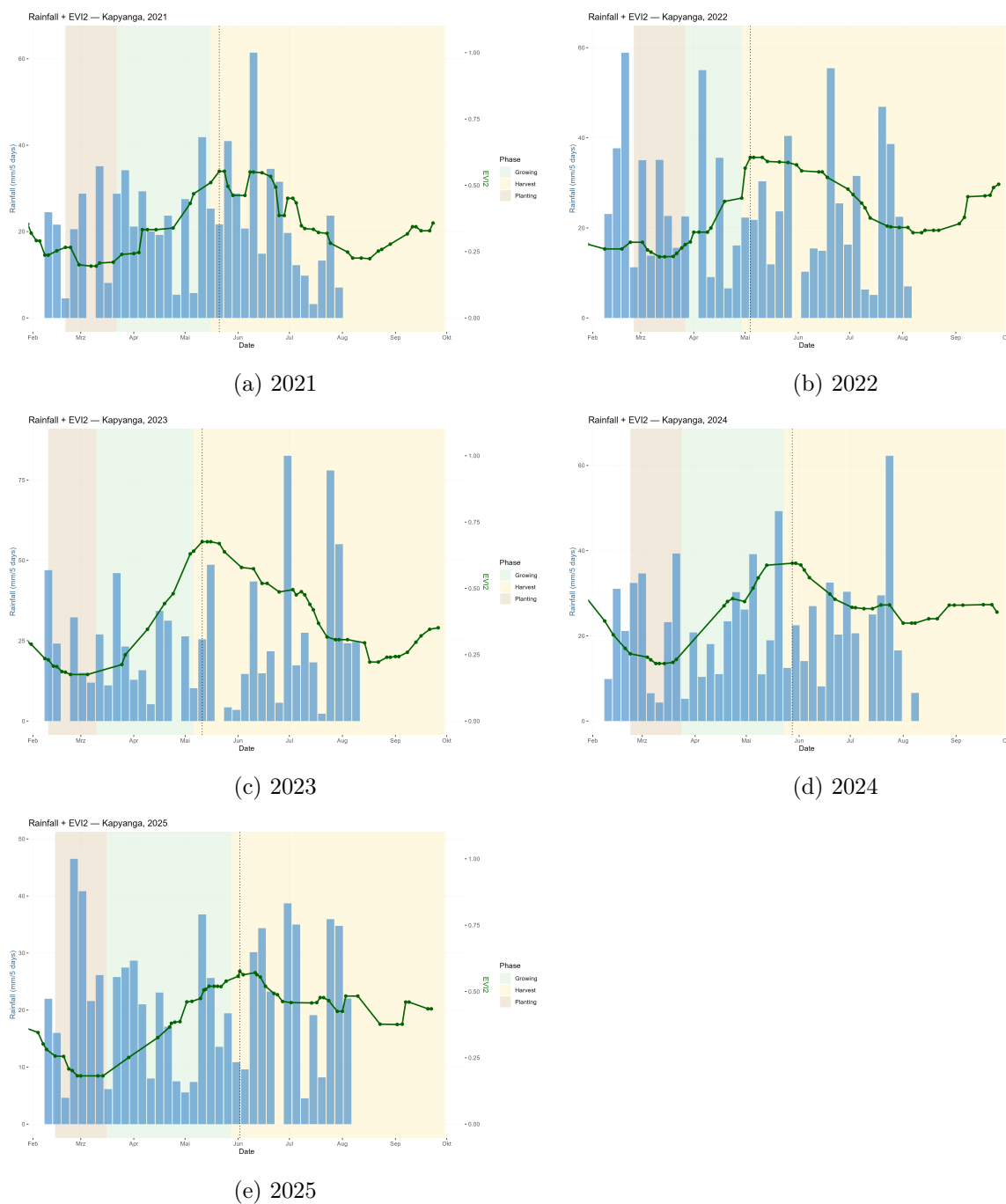


Figure 14: Information card, sheets three to seven: rainfall and vegetation by season. Kapyanga subcounty, 2021–2025.

Notes: The five annual sheets display CHIRPS rainfall in millimeters per five-day period on the left axis (Funk et al., 2015) and EVI2 vegetation on the right axis. Shaded regions indicate the planting, growing, and harvest phases, and the vertical dotted line marks the end of the growing window. These sheets are constructed at subcounty level and are identical across treated villages within Kapyanga. Source: authors' field materials.

C Statistical inference

This appendix documents the inference procedures used throughout the paper. We describe the restricted wild cluster bootstrap, the use of Arm A to benchmark repeated elicitation, Romano–Wolf stepdown adjustment, double selection, the treatment of multiple hypothesis testing, and the clustering structure that determines effective sample sizes. Computational details needed for exact reproduction but not for evaluating the statistical claims are provided in Online Appendix OA-8.

Three features of the design are particularly important for inference. First, the number of independent clusters is limited. The full experiment contains 60 villages, the main two-arm comparison contains 50 villages, Arm B contains 26 treated villages, Arm C contains 10 treated villages, and the historical record varies across only 11 subcounties. Second, Arm A repeats the same expectation elicitation without receiving information, allowing us to measure the amount of apparent updating generated by repeated elicitation alone. Third, substantial parts of the analysis are exploratory. We therefore distinguish explicitly between confirmatory, exploratory, and descriptive results. Sections C.1, C.2, and C.5 address these issues in turn.

C.1 The wild cluster bootstrap

Few clusters. Conventional cluster-robust inference relies on the number of clusters becoming large. Randomization in this study occurs at the village level, and the relevant comparisons contain 60, 50, 34, or 26 village clusters, with only 10 treated villages in Arm C. The historical record itself varies across only 11 subcounties. At these cluster counts, asymptotic approximations can over-reject. We therefore use the restricted wild cluster bootstrap of Cameron et al. (2008), which constructs the reference distribution under the null using the same clustering structure as the observed data.

Bootstrap procedure. For each hypothesis test, we first estimate the unrestricted specification with cluster-robust standard errors and calculate the studentized statistic $t = \hat{\beta}/\widehat{\text{se}}$. We then estimate the restricted specification on exactly the same estimation sample, imposing the null by excluding the tested regressor. Let $\hat{y}_i^0$ and $\hat{u}_i^0$ denote the fitted values and residuals from this restricted model. For each bootstrap replication b , one random weight $w_{g(i),b}$ is assigned to each cluster, so all observations within a village receive the same weight, and the bootstrap outcome is constructed as

$$y_{ib}^* = \hat{y}_i^0 + w_{g(i),b}\hat{u}_i^0.$$

We then re-estimate the unrestricted specification on each bootstrap sample, recompute the cluster-robust standard error using the same clusters, and store the studentized statistic t_b^* . Studentization allows the bootstrap statistics to remain comparable when residual variances

differ across replications. The two-sided bootstrap p-value is

$$p = \frac{1 + \sum_{b=1}^{B_{\text{eff}}} \mathbf{1}\{|t_b^*| \geq |t|\}}{B_{\text{eff}} + 1},$$

with analogous signed comparisons for one-sided tests. Replications that do not produce a finite statistic are excluded, and B_{eff} denotes the number of valid replications. Inference is not reported if fewer than 99 valid replications remain.

Bootstrap settings. We use $B = 9,999$ replications and a fixed random seed of 20260411. Two weight distributions are used. Rademacher weights take values -1 and $+1$ with equal probability. The six-point weights of Webb (2023) take values

$$\{-\sqrt{3/2}, -1, -\sqrt{1/2}, \sqrt{1/2}, 1, \sqrt{3/2}\}.$$

Rademacher weights are used for the comparisons of Arm B with Arm A, which contain 26 treated villages. Webb weights are used for all contrasts involving Arm C, which contains 10 treated villages, for the compression regression whether it is clustered on the 60 villages or on the 11 subcounties, and for the memory-predictor screen. Table 4 lists the weights used for each result. The finite-sample adjustment in the p-value formula implies a minimum reportable value of $1/10,000$, so no bootstrap p-value below 0.0001 is reported.

Results using bootstrap inference. Table 4 summarizes the results for which wild cluster bootstrap inference is reported.

For the compression slope, the restricted bootstrap gives a two-sided p-value of 0.0091 using Webb weights over the 60 village clusters. An independent computational implementation gives $p = 0.0106$, while the asymptotic village-clustered p-value is 0.0056. Because the two implementations fall on either side of 0.01, Section 5.1.1 reports the value without describing the slope as significant at the one percent level. When the same regression is instead clustered at the subcounty level, where the historical record varies, the restricted bootstrap p-value rises to 0.1759 over 11 clusters. Section C.6 discusses the implications of this coarser level of independent weather variation.

C.2 Placebo testing against Arm A

The repeated elicitation in Arm A provides an empirical benchmark for changes in reported beliefs that occur even when no information is supplied. This design feature is particularly important for statistics constructed from pre-to-post belief changes.

C.2.1 The repeated control elicitation

Arm A completes the same forward expectation elicitation twice, with no information card and no debrief between the two elicitations. All 274 Arm A respondents completed both rounds. Every within-Arm-B updating statistic therefore has a direct counterpart among

Result	Cluster	G	Weights	B
Recall-on-record slope, $\beta = 0.0948$	village	60	Webb	9,999
same estimate, clustered at the level at which the record varies	subcounty	11	Webb	9,998
Updating family, 11 outcomes	village	50	Rademacher	9,999
Learning elasticity ϕ_1 and arm interaction	village	50	Rademacher	9,999
Receptivity factor, within-arm and interacted	village	50	Rademacher	9,999
Farm ITT, Arm B versus Arm A	village	50	Rademacher	9,999
Farm ITT, contrasts involving Arm C	village	34, 60	Webb	9,999
Memory-predictor survivors, 37 cells	village	50	Webb	9,999
Predictor-block joint tests, 56 cells	village	50	—	9,999
Adaptation composition and financing families	village	50	Rademacher	9,999
Index regressions at the resolution of the record	subcounty	11	—	1,999

Table 4: Results carrying wild cluster bootstrap p-values.

Notes: G is the number of clusters in the estimation sample. Webb six-point weights are used for contrasts involving Arm C, for the compression regression at both clustering levels, and for the memory-predictor screen. The mixture-weight comparison in Table 6 instead uses a village cluster bootstrap with 1,999 draws and is the one inference result in the paper that does not use the restricted wild cluster procedure described above. Source: authors' computation.

respondents who received no information, including movement toward the card, movement toward the historical record, changes in expected planting onset, the magnitude of revision, changes in stated confidence, and parameters constructed from pre-to-post change scores.

We use this repeated control elicitation as the empirical benchmark for the amount of movement generated by the measurement process itself. In Arm A, 57.3 percent of respondents meaningfully change their ten-token allocation between elicitations. The mean absolute shift is 7.71 tokens out of ten, stated confidence rises by 0.208 Likert points, and the distance between the elicited belief and the card that these respondents never saw increases by 0.037 in earth-mover distance. Repeating the elicitation within the same interview can therefore generate substantial apparent updating even in the absence of information.

C.2.2 Treatment effects net of the Arm A comparison

Table 5 reports the three treatment effects emphasized in Section 5.4.1, each expressed relative to the corresponding change in Arm A.

The first two rows of Table 5 are estimated as differences in the raw pre-to-post change between Arms B and A, without fixed effects, so the treatment contrast decomposes exactly into the two arm means shown in the table. These specifications give an earth-mover-distance effect of -0.1338 (SE 0.0497, clustered $p = 0.0097$, restricted bootstrap $p = 0.0123$) and a ranked-probability-score effect of -0.1509 (SE 0.0506, clustered $p = 0.0044$, restricted bootstrap $p = 0.0047$). Specifications including subcounty fixed effects give closely similar

Pre-to-post change	Arm A	Arm B	DiD	SE	p
Earth-mover distance to the card	+0.0369	−0.0969	−0.1338	0.0497	0.0097
Ranked probability score against realized 2025	+0.0346	−0.1163	−0.1509	0.0506	0.0044
Believed planting onset, days [†]	+0.552	−2.039	−2.591	0.862	0.0042

Table 5: Treatment effects benchmarked against repeated elicitation in Arm A.

Notes: $N = 567$, of whom 274 are in Arm A and 293 in Arm B, across 50 villages and 11 subcounties. Lower values indicate improvement for both distance measures, so a negative difference-in-differences corresponds to movement toward the card or the historical record. For planting onset, the change within Arm A is statistically indistinguishable from zero ($p = 0.452$), while the change within Arm B is not ($p = 0.00031$). Because the first two outcomes drift away from the card in Arm A, the within-Arm-B changes of -0.0969 and -0.1163 would understate the corresponding treatment effects by roughly 28 percent if repeated elicitation were ignored. For planting onset, the treatment contrast is -2.5913 days with a village-clustered standard error of 0.8624 and $t = -3.0049$, giving $p = 0.0042$ under the finite-sample clustered correction and $p = 0.0027$ under the normal approximation. [†]Onset days are measured on the fielded calendar with interval midpoints from 49 to 111. Appendix B.4 documents the construction of the day scale. The corresponding effect on the interval scale is -0.3776 (SE 0.1270 , $p = 0.0046$). Source: authors’ computation.

estimates of -0.1399 (SE 0.0438) and -0.1497 (SE 0.0475), with bootstrap p-values close to 0.005 . The two specifications therefore agree in sign, magnitude, and inference. Romano–Wolf adjustment in Section C.3 is applied to the fixed-effects family.

For expected planting onset, the village-clustered p-value is 0.0042 for the -2.591 -day treatment effect on the fielded calendar and 0.0046 for the corresponding -0.3776 -interval effect, based on 50 village clusters and 567 respondents. The normal approximation gives p-values of 0.0027 and 0.0029 , respectively. Both specifications therefore place the onset effect below one percent under either convention. The onset contrast is not a member of the eleven-outcome updating family in Appendix C.3, so it has no Romano–Wolf adjustment within that family. Its wild cluster bootstrap p-value is 0.0055 , and in a twelve-outcome family that adds onset to the eleven updating outcomes, estimated with subcounty fixed effects, its q-value is 0.097 (Appendix H.6).

C.2.3 Designs without a repeated control elicitation

Several parameters that appear informative when estimated within Arm B are also reproduced in Arm A. Table 6 reports three examples.

C.2.4 Mechanical revision

The three patterns in Table 6 can arise from the same arithmetic feature of change-score regressions that condition on the elicited initial belief.

Write the elicited initial probability for bin b as $p^{\text{pre}} = \pi + \varepsilon$ and the second elicitation as $p^{\text{post}} = \pi + \nu$, where π is the respondent’s underlying belief and ε and ν are independent elicitation errors with variance σ_ε^2 . Under the null of no information, $\mathbb{E}[\nu \mid \varepsilon] = 0$, so the

Quantity	Arm B	Arm A	Arm comparison
Learning elasticity ϕ_1	0.419	0.374	+0.045 (SE 0.0543), $p = 0.41$
Mixture weight on the card, ξ , OLS reconstruction	0.470	0.186	—
Mixture weight on the card, ξ , cluster-bootstrap reconstruction	0.470	0.412	+0.063 (SE 0.0679), $p = 0.35$
Receptivity factor on meaningful revision	+0.0585	+0.0727	−0.0164 (SE 0.0212), $p = 0.43$

Table 6: Parameters reproduced by repeated elicitation in Arm A.

Notes: The learning elasticity is the coefficient on the card-minus-prior gap in a bin-level regression of the change in bin probability with respondent fixed effects, estimated on 2,637 bin-cells in Arm B and 2,466 in Arm A. Arm A reproduces 89.3 percent of the Arm B estimate and has a t statistic of 9.59. The mixture-weight estimates are reported under two reconstructions. The ordinary least-squares reconstruction gives an Arm A coefficient of 0.186, while the village cluster-bootstrap reconstruction gives 0.412 in Arm A against 0.470 in Arm B and an arm difference statistically indistinguishable from zero. The receptivity factor has $\alpha = 0.748$ and $\omega = 0.901$, is orthogonal to education, and predicts meaningful revision in both arms, with a somewhat larger coefficient in Arm A. All arm comparisons use 50 village clusters. Source: authors’ computation.

second elicitation is a second noisy measurement of the same underlying belief. The change score and the initial distance from the card are then

$$\delta = p^{\text{post}} - p^{\text{pre}} = \nu - \varepsilon, \quad \text{gap} = I_v - p^{\text{pre}} = (I_v - \pi) - \varepsilon, \quad (11)$$

where I_v denotes the information shown on the card. Both constructed variables contain the same term, $-\varepsilon$. It follows that

$$\text{Cov}(\delta, \text{gap}) = \sigma_\varepsilon^2$$

and

$$\text{Var}(\text{gap}) = \text{Var}(I_v - \pi) + \sigma_\varepsilon^2.$$

The corresponding population regression coefficient is therefore

$$\phi = \frac{\sigma_\varepsilon^2}{\text{Var}(I_v - \pi) + \sigma_\varepsilon^2} > 0. \quad (12)$$

Under these assumptions, a positive relationship between revision and the initial card gap arises mechanically even when no information is received, and the relationship becomes stronger as elicitation noise increases. The mixture-weight specification captures a closely related object. Regressing p^{post} on p^{pre} and I_v assigns weight to I_v whenever measurement error attenuates the coefficient on the initial elicitation. That attenuation depends on σ_ε^2 , not necessarily on learning. The receptivity result can arise through the outcome rather than the regressor. Any characteristic associated with greater token reshuffling will increase $|\delta|$ and can therefore predict measured revision even without treatment-specific learning.

Equation (12) also shows why the repeated control elicitation is useful. The mechanical component is not a fixed additive bias whose magnitude can be known in advance. It depends

on elicitation noise relative to the dispersion of the true distance between beliefs and the signal. Arm A provides an empirical estimate of this component. In this experiment, it reproduces about ninety percent of the within-Arm-B learning elasticity.

Expected planting onset requires a different benchmark because the information card contains no parish-specific onset signal and therefore provides no analogous onset gap. The relevant mechanical force is regression to the mean in the initial elicitation. In Arm A, the slope of revision on initial expected onset is -0.2835 (SE 0.0381 , $p = 1.0 \times 10^{-13}$). The corresponding slope in Arm B is -0.4317 (SE 0.0595 , $p = 4.0 \times 10^{-13}$), giving an arm difference of -0.1482 (SE 0.0700 , $t = -2.117$, $p = 0.034$). Recomputed on the fielded calendar, the arm difference is -0.1486 (SE 0.0705 , $t = -2.107$, $p = 0.040$). The treatment-relevant object is therefore the difference between arms rather than the within-arm slope itself. This is the continuous counterpart of the tercile analysis in Section 5.4.2. Because a regression slope of revision in days on an initial belief in days is invariant to affine transformations of the day scale, the two versions agree to three decimal places even though the levels of the calendar measures differ.

C.2.5 The positional result

Regression to the mean pulls extreme initial measurements inward from both directions. The pattern in Section 5.4.2 is instead asymmetric. Farmers whose initial expected onset is already early, with a mean of day 51.7, show essentially no treatment response, while those in the middle and late terciles move 4.40 and 4.10 days earlier. Moreover, each treatment effect is estimated relative to Arm A within the same initial-belief tercile. Mechanical mean reversion is therefore differenced out within each part of the initial-belief distribution rather than only in the aggregate.

C.2.6 Netting out mechanical revision and residual contamination

As an additional diagnostic, we estimate the relationship between initial beliefs and mechanical revision using Arm A alone, both at the bin level and at the individual level. These relationships are then used to predict the revision that would be expected mechanically for each respondent in Arm B. Excess revision is defined as observed revision minus this predicted mechanical component.

This adjustment has an important asymmetry. Because the mechanical relationship is estimated within Arm A, the resulting Arm A excess measure is orthogonal by construction to the regressors used to generate the prediction, whereas the Arm B excess measure is not. For predictors correlated with these mechanical regressors, this construction tends to shrink the Arm A coefficient and can therefore increase the apparent difference between arms. The resulting bias works against finding equal coefficients across arms, making the null arm differences reported below conservative. We also examine the remaining relationship between each excess measure and the initial elicitation, since removing the conditional mean estimated in Arm A does not eliminate all first-elicitation noise. The corresponding diagnostics are reported in Online Appendix OA-8.

C.2.7 Implications for information experiments

More generally, information experiments that relate belief revision to the initial belief-to-signal gap, estimate mixture weights on a signal, or ask which characteristics predict the magnitude of revision face a mechanical component whenever the same noisy elicitation enters both sides of the estimating equation. Without a group that repeats the same elicitation in the absence of information, this component cannot be separately observed. A repeated elicitation in the control group provides a direct empirical benchmark for it.

C.3 Romano–Wolf stepdown

We use Romano–Wolf stepdown adjustment for pre-defined families of related hypotheses. Shared cluster draws are used across hypotheses so that dependence among the test statistics is preserved even when estimation samples differ. The computational construction and validation of the procedure are documented in Online Appendix OA-8. Here we report the hypothesis families and the resulting adjusted inference.

Families and results. Table 7 lists every family to which Romano–Wolf adjustment is applied.

Family	J	G	B	Weights	Survivors at $q < 0.10$
Updating difference-in-differences	11	50	9,999	Rademacher	2; none at $q < 0.05$
Updating screen, amount, Arm B	4×22	26	9,999	Rademacher	0 in each of the four
Updating screen, onset, Arm B	3×22	26	9,999	Rademacher	1 in one of the three
Updating screen, Arm A placebo	3×22	24	9,999	Rademacher	1 in one of the three
Memory-quality predictor screen	420	50	4,999	Rademacher	37
Adaptation composition	10	50	9,999	Rademacher	0
Adaptation financing channels	8	50	9,999	Rademacher	0
Demographic selection into recall	5	60	999	Rademacher	2
Prior gap on adaptation spending and channels	2	50	9,999	Rademacher	2
Memory and belief measures and adaptation outcomes	25	50/60	9,999	Rademacher	6; recalled extreme on spending $q = 0.0002$

Table 7: Romano–Wolf hypothesis families and adjusted results.

Notes: J is the number of hypotheses in the family, G the number of village clusters, and B the number of bootstrap replications using shared cluster draws. The memory-quality screen consists of 105 candidate predictors crossed with four pre-specified outcomes, and the full 420-cell grid defines the family. The updating screen crosses 22 pre-treatment predictors with four amount outcomes and three onset outcomes, with each of the seven outcomes corrected as a separate family. The Arm A analyses repeat the corresponding screens within the group that received no information. Source: authors’ computation.

The updating family. Among the eleven pre-to-post belief changes, three have unadjusted p -values below 0.05: earth-mover distance at $p = 0.0061$, ranked probability score at $p = 0.0060$, and stated rainfall confidence at $p = 0.0179$. The corresponding adjusted values are $q = 0.0597$, 0.0625 , and 0.1318 , so the first two survive at ten percent and none at five percent. The earth-mover distance is labeled the primary outcome of this family because it measures distance to the object displayed on the card, but that designation dates from the re-analysis of August 2026, after within-arm movement toward the card had been estimated, and the earth-mover distance is not a registered outcome. The ranked probability score is among the registered belief outcomes (AEA registration H3; Appendix E), and its adjusted value is $q = 0.0625$.

The 22-predictor updating screen. Across the four weather-amount outcomes, none of the 22 predictors survives at $q < 0.10$ in Arm B in any family. The smallest adjusted values are $q = 0.1068$ for memory entropy when predicting bin-level excess revision and $q = 0.1218$ for the absolute initial gap when predicting excess revision magnitude. The corresponding Arm A placebo screen is informative about these associations. The absolute initial gap reaches $q = 0.061$ in Arm A, compared with $q = 0.122$ in Arm B. The stronger association in the untreated arm indicates that this predictor captures a feature of repeated elicitation rather than a treatment-specific response.

The only adjusted survivor elsewhere in the screen produces the same qualitative conclusion. For excess earliness of the onset belief in Arm B, the receptivity factor has a coefficient of -1.25 days per standard deviation (SE 0.34 , $t = -3.72$) and $q = 0.0519$. In the corresponding Arm A placebo family, the coefficient is -1.36 days (SE 0.47 , $t = -2.92$), which is larger in magnitude, although its adjusted value is $q = 0.2033$ because another statistic leads that family. Both coefficients use the fielded calendar. Together with Table 6, this pattern is consistent with the receptivity factor capturing a general propensity to revise or reshuffle the elicited distribution rather than a treatment-specific propensity to absorb information. Across the 154 predictor-by-outcome arm interactions, which directly test differential response to the card, no adjusted value falls below 0.10.

Interpretation of the stepdown adjustment. Romano–Wolf adjustment is most naturally interpreted for hypothesis families defined before the corresponding estimates are examined. We therefore use it as a confirmatory device for pre-registered and otherwise pre-committed families. We also report it for the 22-predictor updating screen, where the adjusted results are useful for comparing the treated and placebo arms, but we do not treat an ex post exploratory family as though it had been pre-specified.

The associations between memory, beliefs, and planned adaptation in Section 5.3.2 lie outside the registered families and are associational rather than causal. We apply Romano–Wolf adjustment over their complete grid, the 25 specifications that cross five memory and belief measures with five adaptation outcomes (Appendix D.5), rather than over the cells the main text reports. The main-text association, between recalling a past extreme and planned adaptation spending, survives this adjustment with $q = 0.0002$, and it also survives rich pre-determined controls and enumerator and interview-date fixed effects (Appendix H.7). Six of the 25 cells have $q < 0.10$; the main text interprets only the recalled-extreme association,

because the association with a stated expectation of an extreme does not survive enumerator fixed effects and the insurance cells are too thin to interpret.

C.4 Double selection

Purpose. We use double selection (Belloni et al., 2014) to reduce discretion over the control set in the heterogeneity analyses. In both applications, the question is whether a pre-treatment characteristic predicts an outcome after allowing the remaining candidate characteristics to enter flexibly through the selection procedure. Stepwise selection is not used. The candidate sets, penalty choices, and estimability rules are documented in Online Appendix OA-8.

Results. Double selection changes little and selects few additional controls. For excess revision magnitude in Arm B, the absolute initial gap is estimated at +0.1487 with $p = 0.0067$ and no selected controls. Memory entropy is estimated at +0.1427 with $p = 0.00047$ and two selected controls, while the receptivity factor is estimated at +0.1124 with $p = 0.0146$ and three selected controls. For excess earliness of expected onset, the receptivity factor is the leading target, with an estimate of -1.13 days (SE 0.38), $p = 0.0031$, and a Benjamini-Hochberg adjusted value of $q = 0.068$, with three controls selected.

The placebo comparisons determine how these estimates should be interpreted. The variables selected most clearly by double selection are the absolute initial gap and the receptivity factor, but both also predict revision in Arm A. The absolute initial gap is more statistically pronounced in the untreated arm, while the receptivity coefficient is larger in magnitude there. Double selection therefore addresses discretion in the choice of controls but cannot remove mechanical dependence built into the construction of the revision outcome. In this setting, the repeated Arm A elicitation provides the more relevant discipline for distinguishing treatment-specific updating from general re-elicitation effects.

C.5 Multiple-hypothesis accounting

Table 8 provides a complete inventory of the main hypothesis families, the multiplicity adjustments applied to them, and the inferential status of the corresponding claims.

The general rule is to apply multiplicity adjustment when a hypothesis family was defined before estimation and to use the repeated Arm A elicitation whenever inference is based on a change score. Where a family was not defined in advance, as for the associations of Section 5.3.2, the adjustment runs over the complete grid of specifications that were estimated, so that the reported association is not selected from it.

C.6 Clustering and effective sample size

The most important constraint on statistical precision comes from the geographic resolution at which the historical record varies.

Family	Tests	Correction	Status of the claim
Registered family $\{H1, H3, H4\}$	3	Romano–Wolf, registered	Confirmatory in intent but not executable as written: $H2$ is suspended, $H1$ is degenerate, and $H3$ has an effective n of 11. Not reported in this version; see Appendix E.
Farm-outcome ITT	4×3	Wild cluster bootstrap; no stepdown across outcomes	Registered as AEA H4 and reclassified as exploratory by the August 2026 addendum. Null; reported with confidence intervals rather than as evidence of absence.
Updating difference-in-differences	11	Romano–Wolf on shared draws	Exploratory. Related to AEA H3, but the eleven-outcome family and its primary outcome were fixed after fieldwork. Two outcomes survive the stepdown at ten percent and none at five percent.
Updating predictor screen	22×7	Romano–Wolf by outcome family, Benjamini–Hochberg, and Arm A placebo families	Exploratory and reported as a bounded null with a stated minimum detectable effect.
Memory-quality predictor screen	420	Romano–Wolf over the full grid and Holm adjustment over 56 block tests	Exploratory. The family is the complete search rather than the subset of surviving coefficients.
Memory, beliefs, and planned adaptation	25	Romano–Wolf over the full grid; wild cluster bootstrap	Exploratory and associational. Recalling a past extreme predicts planned spending with $q = 0.0002$ over the grid and survives rich controls and enumerator and interview-date fixed effects (Appendix H.7).
Adaptation composition and financing	$10 + 8$	Romano–Wolf by family	Exploratory and null.
Expected onset	1	Wild cluster bootstrap; Romano–Wolf in a twelve-outcome family with the eleven updating outcomes	Exploratory. Bootstrap $p = 0.0055$, $q = 0.097$ (Appendix H.6).
Post-only arm (AEA H6)	5×3	None; Webb bootstrap and matched-pair randomization inference	Registered; reported in Appendix H.4. Imprecise with ten treated villages.

Table 8: Confirmatory, exploratory, and descriptive inference across the paper.

Notes: “Tests” counts coefficients of interest rather than regressions. The updating predictor screen estimates each of its 154 predictor-by-outcome cells in Arm B, in Arm A as a placebo, and as an interaction between the two arms. The memory, belief, and adaptation grid is corrected over all 25 cells, not over the cells reported in the main text. Sections 5.1 through 5.4 are exploratory except where a registered hypothesis is explicitly identified. Source: authors’ construction.

The historical record varies at the subcounty level. The weather panel contains 42 parish rows, but there is only one distinct picture classification and one peak-greenness value for each subcounty-year. Across the five seasons from 2021 to 2025, the nominal 210 parish-year observations therefore contain only 55 independent weather observations, corresponding to 11 subcounties observed for five years. Treating the 42 parishes as independent weather realizations would overstate precision, so the paper reports 11 subcounties as the effective number of independent geographic units on the weather side.

The implication is visible in the main compression estimate. The recall-on-record slope in Section 5.1.1 is 0.0948. With respondent fixed effects and standard errors clustered over 60 villages, the standard error is 0.0342 and $t = 2.77$. When inference is instead clustered on the 11 subcounties, which is the level at which the objective regressor varies, the point estimate is unchanged but the standard error rises to 0.0566, giving $t = 1.67$, an asymptotic p-value of 0.094, and a Webb-weighted restricted bootstrap p-value of 0.176. These are tests against zero. The hypothesis that matters for compression is a slope of one, and the same bootstrap rejects it with $p = 0.0002$. Inverting the test gives a 95 percent confidence interval of $[-0.03, 0.22]$. Adding season indicators, which absorb anything common to the order in which seasons were recalled, gives a slope of 0.069 with a village-clustered standard error of 0.033. Restricting the analysis to the clean pre-information sample in Arms A and B gives a slope of 0.0756 with a village-clustered standard error of 0.0389. The point estimate is therefore reasonably stable across these specifications, while its precision depends strongly on recognizing that only 11 independent geographic weather histories underlie the historical record.

The rainfall data provide little independent farm-level variation. The 686 farms occupy 17 distinct CHIRPS grid cells, with a median of 36 farms per cell. Only five of the 11 subcounties contain two or more grid cells, while the remaining six contain a single cell. Within those six subcounties, every farm is therefore assigned the same rainfall measure by construction. This is why the addendum’s headline repair, a farm-cell rainfall anomaly with subcounty fixed effects, could not be implemented and its pre-specified fallback applies (Appendix E). It also explains why the registered primary specification is degenerate. A rainfall z -score that is constant within subcounty is fully absorbed by subcounty fixed effects.

Consequences for effective degrees of freedom. On the weather side, the effective sample contains 55 subcounty-year cells rather than 3,430 farm-year observations. In the backtest of Section 5.2.2, which is evaluated on a nominal 220-row panel, the intraclass correlation by year is 0.469 for the leading model comparison. This implies a design effect of 5.69 and an effective sample size of 38.6 rather than 220. Diebold–Mariano comparisons are therefore reported using both subcounty clustering with $G = 11$ and year clustering, while leave-one-subcounty-out cross-validation uses 11 folds. Bootstrap inference for index regressions at this resolution uses 1,999 replications over 11 clusters, and all corresponding coefficients are statistically null.

Results affected by the resolution of the record. This geographic constraint applies whenever the regressor of interest is the historical record. It therefore affects the compression slope, the signed dry lean, the flatness statistics constructed from the 55 subcounty-year cells, the backtest and its cross-validation, and the registered hypotheses comparing a belief index with a rainfall anomaly.

The same constraint does not govern the randomized information experiment. Treatment is assigned at the village level, and the specifications in Section 5.4 regress within-respondent belief changes on village-level assignment. The relevant cluster count is therefore the 50 villages represented in Arms A and B, of which 26 are treated, and the geographic resolution of the weather record does not determine treatment-effect inference. The same reasoning

applies to the farm-outcome ITT. The pooled comparison uses 60 villages and contrasts involving Arm C use 34 villages. The limited power of those farm-outcome tests instead reflects the resolution of the outcome data. Farm identity explains only 0.35 percent of the variation in peak greenness, while the CHIRPS cell explains 63.4 percent, leaving little independent farm-level signal for a village-randomized intervention to affect. Finally, the geographic weather constraint does not determine inference for the belief–investment relationships in Section 5.3.2 or the extreme-event memory results in Section 5.3.1, because the regressors in those analyses vary at the individual level. Those specifications cluster standard errors at the village level and include subcounty fixed effects to account for the common geographic environment.

D Additional robustness

This appendix reports additional analyses supporting the main results. We present the out-of-sample backtest underlying Section 5.2.2, power and equivalence bounds for null results that matter for the interpretation of the paper, measurement checks for the compression and flatness results, the full specification inventory underlying Section 5.3.2, supplementary results on dry-spell memory, two additional figures, and a detailed account of the 2026 growing season and farm-outcome analysis.

D.1 Inference

The inference procedures used throughout the paper are documented in Appendix C. Standard errors are clustered at the village level, using 50 clusters for the main comparison between Arms A and B and 60 clusters where the full experimental sample is used. Wild cluster bootstrap inference uses 9,999 replications, the finite-sample p-value adjustment, and the weights described in Appendix C.1 (Cameron et al., 2008). For contrasts involving Arm C, which contains only 10 villages, we use Webb six-point weights because wild cluster bootstrap inference can be unreliable when few clusters are treated (MacKinnon & Webb, 2017) and Rademacher weights generate few distinct bootstrap samples when clusters are few (Webb, 2023). Multiple testing is addressed using Romano–Wolf stepdown within defined outcome families (Romano & Wolf, 2005), while variable selection where used follows the double-selection procedure of Belloni et al. (2014) rather than stepwise selection.

D.2 The backtest of the perceived weather history

Section 5.2.2 shows that applying the estimated memory distortion removes most of the variation across recent weather histories. A natural validation question is whether measuring weather shocks relative to this constructed perceived history adds predictive information for farm productivity beyond a conventional rainfall anomaly. We test this out of sample.

For a farmer entering season t in subcounty s , the perceived historical reference constructed in Section 5.2.2 is

$$\hat{\mu}_{st} = \frac{1}{5} \sum_{j=t-5}^{t-1} [5 + \beta (b_{sj} - 5) + \gamma], \quad \beta = 0.0948, \quad \gamma = -0.324, \quad (13)$$

where b_{sj} is seasonal CHIRPS rainfall in season j in subcounty s on the nine-picture scale, β is the estimated recall sensitivity and γ the estimated dry bias. Setting $\beta = 1$ and $\gamma = 0$ gives the undistorted reference under accurate memory.

We define a perceived shock as the deviation of realized weather from the perceived historical reference in Equation (13). This is the belief-relative analog of a conventional rainfall anomaly. We then ask whether this measure predicts farm productivity out of sample. The analysis covers 3,430 farm-year observations, corresponding to 686 farms over the five

seasons from 2021 through 2025. Farm productivity is measured using peak vegetation greenness relative to each farm’s own observed history. This outcome does not enter the construction of the perceived-weather index.

Table 9 reports leave-one-subcounty-out prediction. None of the alternative weather-shock measures improves on predicting the sample mean, and the memory-distorted measure performs slightly worse than the conventional rainfall anomaly. This is consistent with the flatness result in Section 5.2.2. Once the historical reference varies very little across places and years, subtracting that reference from realized weather produces an index that is close to an absolute weather anomaly. Correspondingly, the distorted index has a correlation of 0.985 with the conventional rainfall anomaly. The undistorted historical reference, which preserves the variation that would remain under accurate memory, performs marginally better than the other weather measures but still does not outperform the sample mean.

Table 9: Out-of-sample prediction of farm productivity across alternative weather-shock measures.

Shock measure	Leave-one-subcounty-out R^2	Reference
Predicting the sample mean	0.0000	benchmark
Undistorted expectation ($\beta = 1$)	−0.0027	Eq. (13)
Conventional rainfall anomaly	−0.0078	conventional measure
Distorted memory index	−0.0116	Eq. (13)
Distorted drought surprise	−0.0273	Eq. (13)

Notes: Out-of-sample R^2 for farm productivity over 3,430 farm-year observations, corresponding to 686 farms in 55 subcounty-year cells over 2021–2025, using leave-one-subcounty-out cross-validation. Farm productivity is peak vegetation greenness relative to the farm’s own observed history and does not enter the construction of the shock measures. Negative values indicate worse out-of-sample prediction than the sample mean. Source: authors’ computation.

The conclusion is unchanged under alternative cross-validation schemes. Table 10 compares leave-one-year-out prediction over five folds, leave-one-subcounty-out prediction over 11 folds, and a leakage-free leave-one-year-out procedure in which the farm-specific historical productivity benchmark is recomputed within each training sample so that the held-out year does not enter the benchmark. Under all three procedures, the distorted memory index performs slightly worse than the conventional rainfall anomaly, with differences in out-of-sample R^2 of −0.0016, −0.0038, and −0.0008, respectively. No specification outperforms prediction by the sample mean. We also estimate leave-one-subcounty-out models including year fixed effects, with the same qualitative conclusion.

D.3 Power and equivalence bounds for the null results

Several null results play a substantive role in the paper. For these results, a failure to reject zero is not by itself sufficient to establish that economically meaningful differences are absent. We therefore report minimum detectable effects, confidence intervals, and equivalence bounds where appropriate.

Table 10: Out-of-sample R^2 by shock measure and cross-validation scheme.

Folds	Leave-one-year-out 5	Leave-one-subcounty-out 11	Leakage-free leave-one-year-out 5
Predicting the sample mean	0.0000	0.0000	0.0000
Undistorted expectation ($\beta = 1$)	-0.0368	-0.0027	-0.0311
Conventional rainfall anomaly	-0.0412	-0.0078	-0.0351
Distorted memory index	-0.0429	-0.0116	-0.0359
Distorted drought surprise	-0.0406	-0.0273	-0.0375
Distorted minus conventional	-0.0016	-0.0038	-0.0008

Notes: Out-of-sample R^2 for farm productivity over 3,430 farm-year observations from 2021 through 2025. Farm productivity is peak vegetation greenness relative to each farm’s own historical performance. In the leakage-free specification, that benchmark is recomputed separately within each training fold. Negative values indicate worse out-of-sample prediction than the sample mean. The final row reports the difference between the distorted-memory index and the conventional rainfall anomaly. Source: authors’ computation.

Observable characteristics and updating. Across 22 pre-treatment characteristics and four measures of weather-belief updating, the minimum detectable differential effect ranges from approximately 0.22 to 0.25 outcome standard deviations for a one-standard-deviation change in the predictor. The largest estimated differential effects range from approximately 0.16 to 0.21 standard deviations. The experiment therefore has power against relatively large differences in responsiveness across observable farmer characteristics but cannot exclude moderate ones. These figures are per standard deviation of the characteristic; for a binary characteristic such as literacy, whose standard deviation is about half the difference between the two groups, the detectable difference between literate and non-literate farmers is roughly twice as large.

Demographics, updating, and memory quality. The contrast between updating and initial memory quality is particularly informative. For the three belief-updating outcomes, joint tests of education, literacy, age, and household size give p-values of 0.82, 0.31, and 0.30. Across the corresponding cells, minimum detectable effects range from 0.13 to 0.19 standard deviations and equivalence bounds from 0.09 to 0.21 standard deviations. One of the twelve individual coefficients, household size on the change in mean rainfall belief, has $p = 0.028$, but this isolated result is not unusual in a family of twelve tests.

The same demographic characteristics strongly predict memory quality. Joint tests reject zero for recall accuracy measured by the ranked probability score and by a linear scoring rule, with p-values below 0.0001 under alternative fixed-effect specifications, and for memory entropy at $p = 0.007$. Literacy is the strongest individual predictor: respondents who cannot read recall past seasons 0.33 standard deviations less accurately than literate respondents (0.17 standard deviations per standard deviation of literacy), and schooling, recorded in four levels, adds nothing once literacy is held fixed (-0.02 standard deviations per level, $p = 0.75$). By contrast, the signed recall bias toward drier conditions is jointly unrelated

to these characteristics, with $p = 0.50$. Minimum detectable effects across the 16 memory-quality cells range from 0.08 to 0.17 standard deviations, with equivalence bounds between 0.05 and 0.23. The observed literacy gradient in recall accuracy lies within this detectable range and is observed in the data. The contrast in the main text is therefore supported by the power calculations. Human capital is systematically associated with initial memory accuracy, while neither the average dry lean nor short-run responsiveness to information shows a comparable demographic gradient. This is also consistent with the broader memory-quality predictor screen in Table 7, where 37 of 420 cells survive Romano–Wolf adjustment.

Farm-outcome treatment effects. The intention-to-treat estimates for the four primary Arm B farm outcomes are statistically indistinguishable from zero, but the precision differs substantially across outcomes. For standardized peak greenness, the estimate is -0.140 with a 95 percent confidence interval of $[-0.456, +0.176]$, already expressed in standard-deviation units. For the own-history benchmark, the estimate is -0.0033 with an interval of $[-0.0111, +0.0045]$. Relative to the outcome standard deviation of 0.0401, this corresponds to approximately $[-0.28, +0.11]$ standard deviations. For the village-mean benchmark, the estimate is $+0.0003$ with an interval of $[-0.0004, +0.0010]$. For the double-differenced benchmark, the estimate is $+0.0002$ with an interval of $[-0.0002, +0.0007]$, corresponding to approximately $[-0.02, +0.06]$ standard deviations given an outcome standard deviation of 0.0116. Interactions between treatment and the pre-treatment belief gap have bootstrap p-values between 0.29 and 0.75 across the four outcomes. The tightest interval, for the double-differenced benchmark, therefore excludes favorable Arm B effects larger than approximately six percent of an outcome standard deviation. The other outcomes leave considerably wider ranges of economically relevant effects open.

D.4 Measurement checks on the compression and flatness results

The main compression result is not sensitive to the use of a nine-picture elicitation scale. Reclassifying the nine bins into three broader categories, dry, normal, and wet, produces a recall-on-record slope of 0.0537 with a standard error of 0.0313 and a bootstrap p-value of 0.0982 over 2,559 respondent-year observations. Perfect preservation of differences across seasons would still imply a slope of one after this recoding. The lower precision therefore reflects the variation lost through coarsening rather than the disappearance of compression.

The estimate is also insensitive to alternative fixed-effect structures. Replacing respondent fixed effects with village fixed effects gives a slope of 0.0997 (SE 0.0337), while including both respondent and village effects gives 0.0948 (SE 0.0391). These estimates are close to the preferred coefficient of 0.0948 and provide little indication that common village-level survey conditions generate the compression pattern.

The survey export has no dedicated enumerator field, but enumerator codes can be recovered from the respondent identifier for 683 of the 686 interviews (Appendix A). Because each enumerator worked in 56 to 59 of the 60 villages, enumerator effects are separately identified from village effects. Because each respondent was interviewed by one enumerator, the respondent fixed effects in the compression regression already absorb enumerator fixed

effects. Appendix H reports enumerator fixed effects for the association between memory and expectations, the treatment contrasts, and the associations with planned adaptation.

A second concern is that apparent flatness could simply reflect respondents spreading their tokens across many pictures. The elicited distributions show the opposite. Before treatment, forward expectations occupy 1.76 bins on average, with a median of two, and 48.68 percent of the 567 Arm A and Arm B distributions place all ten tokens on a single picture. Recalled distributions occupy between 1.93 and 2.12 bins on average, each with a median of two. Across the 2,637 respondent-year recall observations, 35.0 percent are point masses. Half of respondents, 50.2 percent, give a point-mass recall distribution for at least one season, while 21.3 percent give one for every season they report. Planting-onset beliefs are still more concentrated, occupying between 1.63 and 1.76 intervals on average with medians between one and two. The elicited distributions are therefore generally concentrated. The flatness result in Section 5.2 concerns the limited variation in their means across objective weather histories, not excessive dispersion within individual responses.

The numerical flatness result is calculated over the 55 subcounty-year cells, giving each cell equal weight. The constructed perceived historical reference has a standard deviation of 0.0724 bins and ranges from 4.653 to 4.992. Under accurate memory, the corresponding historical reference has a standard deviation of 0.7644 bins and ranges from 4.760 to 8.339. These are the values reported in Section 5.2.2 and Figure 6. Weighting subcounty-year cells by the number of sampled farms would produce different summary statistics because heavily sampled locations would receive greater influence. The paper uses the unweighted distribution because the object of interest is variation across independent subcounty-year weather histories.

D.5 Specification inventory for memory, beliefs, and planned adaptation

Section 5.3.2 reports the association between remembered extremes and planned adaptation. Table 11 reports the complete grid from which it is drawn: five memory and belief measures crossed with five adaptation outcomes, 25 specifications in all. Each specification includes subcounty fixed effects and controls for cultivated acreage, age, and household size, with standard errors clustered at the village level. The associations are cross-sectional and not causal. To guard against selection from the grid, the q column reports Romano–Wolf q -values computed over all 25 cells. The association between recalling a past extreme and planned adaptation spending is among the strongest in the grid and survives the adjustment ($q = 0.0002$); it also survives rich pre-determined controls and enumerator and interview-date fixed effects (0.50 log points, $p = 0.008$; Appendix H.7).

Two patterns in this grid qualify the summary in the main text. The relationship between memory accuracy and irrigation changes sign across alternative specifications and is therefore not interpreted as a robust pattern. Recalled event severity is negatively associated with both planned spending and fertilizer use, the two severity coefficients estimated precisely; the other three are small and imprecise. Severity therefore does not point in the same direction

Table 11: Belief and memory measures and planned adaptation outcomes.

Regressor	Outcome	Coef.	SE	p	q	n
Expects future extreme	Adaptation spend (log)	0.659	0.175	0.0004	0.006	323
	Fertilizer	0.125	0.042	0.0043	0.060	567
	Improved seeds	0.066	0.051	0.206	0.898	567
	Irrigation	-0.051	0.030	0.089	0.636	567
	Plans insurance	0.048	0.020	0.020	0.226	327
Recalled past extreme	Adaptation spend (log)	0.781	0.151	< 0.0001	0.0002	397
	Fertilizer	-0.014	0.045	0.758	0.944	686
	Improved seeds	-0.056	0.046	0.221	0.898	686
	Irrigation	0.035	0.033	0.294	0.898	686
	Plans insurance	0.018	0.007	0.018	0.217	404
Recalled severity (1–10)	Adaptation spend (log)	-0.108	0.036	0.0037	0.060	333
	Fertilizer	-0.022	0.010	0.038	0.400	550
	Improved seeds	-0.011	0.009	0.200	0.898	550
	Irrigation	0.010	0.009	0.276	0.898	550
	Plans insurance	-0.003	0.002	0.196	0.898	339
Memory RPS (higher = worse)	Adaptation spend (log)	-0.351	0.128	0.0081	0.115	396
	Fertilizer	-0.055	0.027	0.046	0.436	685
	Improved seeds	-0.172	0.026	< 0.0001	0.0001	685
	Irrigation	0.065	0.026	0.017	0.217	685
	Plans insurance	-0.005	0.009	0.555	0.944	403
Memory entropy	Adaptation spend (log)	1.356	0.359	0.0004	0.006	396
	Fertilizer	-0.051	0.094	0.593	0.944	685
	Improved seeds	0.198	0.100	0.052	0.452	685
	Irrigation	-0.090	0.075	0.234	0.898	685
	Plans insurance	0.000	0.027	0.986	0.986	403

Notes: Each row reports a separate regression of the indicated adaptation outcome on the belief or memory measure. Specifications include subcounty fixed effects and controls for cultivated acreage, age, and household size, with village-clustered standard errors. There are 50 clusters when the regressor requires the pre-treatment forward elicitation and 60 otherwise. Adaptation spending and planned-insurance outcomes are defined among respondents planning any adaptation. Recalled severity is the maximum severity assigned to a named past event. Memory RPS is the ranked probability score of recalled weather and increases as recall becomes less accurate. The recalled-extreme rows use all three arms (0.781 on spending); the main text’s 0.77 uses Arms A and B, the sample of the richer specifications in Appendix H.7. p : village-clustered. q : Romano–Wolf stepdown over all 25 cells, from wild cluster bootstrap draws (Rademacher weights, 9,999 replications) shared across cells. The associations are cross-sectional. Source: authors’ computation.

as the other memory measures. The insurance rows rest on eight planning respondents, seven of them interviewed by the same enumerator, and are not interpreted (Appendix H.7).

D.6 Dry-spell salience in memory

We also ask whether seasons containing longer dry spells are remembered as drier after conditioning on the overall realized seasonal weather classification. In the cross section, the relationship goes in the opposite direction, with years containing longer dry spells recalled as somewhat wetter. This association disappears when respondent fixed effects are introduced and is also statistically indistinguishable from zero when inference is clustered at the subcounty level. The cross-sectional pattern therefore appears to reflect differences across locations rather than within-person memory encoding of dry spells.

The support of this test is limited. Longest dry spells in the recalled seasons range only from 5 to 13 days, whereas the 2026 season contains a 25-day dry spell. The null result therefore does not show that unusually long dry spells have no effect on memory. The historical sample simply contains no events close to the magnitude observed in 2026.

D.7 Supplementary figures

The following figures provide additional detail for the extreme-event memory results in Section 5.3.1 and the planned-adaptation results in Section 5.3.2.

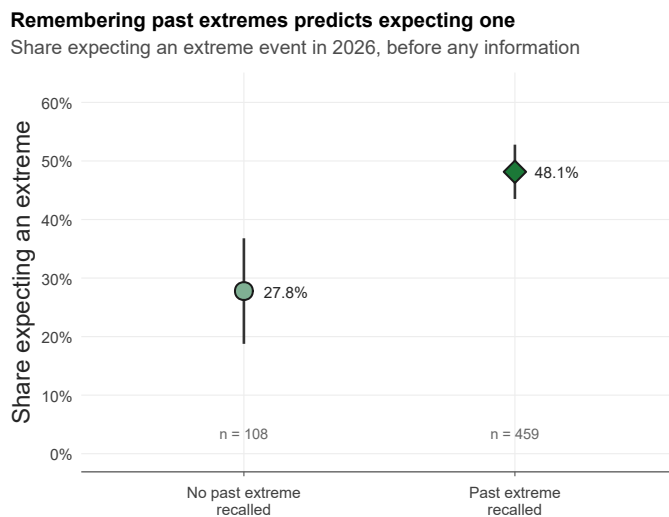


Figure 15: Remembering past extremes predicts expectations of future extremes.

Notes: Points show the share of farmers expecting a major weather event in 2026 among those who did and did not freely recall one from the past, with 95% village-clustered confidence intervals. With subcounty fixed effects, the difference is 19.4 percentage points (SE 4.5 percentage points). Source: authors' computation.

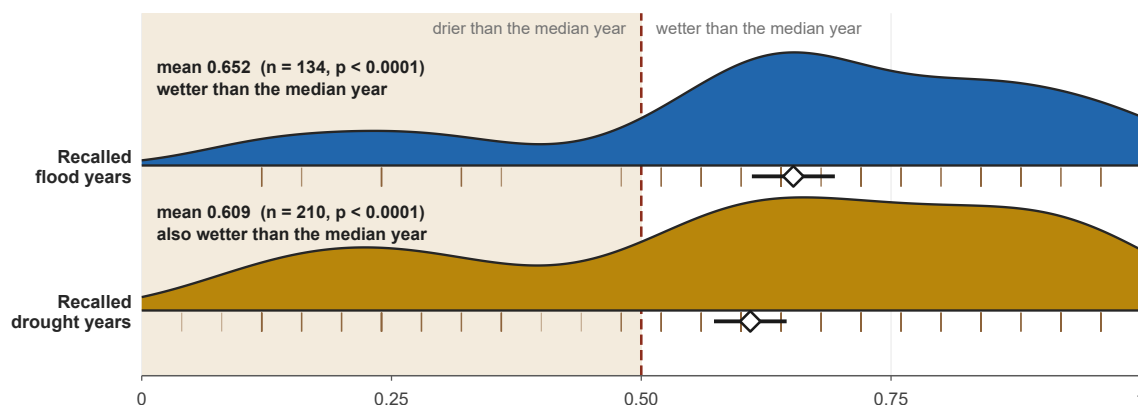
D.8 The 2026 season and the farm-outcome test in detail

The realized season. Rainfall from February through June 2026 was among the wettest in the 25-year record in every subcounty. Because the seasonal window extends beyond June,

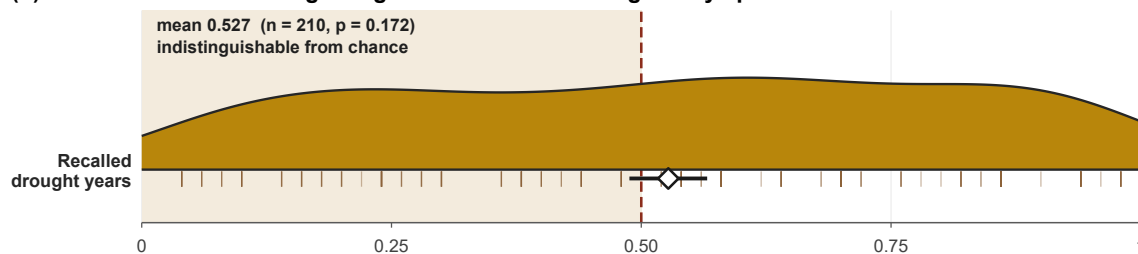
Recalled flood years were wet; recalled drought years were not dry

(a) Recalled extremes against the season's realized rainfall

Within-subcounty percentile of February–September rainfall in the year each farmer named



(b) The same recalled droughts against the season's longest dry spell



Within-subcounty percentile of the recalled year, 2001–2025 (0.50 = the median year)

Figure 16: Recalled floods and droughts differ in their correspondence with the historical record.

Notes: For each farmer who freely named a past extreme event and reported its year, the figure locates that year within the 2001–2025 rainfall distribution of the respondent's subcounty. Panel (a) shows that recalled flood years are relatively wet, with a mean seasonal-rainfall percentile of 0.652 ($n = 134$, $p < 0.0001$). Other flood measures show the same pattern, with mean percentiles of 0.624 for the number of days with at least 20 millimeters and 0.601 for maximum daily rainfall, both $p < 0.0001$. Years recalled as droughts have a mean seasonal-rainfall percentile of 0.609 ($n = 210$, $p < 0.0001$), placing them on the wetter rather than drier side of the local distribution. Panel (b) shows that recalled drought years are also not unusually dry when measured by longest dry spell, with a mean percentile of 0.527 ($p = 0.172$), or by the number of rain days, with a mean percentile of 0.519 ($p = 0.354$). Some 37.1 percent of recalled drought years fall in the upper dry-spell tercile, compared with 33.3 percent mechanically. Reported drought severity is essentially unrelated to the available measures of realized dryness ($r = 0.002$ and $r = 0.063$), while reported flood severity is positively related to realized rainfall. Diamonds show means with 95 percent confidence intervals and ticks show individual recalled events. The benchmark is measured at subcounty-season resolution and may therefore miss drought conditions that were highly localized or concentrated in a crop-sensitive part of the season. Source: authors' computation from the March 2026 survey and CHIRPS (Funk et al., 2015).

cumulative rainfall through 30 June amounted to 76.5 percent of the long-run mean for the full season. Even if the remainder of the season had matched the lowest historical July–August rainfall observed at each location, the resulting full-season rainfall ratio would have been 0.885, above the pre-defined dry-season threshold of 0.85. No study location could therefore be classified as dry under the fixed seasonal rule.

Weather beliefs and planned adaptation

Coefficient on each belief indicator, with subcounty fixed effects and controls for cultivated acres, age and household size; 95% village-clustered confidence intervals. Cross-sectional associations, not identified by the experiment.

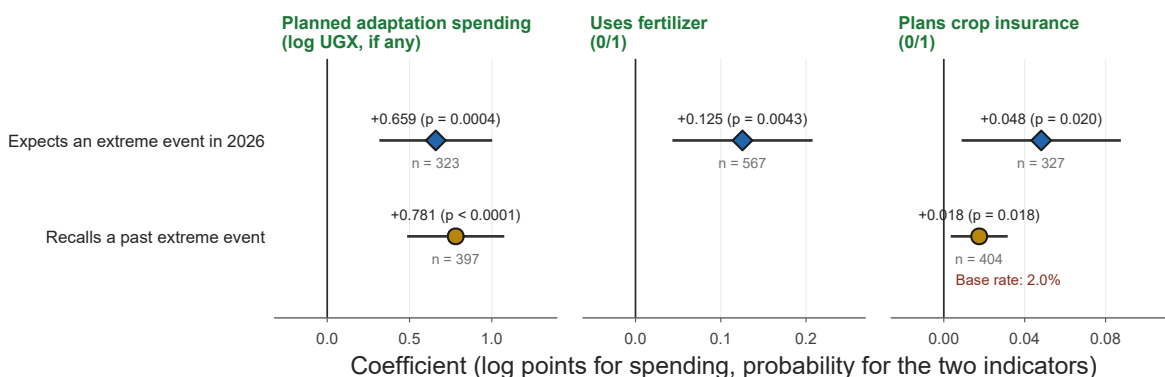


Figure 17: Weather beliefs are associated with planned adaptation and protection.

Notes: The figure reports coefficients on two belief measures, whether the farmer expects an extreme event in the upcoming 2026 season and whether the farmer freely recalled a past extreme event, estimated separately for planned adaptation outcomes. Specifications include subcounty fixed effects and controls for cultivated acreage, age, and household size, with village-clustered standard errors. Whiskers show 95 percent confidence intervals. Expecting an extreme event is associated with 0.659 log points greater planned adaptation spending ($p = 0.0004$, $n = 323$), a 12.5 percentage point higher probability of fertilizer use ($p = 0.0043$, $n = 567$), and a 4.8 percentage point higher probability of planning crop insurance ($p = 0.020$, $n = 327$). Among respondents planning any adaptation, 2.0 percent plan to purchase crop insurance. Recalling a past extreme is associated with 0.781 log points greater adaptation spending ($p < 0.0001$, $n = 397$) and a 1.8 percentage point higher probability of planned insurance ($p = 0.018$). A higher ranked probability score, indicating less accurate recall, is associated with lower planned use of improved seeds (-0.172 , $p < 0.0001$) and lower adaptation spending (-0.351 , $p = 0.0081$), while greater memory entropy is associated with greater spending ($+1.356$, $p = 0.0004$). The insurance panel rests on eight planners, seven of them interviewed by one enumerator, and is too thin to interpret; the associations with expecting an extreme event vanish with enumerator fixed effects, whereas the association of recalling a past extreme with spending survives them (Appendix H.7). These estimates are cross-sectional associations. Source: authors' computation.

The season nevertheless contained an unusually long dry spell. The longest observed spell with daily rainfall below the relevant threshold lasted 25 consecutive days. Across 275 historical subcounty-year observations, the previous maximum was 20 days and the median was nine. The 2026 episode remained three days below the 28-day threshold that would trigger the slightly-dry override described in Appendix B.

Despite its length, the dry spell is not associated with detectable vegetation damage in the available satellite measures. Peak greenness changes by approximately -0.0031 per additional dry-spell day, a relationship statistically indistinguishable from zero. Across the 17 rainfall grid cells, the rank correlation between dry-spell length and vegetation performance is 0.051 and the relationship is not monotonic. The same relationship is also null over the historical range of dry spells from 5 to 13 days. We therefore retain the pre-specified weather classification rather than recalibrating the dry-spell threshold ex post.

By the time farm outcomes were measured, the vegetation cycle was sufficiently advanced to characterize the season. All 686 farms were at least 30 days beyond their observed

vegetation peak. The median peak occurred on 13 May, day 133 of the year, median peak EVI2 was 0.555, and by 5 August median vegetation senescence had covered 57 percent of the decline from peak greenness back toward pre-planting levels.

Farm-level vegetation performance was close to its recent historical average. The median farm’s 2026 peak greenness was -0.0012 relative to its own 2021–2025 history, 47.5 percent of farms performed above their own historical benchmark, and the mean within-farm rank of 2026 among the six seasons from 2021 through 2026 was 3.65, compared with 3.5 for an exactly middle-ranked season. The aggregate realization was therefore broadly typical, although the lower tail was somewhat heavier than in an exactly average year.

Treatment estimates. Interactions between treatment assignment and the pre-treatment belief gap have bootstrap p-values between 0.29 and 0.75 across the four farm outcomes. The corresponding confidence intervals for the main Arm B effects are reported in Section D.3. One coefficient involving Arm C reaches conventional significance for raw peak greenness, with an estimate of -0.610 , but the relationship disappears for each benchmarked outcome. Given that Arm C contains only 10 villages and the pattern is not robust across outcome definitions, we do not interpret this coefficient substantively.

Reported adaptation intentions also decline between the two survey measurements, but the corresponding pattern in Arm A shows that this decline cannot be attributed to the information treatment. The share reporting plans for any adaptation falls by 8.03 percentage points in Arm A, which receives no information, compared with 5.14 points in Arm B and 8.40 points in Arm C. The fact that the largest decline occurs in the untreated arm is consistent with a repeated-question effect rather than a treatment response and provides another example of why repeated elicitation matters for interpreting within-interview changes.

Location of the farm coordinates. Farm outcomes are extracted around the coordinate recorded for each respondent in the survey item “GPS of main farming plot.” The enumerator manual instructed enumerators to start from the interview location and mark the respondent’s main plot on the tablet map. The recorded points nonetheless cluster tightly: the median distance from a respondent’s point to the nearest other respondent’s point is 3.3 meters, the 686 points form 35 groups in which every point lies within 110 meters of every other, and 94 percent of farms share more than 90 percent of their 110-meter buffer with another farm. The coordinates therefore mark interview sites, or points near them, rather than separate plots, and the farm outcomes are in effect village-level outcomes. The survey also contains manually typed coordinates, entered as a fallback (Online Appendix OA-7); they have not been validated as plot locations and are not used.

Limited power of the farm-outcome test. Farm outcomes are measured using satellite vegetation as a proxy for agricultural productivity, in the spirit of Burke & Lobell (2017), who validate satellite-based yield measures for smallholder maize. At the spatial resolution available here, most of the variation in this measure occurs across locations rather than individual farms. In the 2021–2026 farm panel, the CHIRPS rainfall cell explains 63.4 percent of the variation in peak greenness, village explains 6.2 percent, year explains 2.4 percent, farm identity explains 0.35 percent, and reported crop explains essentially none.

This leaves little independent farm-level variation for a village-randomized intervention to affect. Benchmarking each farm against its own history and then differencing this change from the corresponding village-level historical gain substantially reduces common geographic variation. Under this double-differenced measure, the rainfall-cell share falls from 0.92 to 0.005. The remaining variation is correspondingly small, with a standard deviation of 0.0116 compared with 0.0401 for the simpler own-history difference. The farm-outcome nulls should therefore be interpreted together with their confidence intervals rather than as evidence that improved beliefs could not affect outcomes in a season with more consequential weather variation.

E Pre-registration and deviations

The registration record consists of three documents. The AEA RCT Registry entry (AEARCTR-0018045) was filed in March 2026, before fieldwork began on 10 March. It registers six hypotheses together with the design, the randomization, the construction of the weather record, the outcome families, and the inference procedures. An operational pre-analysis plan dated 11 April 2026 defines four confirmatory tests for the post-harvest round in its own numbering. It was written after fieldwork ended on 31 March and after the survey data were exported on 1 April, so it is *ex ante* with respect to the 2026 outcomes but not with respect to the survey data. A post-harvest addendum dated 5 August 2026 reconciled the two documents under an explicit precedence rule before any 2026 outcome data were processed. For every result based on the survey, the March registration is therefore the only document fixed before the data were seen. Table 12 maps each registered hypothesis to where it is reported, or states why it is not.

The March registration also fixes the inference for all estimating equations: village-clustered standard errors, the wild cluster bootstrap with Rademacher weights as primary inference, Conley standard errors, randomization inference for treatment indicators, and Romano–Wolf adjustment within declared families.

Operational H2 is suspended. No post-harvest survey was conducted. There is no instrument, fieldwork contract, or budget line for one, and the approved ethics protocol covers only the single pre-season interview: it does not include a second contact with respondents after May 2026, and no amendment was filed. The funded proposal anticipated remotely sensed vegetation rather than survey-based yield outcomes. We therefore treat the self-reported-yield hypothesis as infeasible and suspended, rather than reconstructing an outcome after observing the satellite results.

Operational H1 required a pre-outcome specification change. The registered specification compares a belief-based index with a rainfall z -score while including subcounty fixed effects. Because the rainfall measure is constant within subcounty, it is fully absorbed by those fixed effects. The addendum’s headline repair keeps the fixed effects and measures the rainfall anomaly in each farm’s own CHIRPS cell, with a pre-specified fallback if too few subcounties contain more than one cell. The 686 farms occupy only 17 CHIRPS cells, with a median of 36 farms per cell, and only five of 11 subcounties contain more than one cell, so the fallback applies: the individual belief-based shock measure is compared with the subcounty rainfall anomaly without subcounty fixed effects, with inference clustered over the 60 villages. This change was fixed before the 2026 outcome data were processed.

Operational H3 has 11 independent geographic units. The objective weather classification varies at subcounty rather than parish level. Although the registered comparison refers to 42 parishes, the weather side contains only 11 independent geographic units. Treating parishes as independent would therefore overstate precision. The paper uses the subcounty as the relevant level of independent variation whenever inference depends on the historical record; Appendix C.6 provides the corresponding effective sample sizes and alternative inference.

Table 12: Registered hypotheses and where they are reported.

Hypothesis	As registered	In this paper
<i>AEA registry entry, March 2026</i>		
H1 Memory accuracy	Recalled distributions deviate from the record; divergence D_i (squared Euclidean) regressed on recency, extremity, and salience exposure.	In modified form: recall-on-record slope, dry lean, recall-age gradient, and ranked probability score (Section 5.1; Appendix F.1); salience through free recall (Section 5.3.1). The divergence regression as registered is not reported.
H2 Memory and forecast accuracy	Lower memory divergence predicts pre-information forecasts closer to the realized season.	Not reported. The test requires placing the 2026 season on the nine-picture scale, which this version does not do. Section 5.2.1 reports a different object, the association between recalled histories and expectations.
H3 Belief updating, B vs. A	Arm B beliefs move toward the record and become more precise; updating elasticity ϕ_1 and mixture weight ξ ; belief family {log score, ranked probability score, entropy}.	Reported as exploratory: change-score contrasts on eleven belief outcomes, including the ranked probability score and entropy (Section 5.4.1; Appendix C.3). Entropy does not move detectably, and the log score is not reported. ϕ_1 and ξ are reproduced by repeated elicitation in Arm A (Table 6).
H4 Farm outcomes, intention to treat	End-of-season EVI2 at farm coordinates as a within-village z -score, with stratum fixed effects; family {EVI2, NDVI, SAR}.	Reported, null (Section 5.4.3; Appendices D.3 and D.8), using peak rather than end-of-season EVI2. NDVI and SAR outcomes are not produced (Appendix G), and the addendum reclassifies the family as exploratory.
H5 Belief-relative shocks	Individual shock S_i predicts agronomic outcomes better than a conventional rainfall anomaly.	In part: an out-of-sample backtest over 2021–2025 with a constructed perceived-history index (Appendix D.2). The individual-level 2026 test is operational H1 below.
H6 Anchoring, C vs. A	Arm C's first-stated beliefs lie closer to the record than Arm A's pre-information beliefs.	Reported as exploratory (Appendix H.4). Stated confidence rises; shifts in the amount and onset beliefs are not detected, with ten treated villages.
<i>Operational plan, 11 April 2026, and addendum, 5 August 2026</i>		
H1 Farm-level horse race	S_i against a rainfall z -score for 2026 farm greenness, with subcounty fixed effects.	Degenerate as registered; re-specified before outcome data were processed (below). Not reported in this version.
H2 Self-reported yield	The same comparison for self-reported maize, bean, and cassava yields.	Suspended (below).
H3 Out-of-sample kernel test	Kernel-predicted 2026 beliefs against subcounty climatology as forecasts of the realized 2026 season.	Not reported in this version. The test has 11 independent units (below).
H4 Confidence calibration	Arm B confidence against the accuracy of the same respondent's forecast for 2026.	Not reported in this version.

Notes: The operational plan numbers its hypotheses independently of the registry: its H1 implements registry H5, and its H4 is a post-harvest test related to registry H3. After the suspension of operational H2, the confirmatory family is $\{H1, H3, H4\}$ in the operational numbering. Source: authors' construction.

Differences between the registered and the fielded design. Three features of the fielded study differ from their description in the March registration. The registration describes a laminated one-page card for each village; the fielded treatment was a seven-sheet printed packet labeled by parish and shared among a village’s respondents (Appendices B.1 and F.7). It describes onset intervals of about five days from mid-February to early April; the fielded calendar has ten intervals of six or seven days from 15 February to 24 April (Appendix B.3). And it places the study on the western slopes of Mount Elgon, whereas the registered districts lie on the plateau west of the slopes (Section 2.2).

Status of the remaining analyses. Analyses in Sections 5.1 through 5.4 that are not mapped to a registered hypothesis in Table 12 are exploratory. The farm-outcome results are under-powered relative to the registered minimum detectable effect, because the main information arm contains only 26 treated villages, so we report them with confidence intervals and power calculations rather than interpreting insignificance as evidence of no effect.

F Mechanism, benchmark resolution, and the scope of the results

This appendix collects the analyses that support the discussion of mechanism, measurement, adaptation plans, generality, and scale in the main text. All of them are exploratory and lie outside the registered family of hypotheses. Inference follows Appendix C. Standard errors are clustered on villages, bootstrap p-values come from the restricted wild cluster bootstrap with 9,999 replications, and results whose regressor is the historical record are also reported with inference on the 11 subcounties.

F.1 Sources of compression

Several accounts predict that recalled magnitudes respond less than one for one to realized magnitudes. Under rational inattention, farmers keep a coarse record because a precise one is costly, and they should keep a finer record when more is at stake. Under Bayesian decoding of a noisy memory trace, the farmer combines what she retrieves with her prior knowledge of local seasons, and recall is drawn toward that prior by more when the trace is noisier. Under availability, salient hazards are recalled too readily, which produces asymmetries between droughts and floods. Under anchoring on the response scale, answers gravitate toward the middle picture irrespective of local conditions. The first three imply the same linear form for the mean of recall, so they can be told apart only through comparative statics. Table 13 sets out the predictions we can test and what we find.

The evidence narrows the set of explanations without selecting one. Anchoring on the middle picture and a pure grid artifact are rejected. Recall is drawn toward one common point on the dry side of optimal, which is what a shared notion of a normal season would produce. Stakes and the cost of processing do not change the slope, which is difficult to reconcile with a simple rational inattention account, although the power of these tests is limited. The association between diffuse recall and stronger compression is the pattern a noisy trace would generate, but part of it is arithmetic on a bounded scale. We therefore describe the friction as noisy memory drawn toward a common reference point and leave its optimality open.

F.2 Benchmark resolution and compression

The historical record varies by subcounty and season. If a farmer's own plot experienced something different, the record measures her experience with error, and the slope in Equation (8) is attenuated even if her memory is accurate. With classical error the estimated slope equals true recall sensitivity times the reliability of the benchmark, so the record would need a reliability of 0.09 for perfect memory to produce our estimate, and of 0.19 for a true recall sensitivity of one half. Three checks bound the role of this channel.

Table 13: Testable predictions of accounts of compressed memory and the evidence.

Account	Prediction	Evidence	Reading
Rational inattention	Less compression when more is at stake	Slope interaction with log cultivated area +0.011 (SE 0.031); detectable effect 0.086. Household heads compress less (+0.173, SE 0.060), but headship is entangled with gender	Mostly null
Rational inattention	Less compression when processing is cheaper	Literate -0.037 (SE 0.054); any schooling -0.071 (SE 0.059); gains above 0.07 excluded	Not supported
Noisy memory	Reference point is the local average season	Between-subcounty slope 0.22 (SE 0.27); a value of one is rejected ($p = 0.012$); equality with the within slope is not ($p = 0.67$)	Common reference point
Scale anchoring	Reference point is the middle picture	Implied center 4.37, interval [3.99, 4.75] with inference on subcounties; point-mass answers fall on picture five (22 percent) no more often than on picture four (22 percent)	Rejected
Noisy memory	More compression when recall is more diffuse	Slope interaction with own entropy -0.056 (SE 0.027); with share of multimodal answers -0.248 (SE 0.078)	Weakly supported
Noisy memory, availability	Symmetric or asymmetric shrinkage	Slope below picture five 0.156, above 0.021, difference -0.136 (SE 0.143)	Symmetry not rejected
Availability	Extreme seasons more likely to be remembered at all	Effect of distance from optimal on reporting a memory +0.001 (SE 0.012)	Not supported
All	Recent seasons recalled with less compression	Slope per year of recall distance -0.0001 (SE 0.024). The share able to recall a season falls by 9.4 points per year	Extensive margin only
Response scale	Compression is an artifact of the nine-picture grid	Three-category slope 0.054 (SE 0.031) against a benchmark of one	Rejected

Notes: Each row reports the interaction of the recall-on-record slope of Equation (8) with the indicated moderator, or the indicated test, on 2,559 respondent-season observations with respondent fixed effects and village-clustered standard errors. The reference point is the fixed point of the fitted line. The between-subcounty slope comes from a specification that separates each subcounty's mean realized season from deviations around it, with wild bootstrap inference on the 11 subcounties. Source: authors' computation.

Table 14: Benchmark resolution and recall sensitivity.

	Estimate	SE or interval
<i>A. Agreement among neighbors (no benchmark required)</i>		
Share of within-respondent recall variance that is idiosyncratic	0.911	[0.873, 0.930]
Share common to the subcounty and season	0.075	[0.069, 0.127]
Share common to the village and season, beyond the subcounty	0.014	< 0.02
Implied ceiling on recall sensitivity	0.40	[0.35, 0.48]
<i>B. Correcting for error in the card classification</i>		
Slope on the card classification (paper)	0.095	0.034
Slope on the rainfall-only classification	0.098	0.025
Card classification instrumented by the rainfall-only classification	0.272	0.075
same, clustered on subcounties	0.272	0.115
<i>C. A village-level benchmark</i>		
Recalled distance from optimal on village vegetation index	−0.027	0.027
Card classification distance from optimal on the same index	−0.103	0.042

Notes: Panel A decomposes the recalled mean picture, for 2,637 respondent-seasons, into respondent, subcounty-season, village-season and residual components in a linear mixed model. Intervals come from 999 bootstrap draws of villages; for the village-season share the bootstrap distribution piles up at zero and we report an upper bound. The ceiling is the square root of the shared variance divided by the within-subcounty variance of the record. Shared interview conditions also load on the shared components, so the ceiling is an upper bound. Panel B uses respondent fixed effects and village-clustered standard errors; the first-stage F statistic is 820. Panel C uses peak Sentinel-2 vegetation extracted at the village, standardized within village over 2021–2025, with respondent fixed effects. Source: authors’ computation.

Panel A exploits the fact that neighbors experienced the same weather. Whatever the true local season was, it is common to the farmers of a village, so it can account for at most the shared part of their recalls. That part is 9 percent of the variation in how a farmer recalls different seasons, and almost none of it arises below the subcounty. A finer benchmark therefore has little left to explain, and even if the entire shared component were weather, recall sensitivity could not exceed about 0.40. This ceiling treats the within-subcounty variance of the record as the variance of the weather farmers experienced; if the record itself measures that weather with error, the ceiling is higher. Panel B treats the two classifications as repeated measurements. The instrumented slope of 0.27 indicates that error in the card classification attenuates the estimate, but not by enough to bring it near one. Panel C shows that recalled crop conditions do not follow village-level vegetation, which the card classification does follow. These checks cannot rule out variation in experience across plots within a village, for instance from drainage or soil. With these caveats, they place recall sensitivity well below one; Appendix H.1 corrects the ceiling for error in the record, which raises it to 0.68, and compares recall with continuous rainfall, which gives roughly 0.3. A comparison is instructive. Patel (2025) regresses Bangladeshi farmers’ beliefs about the salinity of their soil over the current season on a sensor-based measure for their own plots and obtains a slope of 0.79 with an R^2 of 0.03. Beliefs about the current season’s conditions, benchmarked at the plot, thus track the truth on average but with a great deal of noise. Our ceiling of 0.40 rests on agreement among neighbors rather than on a finer benchmark,

which suggests that the much lower slope for remembered seasons is not only a matter of resolution.

F.3 An illustration of the value of information

We illustrate what is at stake with the protection rule of Equation (5). A farmer pays P to avoid a loss L if the season falls in an adverse tail and protects when her perceived probability of that tail exceeds P/L . We compare the perceived probabilities in the elicited pre-information beliefs of the 567 respondents in Arms A and B with the frequency of the same events in two records, the 2001–2025 rainfall record expressed on the nine-point scale, and the card classification over its 55 subcounty-season cells. The exercise is illustrative. It abstracts from risk aversion and basis risk and treats the token allocation as the farmer’s probability.

Table 15: Perceived and recorded tail probabilities.

	Picture 3 or drier	Picture 7 or wetter
Frequency in the rainfall record, 2001–2025	0.26	0.30
Frequency in the card classification, 2021–2025	0.13	0.05
Mean perceived probability for 2026	0.11	0.06
Share of farmers placing no tokens on the event	0.75	0.90
Mean recalled mass on the event, 2021–2025	0.33	0.14
Share not protecting although the rainfall record warrants it, $P/L = 0.15$	0.76	0.90
same, $P/L = 0.25$	0.32	0.55
Option value of protection forgone with an accurate five-season memory	0.56	0.59
with the estimated memory distortion	1.00	1.00

Notes: Perceived probabilities are token shares in the first elicitation of expectations for 2026. The last two rows apply the decision rule to a reference distribution formed from the five preceding seasons of the rainfall record, either as realized or after applying the compression slope and dry lean of Section 5.1, without recall noise, and report the share of the gain from optimal protection that is lost, integrated over cost–loss ratios. Allowing for recall noise lowers the loss under distorted memory to 0.60 for the dry tail and 0.94 for the wet tail, and the dry-tail figure is sensitive to how the dry lean is specified. Source: authors’ computation.

Three points emerge from Table 15 and Figure 18. First, expectations are concentrated on the central pictures. Picture five receives 45 percent of all tokens against a frequency of 20 percent in the card classification and 14 percent in the rainfall record, and three quarters of farmers assign no probability to a clearly dry season. Measured against the rainfall record, most farmers would decline protection whenever it costs less than about a quarter of the loss it prevents. Second, this verdict depends on the record. The card classification rarely assigns seasons to the tails, and against it the average perceived tail probabilities are close to the recorded frequencies, although that comparison rests largely on one season. Third, the narrowness of expectations is not inherited from a memory that lacks extremes. Recalled

histories place a third of their mass on picture three or drier. What compression removes is the link between extremes and the seasons in which they occurred. In this illustration a short memory also matters, since five accurately remembered seasons already forgo more than half of the option value of protection. Repeated elicitation in Arm A indicates that a substantial part of the dispersion in perceived tail probabilities is noise, with a correlation between the two measurements of 0.40 to 0.57, so the individual-level figures are upper bounds.

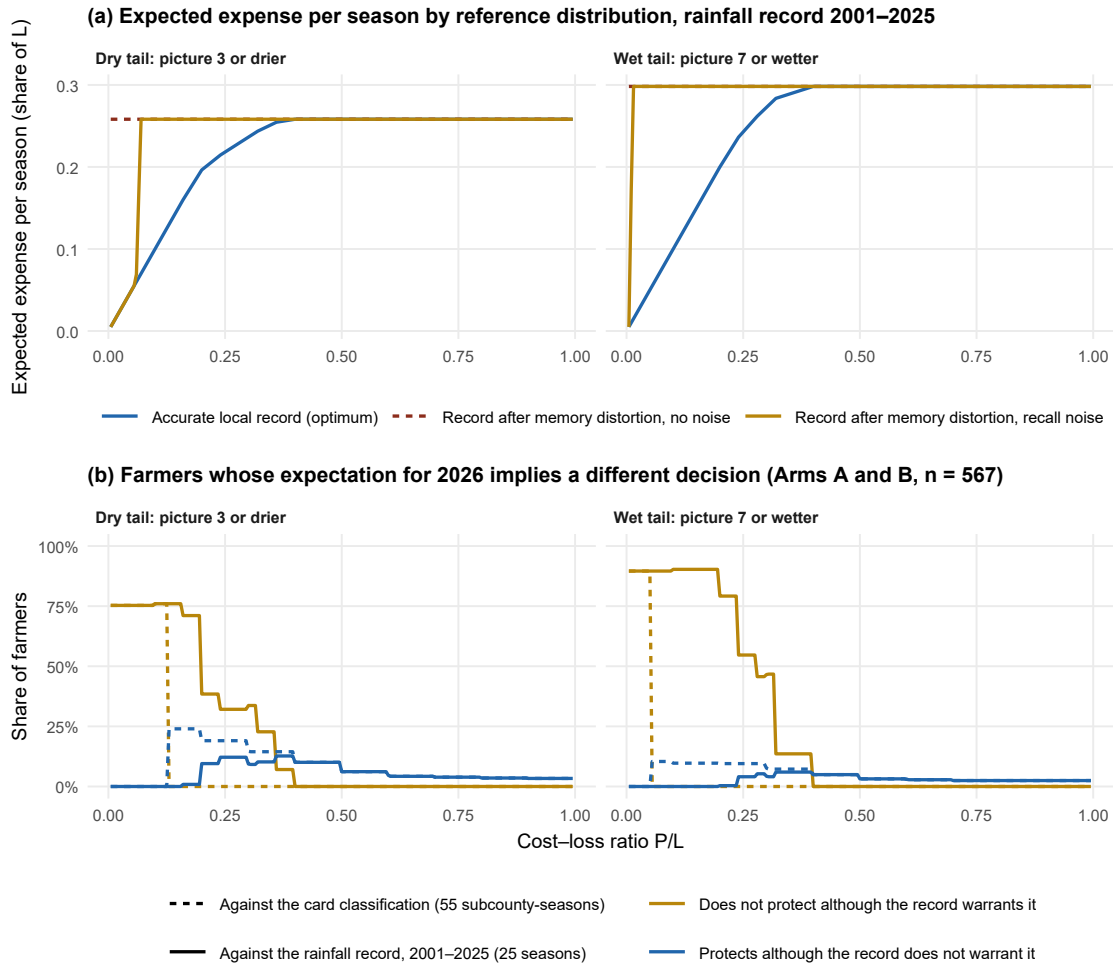


Figure 18: Protection decisions under perceived and recorded tail risk.

Notes: Panel (a) shows the expected expense per season of a farmer who applies the cost-loss rule with a reference distribution formed from the accurate local record, or from the record after the estimated memory distortion, with and without recall noise. Panel (b) shows, at each cost-loss ratio, the share of the 567 respondents whose elicited belief implies a different decision from the one the record warrants. Source: authors' computation.

F.4 Information and planned adaptation

Table 16 reports the estimates behind Section 5.4.3. The signed test, whose outcome is planning any adaptation, pools Arms A and B and interacts assignment to Arm B with a

measure of how far the card pushes the farmer toward more adverse conditions, constructed only from the card and the farmer’s first elicitation. The slope in Arm A, where no card was shown, absorbs whatever is mechanical.

Table 16: The information treatment and planned adaptation.

	Estimate	SE	Bootstrap p
<i>A. Average effects, Arm B minus Arm A</i>			
Plans any adaptation, change (share)	+0.029	0.042	0.49
Plans any adaptation, final level given initial level	+0.027	0.036	0.47
Planned spending, change (log)	+0.060	0.543	0.91
Expects an extreme event in 2026, change (share)	−0.031	0.032	0.35
Ten adaptation types, smallest adjusted p			0.53
Eight financing channels, smallest adjusted p			0.66
<i>B. Signed test: Arm B $\times$ push toward adverse conditions (per SD)</i>			
Push in mean picture	+0.043	0.042	0.33
Push in mass on picture 4 or drier	+0.082	0.040	0.05
Push in mass on picture 3 or drier	+0.067	0.035	0.07
Push in expected distance from optimal	+0.083	0.041	0.06
<i>C. First stage: belief revision on Arm B $\times$ push</i>			
Push in mean picture	+0.109	0.112	0.36
Push in mass on picture 4 or drier	+0.220	0.074	0.01

Notes: $N = 566$ respondents in 50 villages. The minimum detectable average effect on planning any adaptation is about 12 percentage points. In Panel B the smallest Romano–Wolf adjusted p -value is 0.22 within the eight outcomes examined for a given definition of the push and 0.54 across all 32 tests. About three quarters of the largest estimate reflects a difference between arms in the relation between the push and plans before the card was shown. The first-stage F statistics are 1.0 and 8.9, so no instrumental variables estimate is reported. Source: authors’ computation.

F.5 Prior access to information

If compression reflected a lack of information in these particular villages, farmers with better access to information should compress less, and the results would say little about other settings. We construct an index from six items elicited before the memory module: access to extension, receipt of any weather information, receipt of information on past weather, receipt of forecasts, receipt of forecasts specific to the area, and naming a formal source among the factors considered when forming expectations. Some 62 percent of respondents receive forecasts, 94 percent of them by radio and 20 percent at the village level, 31 percent receive some information on past weather, and one third receive neither forecasts nor weather information. Table 17 relates the standardized index to memory, initial beliefs, and updating.

Table 17: Information access, memory, and updating.

Outcome (per SD of the information index)	Estimate	SE	Adjusted q
Memory accuracy, ranked probability score (lower is better)	−0.072	0.024	0.04
Memory entropy	+0.033	0.007	< 0.01
Initial distance to the card	−0.071	0.024	0.04
Compression slope, interaction	+0.046	0.032	0.56
Treatment effect on distance to the card, interaction	+0.069	0.055	0.57
Treatment effect on score against 2025, interaction	+0.076	0.072	0.59
Treatment effect on expected onset (days), interaction	−0.593	0.731	0.59

Notes: Regressions include subcounty fixed effects and controls for gender, age, household size, literacy, and education, with village-clustered standard errors. Adjusted values are Romano–Wolf over the ten primary hypotheses of this analysis. Treatment interactions pool Arms A and B. The association with the initial distance to the card weakens to −0.049 (SE 0.026) when the index excludes the two items nested in others, while the other rows are not sensitive to the construction of the index. Source: authors’ computation.

Better informed farmers remember more accurately and begin somewhat closer to the record, but we cannot reject that they compress differences between seasons as much as others and respond to the card as others do. The confidence interval for the slope interaction implies a slope of at most 0.31 for a farmer two standard deviations above the mean of the index, still far from one.

F.6 Other forms of protection and the information environment

Table 18 describes how farmers plan to respond to their expectations and how they plan to pay for it. The survey contains no module on transfers, remittances, risk-sharing networks, livestock, or savings, so it cannot measure informal insurance directly.

Farmers form rainfall expectations mainly from their own experience and from one another. Asked which sources they use, 61 percent of respondents name personal memory of past seasons and 32 percent discussions with other farmers, while local leaders and government or extension sources are each named by fewer than 13 percent. Formal forecasts do reach the area: some 62 percent report access to some rainfall forecast, mostly over the radio, but only a fifth of those receive forecasts at the village level. Farmers plan to grow 3.4 crops on average. The financing pattern in Table 18 is in line with national data, in which 37 percent of adults belong to a savings group and almost none hold crop insurance (Financial Sector Deepening Uganda, 2018), and with evidence that savings groups, although widely used, leave average consumption largely unchanged (Karlan et al., 2017).

F.7 The cost of delivering the information

The cards were produced by an automated pipeline from public CHIRPS and Sentinel-2 data, so the cost of generating an additional packet is negligible. Table 19 reports the marginal cost of printing and presenting a packet under low, central, and high assumptions. The

Table 18: Planned adaptation, financing, and sources of information.

<i>Planned adaptation (N = 404)</i>		<i>Financing (N = 404)</i>	
Changing crop varieties	0.64	Community savings group	0.49
Modifying fertilizer use	0.60	Own savings	0.38
Adjusting planting dates	0.38	Sale of assets	0.37
Diversifying crops	0.25	Loans from family or friends	0.16
Investing in irrigation	0.19	Bank loan	0.12
Accessing credit	0.15	Government subsidy or grant	0.07
Attending training	0.13	Microfinance institution	0.04
Seeking off-farm employment	0.03	Any external source	0.72
Purchasing insurance	0.02		
<i>Whom farmers consult (N = 686)</i>		<i>Reactions to the information (N = 412)</i>	
Discusses rainfall expectations with others	0.88	Historical rainfall information helpful for planning	0.98
with neighboring farmers	0.64	Would like similar information in future	0.99
with family	0.50	Information differed from own memory	0.51
Trusts experienced farmers most	0.63	Wished to adjust expectations, Arm B	0.59
Trusts agricultural experts most	0.27	Very or extremely surprised	0.67

Notes: Shares of respondents. Reactions were recorded in Arms B and C. Planned adaptation and financing are multiple-response items asked of the 59 percent of respondents who planned to change their practices in light of their rainfall expectations, before any information was shown. The helpfulness item asks about historical rainfall information in general and not about the cards specifically, and stated demand of this kind is prone to courtesy bias. Source: March 2026 survey.

figures are assumptions about a delivery model, not accounting data from the study, whose fieldwork cost of about GBP 30 per respondent reflects the survey and not the intervention.

Table 19: Marginal cost of the information packet per farmer (GBP).

	Printing	Delivery	Total
Added to an existing visit, low	0.04	0.21	0.24
Added to an existing visit, central	0.18	0.42	0.60
Added to an existing visit, high	5.60	0.94	6.54
Stand-alone village meeting, central	0.37	3.50	3.86

Notes: The central case assumes seven color sheets at GBP 0.30 each shared by eleven farmers, as fielded, and ten minutes of an enumerator’s time at a daily rate of GBP 20. The low case assumes cheaper printing and thirty farmers per packet. The high case assumes a laminated copy for each farmer and a share of travel costs. One pound corresponds to about USD 1.33. Source: authors’ computation.

Table 20 scales the effects of an onset forecast in Burlig et al. (2024) by the ratio of the shifts in expected onset in our experiment and theirs, and by a further factor ρ : the share of the Indian response that an equal belief shift would produce here. The central-case cost of about USD 0.80 per farmer is covered at $\rho \simeq 0.13$ within one month. The calculation treats the onset shift as if it had the value of a shift induced by an accurate forecast of the coming season. That is not established here: the rainfall benchmark captures early rain rather than the planting rains (Appendix H.6), a record informs the distribution a season is drawn from rather than the season itself, our beliefs are measured within minutes, their consumption estimate is not robust to their own adjustment for multiple outcomes, and the settings differ. The table therefore states the effect size the card would have to reach, not an effect it achieves.

Table 20: Downstream effects implied by a belief shift of the size we estimate, under a stated share ρ of the Indian response.

Share of the Indian response, ρ	0.10	0.25	0.50	1.00
Index of economic well-being (SD)	0.006	0.014	0.028	0.056
Food consumption per household member (percent)	0.6	1.5	3.1	6.2
Consumption gain per household in one month (USD)	0.63	1.58	3.16	6.33

Notes: Entries are $\rho \times (2.59/2.76) \times$ the effects of an onset forecast on the well-being index and on food consumption per member in Burlig et al. (2024), where 2.59 and 2.76 days are the shifts in expected onset in our experiment and theirs. The last row multiplies by the mean household size in our sample, 7.75. Entries are a calibration under the stated assumption, not an estimate. Source: authors’ computation.

G Variable definitions

This appendix defines the variables used in the paper, grouped by the object they measure.

G.1 Belief variables

Variable	Definition
Recalled distribution	Allocation of ten tokens across the nine ordered weather pictures for a named past season from 2021 through 2025. Bin 1 represents extreme drought, bin 5 the center of the scale, and bin 9 severe flooding.
Mean recalled bin	Token-weighted mean of the recalled distribution, $\bar{r}_{it} = \sum_b b \frac{n_{itb}}{10},$ where n_{itb} is the number of tokens respondent i assigns to bin b for season t.
Forward belief	The same nine-picture elicitation for the upcoming 2026 season. It is elicited before the information treatment in Arms A and B and after the information treatment in Arm C.
Mean forward bin	Token-weighted mean of the forward-belief distribution.
Memory entropy	Shannon entropy of the recalled token allocation, averaged across the seasons recalled by the respondent. Higher values indicate a more diffuse recalled distribution.
Sharpness	Concentration of the recalled distribution, defined as the complement of entropy under the normalization used in the paper. Higher values indicate a more concentrated allocation.
Probability assigned to the realized bin	Share of the respondent's ten tokens assigned to the objectively classified weather bin for that season.
Local concentration	Share of tokens assigned to the realized weather bin or one of its two immediately adjacent bins.
Memory accuracy	Ranked probability score of the recalled distribution relative to the realized weather bin. Lower values indicate more accurate recall, so coefficient signs are interpreted accordingly.
Signed bias	Mean recalled bin minus the realized bin. Negative values indicate recall that is drier than the historical record, while positive values indicate recall that is wetter.

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Variable	Definition
Recall distance	Number of years between the survey year and the season being recalled.
Onset belief	Token-weighted mean of the ten planting-onset intervals, expressed as day of year using the midpoints of the fielded calendar. The calendar runs from 15 February through 24 April, with midpoints at days 49, 56, 62.5, 69, 76, 83, 90, 97, 104, and 111. Appendix B.4 describes the construction in detail.
Recalled extreme event	Indicator and associated characteristics of an extreme weather event freely recalled by the respondent, including hazard type, year, and reported severity. Respondents are not prompted with particular years.
Expects a future extreme	Indicator equal to one if the respondent reports expecting any extreme weather event during the upcoming 2026 season.
Confidence	Self-reported confidence in the elicited weather expectation, measured on the survey Likert scale.

G.2 Objective weather and outcome variables

Variable	Definition
Realized weather bin	Classification of each season into the same nine-picture scale used in the belief elicitation, based on CHIRPS rainfall and satellite vegetation. The classification varies at the subcounty-year level, yielding 55 independent cells across 11 subcounties and five seasons. Appendix B.1 gives the complete classification rule.
Season rainfall	Total CHIRPS rainfall over the defined growing-season window.
Heavy-rain days	Number of days during the season with at least 30 mm of rainfall.
Maximum daily rainfall	Largest single-day rainfall total during the season.
Longest dry spell	Longest sequence of consecutive days during the season with rainfall below 1 mm. The same threshold enters the dry-spell rule used in the weather classification.
Within-subcounty percentile	Rank of a season's weather measure within the corresponding subcounty's 2001–2025 historical distribution, rescaled to the unit interval.
Rainfall anomaly (z)	Seasonal rainfall relative to the subcounty's historical mean, expressed in standard-deviation units.

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Variable	Definition
Peak EVI2	Maximum two-band enhanced vegetation index during the relevant growing period, measured within a 110-meter buffer around the coordinate recorded for the respondent's main plot, which in practice marks the interview site (Appendix D.8).
Own-history benchmark	Peak EVI2 relative to the same farm's peak-EVI2 history over 2021–2025. Construction requires at least three usable historical seasons.
Double-differenced benchmark	Change in a farm's peak EVI2 relative to its own history, net of the corresponding village mean change. This transformation reduces the share of variance attributable to the rainfall grid cell from 0.92 to 0.005. Its standard deviation is 0.0116, compared with 0.0401 for the unadjusted own-history difference.

NDVI, a registered secondary outcome, is not used anywhere in the paper: the available farm-level NDVI extraction is numerically unstable, with values above one. EVI2, which is used for the farm-outcome analysis, is unaffected.

G.3 Adaptation and investment variables

Variable	Definition
Adaptation spending	Log planned expenditure on adaptation in Ugandan shillings. The amount is observed only among respondents who report planning some form of adaptation, so spending regressions condition on this group.
Uses fertilizer	Indicator for planned fertilizer use. The sample mean is 57.3 percent.
Uses improved seeds	Indicator for planned use of improved seeds. The sample mean is 52.6 percent.
Uses irrigation	Indicator for planned irrigation use. The sample mean is 22.6 percent.
Plans insurance	Indicator for planning to purchase crop insurance for the upcoming season. Among respondents planning any adaptation, the sample mean is 2.0 percent. The survey does not establish whether an index or indemnity insurance product was commercially available to each respondent in the study area during 2026.

G.4 Treatment and updating variables

Variable	Definition
Arm A	Control arm. Respondents complete the same forward-belief elicitation twice with no information treatment between the two measurements. The arm contains 274 respondents in 24 villages.
Arm B	Information arm. Respondents complete the forward-belief elicitation, receive the historical weather information treatment, and then complete the forward elicitation again. The arm contains 293 respondents in 26 villages.
Arm C	Post-only information arm. Respondents receive the historical weather information before completing a single forward-belief elicitation. No pre-treatment belief is observed. The arm contains 119 respondents in 10 villages.
Earth-mover distance to the historical record	Wasserstein distance between the respondent's elicited forward distribution and the weather distribution summarized by the information card. Lower values indicate greater similarity to the historical record shown on the card.
Ranked probability score against realized 2025 weather	Ranked probability score of the elicited forward distribution evaluated against the realized 2025 weather bin. Lower values indicate a distribution closer to the realized benchmark.
Accuracy gain	Pre-treatment distance or scoring-rule value minus its post-treatment value. Positive values therefore indicate movement toward the relevant historical benchmark.
Absolute prior gap	Absolute distance between the mean of the information-card distribution and the respondent's mean pre-treatment belief. Because the initial elicitation enters both this gap and any subsequent change score, regressions using this quantity are subject to the mechanical relationship discussed in Appendix C.2.
Prior onset tercile	Tercile of the respondent's pre-treatment expected planting onset. Mean initial beliefs are days 51.7, 63.7, and 80.4 in the early, middle, and late terciles, respectively. All lie within the feasible support of the fielded calendar, [49, 111]; see Appendix B.4.

H Additional robustness: estimators, benchmarks, and enumerator effects

This appendix reports further checks on the main results. All of them were run after the first version of the paper and are exploratory; none belongs to a registered family. Unless stated otherwise, standard errors are clustered by village, wild cluster bootstrap p-values use 9,999 replications (Rademacher weights for comparisons of Arms A and B, Webb weights for comparisons involving Arm C or 11 subcounty clusters), randomization-inference p-values use 4,999 assignments drawn under the study’s rerandomization rule, and Romano–Wolf q-values are computed within the family stated in each table.

H.1 Recall sensitivity

Table 25 collects the checks on the compression estimate of Section 5.1.1. The slope is similar in every season: estimated separately, the five season-specific slopes lie between 0.03 and 0.16 and their equality is not rejected ($p = 0.77$ with subcounty clustering, 0.90 with village clustering). Among the 400 respondents who recall all five seasons the slope is 0.096, and reweighting them for selection into complete histories gives 0.090. Misdated seasons would load positively on the record of adjacent seasons; adding the records of seasons $t - 1$ and $t + 1$ gives coefficients of -0.079 and -0.025 , and the three coefficients sum to -0.05 , so misdating does not hide a larger recall sensitivity. Nor does carry-over from the previous answer: in adjacent pairs of the elicitation sequence the slope is, if anything, higher (0.109) than in non-adjacent pairs, and identical token allocations are less frequent.

The slope on the nine-picture classification may understate recall sensitivity if the classification measures the seasons with error. Panel B replaces the classification with continuous seasonal rainfall. Within subcounties, recall moves 0.151 pictures per standard deviation of the CHIRPS normal score and the classification moves 0.537, a ratio of 0.28 with an Anderson–Rubin 95 percent interval of $[0.10, 0.63]$; a ratio of one is rejected ($p = 0.005$) with 11 subcounty clusters. The classification varies within subcounties mainly with rainfall (correlation 0.42 with the normal score) and hardly with peak vegetation (0.12), so the slope is not driven by small crossings of vegetation thresholds. Because the rainfall benchmarks share CHIRPS pixels with the classification, correlated errors bias the ratio toward the classification slope, so 0.28 may understate recall sensitivity, and the rejection of a slope of one inherits this caveat.

Panel C bases inference on the level at which the record varies. Reassigning the subcounty histories gives $p = 0.108$ against zero and a 95 percent interval of $[-0.02, 0.19]$, so at this level the slope is not distinguishable from zero, while a slope of one is never reached in 99,999 draws. Shuffling seasons within each subcounty’s own history gives $p = 0.049$. Ordinal agreement between recalled and recorded seasons (Somers’ d of 0.090) is what linear compression plus recall noise implies (0.075): the order of seasons survives in recall no better than the distances between them.

Table 25: Robustness of recall sensitivity.

	Estimate	SE village	SE subcounty
<i>A. Samples and specifications (respondent fixed effects)</i>			
All respondent-seasons (Section 5.1.1; $N = 2,559$)	0.095	0.034	0.057
Season-specific slopes: 2021 / 2022 / 2023	0.038 / 0.115 / 0.157		0.080 / 0.155 / 0.087
Season-specific slopes: 2024 / 2025	0.093 / 0.032		0.071 / 0.128
Balanced panel, all five seasons recalled ($N = 2,000$)	0.096	0.037	0.058
Balanced panel, inverse-probability weighted	0.090	0.038	0.061
Record of t , with records of $t - 1$ and $t + 1$ (2022–2024; $N = 1,425$)	0.053	0.079	0.045
coefficient on record of $t - 1$ / $t + 1$	−0.079 / −0.025	0.036 / 0.075	0.037 / 0.020
	Ratio	SE subcounty	AR 95% CI; $p(\beta = 1)$
<i>B. Continuous rainfall benchmarks (recall slope / record slope)</i>			
CHIRPS seasonal normal score	0.281	0.116	[0.10, 0.63]; 0.005
Seasonal rainfall (mm)	0.274	0.119	[0.09, 0.65]; 0.009
Rainfall-only classification (Appendix F.2)	0.272	0.115	[0.08, 0.61]; 0.005
			Result
<i>C. Permutation of the 11 subcounty histories (99,999 draws)</i>			
p against $\beta = 0$; permutation-inverted 95% interval			0.108; [−0.02, 0.19]
p against $\beta = 1$			< 0.00001 (no draw reached)
Somers’ d , recalled against recorded order (implied by β plus noise)			0.090 (0.075)
<i>D. Neighbor-agreement ceiling on recall sensitivity</i>			
Record measured without error (Table 14)			0.40 [0.35, 0.48]
Record reliability 0.35, implied by the OLS/IV ratio in Panel B			0.68 [0.60, 0.81]
Record reliability at which the ceiling reaches one			0.16

Notes: Panel A: regressions of the recalled mean picture on the record’s classification with respondent fixed effects; the season-specific slopes also include season fixed effects. The weights in the fifth row are inverse probabilities of recalling all five seasons, estimated on age, sex, education, household size, cultivated area, arm, and subcounty. Panel B: continuous benchmarks are scaled per within-subcounty standard deviation; the ratio of the recall slope to the record slope equals the instrumental-variables estimate of recall on the record with the benchmark as instrument; Anderson–Rubin (AR) intervals use $t(10)$ on subcounty clusters. Panel C: the 11 five-season histories (nine of them distinct) are reassigned across subcounties. Panel D: the ceiling divided by the square root of the record’s reliability; brackets are 95% intervals. Source: authors’ computation.

Panel D corrects the neighbor-agreement ceiling of Appendix F.2 for error in the record. That ceiling assumes the record measures the weather neighbors shared without error; with the reliability of 0.35 implied by Panel B it rises to 0.68 [0.60, 0.81]. The ceiling reaches one only if the record’s reliability is below about 0.16 (0.23 for the upper confidence limit); the within-subcounty correlation between the record and CHIRPS rainfall implies 0.18 if CHIRPS itself were error-free, so under that most pessimistic assumption the ceiling does not exclude one. The ceiling is in any case an upper bound, because shared interview conditions also load on the shared component of recall. Taken together, recall sensitivity is about 0.1 on the classification and between a quarter and a third (0.27–0.28) once noise in the record is allowed for, and it is well below one under every estimate of the record’s reliability that the data provide.

H.2 Memory and expectations

Table 26 reports the association λ of Section 5.2.1 under alternative specifications. The baseline is 0.4351 (standard error 0.0717). The estimate lies between 0.40 and 0.45 with enumerator and interview-date fixed effects, village fixed effects, respondent controls, controls for response style in the recall exercise (share of point masses, entropy, share on the central picture), and inverse-probability weights for having a complete history; every wild cluster bootstrap p-value is below 0.003. In less selected samples, which average fewer recalled seasons and therefore more recall noise, the estimate falls to between 0.28 and 0.38, as attenuation would predict; imputing missing seasons keeps it at 0.41.

The association is carried by differences in the level of remembered weather between respondents. Controlling for the respondent’s own five-season mean, the deviation of her recall for 2024 and 2025 from that mean has a coefficient of 0.036 ($p = 0.70$). The paper therefore reads λ as an association between remembered and expected levels, not as evidence that recent seasons carry more weight; with a linear specification, equal weights on the five seasons are not rejected ($p = 0.45$). A cross-instrument placebo does not show the positive co-movement that generic scale use would produce: expected amounts on recalled onset give -0.038 (standard error 0.025) and expected onset on recalled amounts -0.194 (0.103).

H.3 Estimators for the information experiment

Table 27 reports the treatment contrasts of Section 5.4.1 for three outcomes of the updating family and for expected onset, under three estimators and several sets of fixed effects. The change score of the main text is robust to strata, pair, enumerator, and interview-date fixed effects: the effects on the distance to the card and on the ranked probability score stay between -0.13 and -0.17 , with randomization-inference p-values at or below 0.02 and q-values between 0.04 and 0.11. Local concentration, the share of tokens on or next to the realized picture, falls slightly under every estimator (-0.02 to -0.03 ; wild cluster bootstrap p-values between 0.004 and 0.18, q-values between 0.05 and 0.65). The remaining seven outcomes of the updating family (confidence in onset, shift, share moving, mass on the extremes, mean picture, entropy, sharpness) are not significant at the 5 percent level under any specification.

Table 26: Robustness of the association between remembered weather and expectations.

Specification	λ	SE	N
Subcounty fixed effects (Section 5.2.1)	0.435	0.072	330
+ enumerator fixed effects	0.449	0.068	330
+ interview-date fixed effects	0.444	0.072	330
+ enumerator and interview-date fixed effects	0.452	0.070	330
Village fixed effects	0.401	0.072	330
Subcounty fixed effects + respondent controls	0.438	0.076	330
Subcounty fixed effects + response style, enumerator and date fixed effects	0.442	0.072	330
Inverse-probability weights for a complete history	0.433	0.069	330
Equal weights on the five seasons	0.409	0.067	330
At least one season recalled, equal weights	0.277	0.057	566
At least two seasons recalled, equal weights	0.314	0.054	504
At least three seasons recalled, equal weights	0.351	0.061	416
At least four seasons recalled, equal weights	0.382	0.065	363
All respondents, missing seasons imputed	0.413	0.063	566
Arm A only	0.324	0.113	169
<i>Level against recency (one regression)</i>			
Respondent’s own five-season mean	0.415	0.068	330
Deviation of 2024–2025 recall from that mean	0.036	0.084	330

Notes: Dependent variable: the mean of the respondent’s pre-information expectation for 2026 on the nine-picture scale. Arms A and B. Response-style controls are the share of point-mass answers, entropy, and the share of tokens on the central picture in the recall exercise. Inverse-probability weights come from a logit of having a complete five-season history on ten pre-determined covariates and onset response style; this row has no bootstrap p-value. The row with at least one season recalled adds fixed effects for the number of seasons recalled. Source: authors’ computation.

The Romano–Wolf q-values cited in the main text (0.060 and 0.063, and 0.097 for onset) come from the families estimated with subcounty fixed effects (Appendix C.3); for the change score without fixed effects the table gives 0.085, 0.049, and 0.049.

ANCOVA roughly halves the effects on the two distance measures (-0.072 to -0.082 for the distance to the card; q-values 0.11 to 0.29), and with the second elicitation alone they are not detected. Both estimators are informative here, and ANCOVA is the conservative one. Arm B started further from the card than Arm A (by 0.19 standard deviations before the card; randomization-inference $p = 0.004$, although an omnibus test over ten pre-card belief levels does not reject balance, $p = 0.31$). With a test–retest correlation of about 0.5 in Arm A, the coefficient on the first elicitation is attenuated, so ANCOVA removes only about half of any persistent baseline gap and is biased toward zero when Arm B starts further away; the change score removes the whole gap, including any transient part. Interview-date fixed effects are also demanding: each village was interviewed on a single day, so they absorb almost all variation between villages and identify the effect only from days on which both arms were interviewed. Enumerator fixed effects were not randomly assigned either, and the mix of enumerators differs between the arms (joint $p = 0.003$), so they too absorb part of the between-village contrast on which the effect rests. Stated confidence rises under every estimator, and expected onset moves earlier under all of them, most weakly with interview-date fixed effects (Appendix H.6).

Table 27: Treatment effects on beliefs under alternative estimators.

	Distance to card	RPS against 2025	Confidence, amount	Expected onset (days)
Arm A pre-card SD	0.575	0.696	1.119	13.57
Pre-card difference, B – A (SD units; RI p)	+0.19 (0.004)	+0.13 (0.108)	–0.03 (0.685)	+0.04 (0.501)
Change score, no FE (main text)	–0.134 (0.050) [.011; .002; .085]	–0.151 (0.051) [.004; .005; .049]	+0.178 (0.079) [.030; .022; .217]	–2.59 (0.86) [.005; .009; .049]
Change score, strata FE	–0.149 (0.044) [.005; .001; .048]	–0.158 (0.048) [.006; .005; .058]	+0.163 (0.070) [.047; .057; .294]	–2.68 (0.87) [.008; .012; .084]
Change score, pair FE	–0.131 (0.033) [.002; .007; .038]	–0.149 (0.043) [.011; .020; .104]	+0.152 (0.056) [.060; .069; .359]	–2.94 (0.71) [.003; .009; .027]
Change score, strata + enumerator FE	–0.148 (0.045) [.005; .002; .056]	–0.157 (0.051) [.008; .007; .081]	+0.179 (0.071) [.032; .050; .215]	–2.74 (0.89) [.008; .010; .081]
Change score, strata + date FE	–0.148 (0.043) [.006; .002; .096]	–0.173 (0.051) [.011; .005; .107]	+0.153 (0.076) [.134; .094; .512]	–1.95 (0.79) [.062; .047; .372]
ANCOVA, strata FE	–0.081 (0.033) [.030; .016; .198]	–0.101 (0.037) [.018; .012; .140]	+0.145 (0.052) [.014; .039; .130]	–2.45 (0.72) [.002; .014; .038]
ANCOVA, pair FE	–0.072 (0.021) [.014; .034; .107]	–0.095 (0.027) [.012; .024; .097]	+0.132 (0.049) [.053; .065; .298]	–2.79 (0.65) [.002; .007; .015]
ANCOVA, strata + enumerator FE	–0.074 (0.033) [.046; .017; .289]	–0.095 (0.039) [.032; .017; .221]	+0.158 (0.055) [.011; .035; .112]	–2.42 (0.66) [<.001; .008; .021]
ANCOVA, strata + date FE	–0.082 (0.031) [.035; .018; .241]	–0.115 (0.038) [.018; .011; .172]	+0.138 (0.053) [.044; .055; .244]	–1.95 (0.69) [.029; .040; .211]
Post-only, pair FE	–0.037 (0.022) [.251; .232; .848]	–0.061 (0.031) [.190; .174; .777]	+0.119 (0.054) [.131; .128; .666]	–2.53 (0.77) [.018; .025; .173]

Notes: Arms A and B, 567 respondents (566 with interview-date fixed effects) in 50 villages. Each cell reports the Arm B coefficient with its village-clustered standard error and, below it in brackets, the wild cluster bootstrap p -value, the randomization-inference p -value under the rerandomization rule, and the Romano–Wolf q -value within a family of twelve outcomes (the eleven updating outcomes of Appendix C.3 and expected onset), computed separately for each row. Change score: the change between the two elicitations. ANCOVA: the second elicitation, controlling for the first. Post-only: the second elicitation alone. Strata are the 17 subcounty-by-elevation cells of the design; pairs are the matched pairs. The pre-card difference row uses strata fixed effects. Source: authors’ computation.

H.4 The post-only arm

The registration (hypothesis H6) compares the first-stated beliefs of Arm C, elicited after the card, with the pre-card beliefs of Arm A. Table 28 reports this comparison, a comparison with the pooled first elicitations of Arms A and B (also pre-card and identical in protocol, which lets all ten Arm C matched pairs contribute), and a comparison with Arm B’s second elicitation. The card raises stated confidence between subjects: against the pooled first elicitations, confidence in the amount rises by 0.39 points (standard error 0.10; Webb $p = 0.016$; randomization-inference $p = 0.004$), and Arm C’s confidence equals Arm B’s after the card. Shifts in the amount and onset beliefs are not detected: the distance to the card changes by -0.060 (0.055; randomization-inference $p = 0.54$) and expected onset by -0.04 days (1.26). With ten treated villages these contrasts are imprecise; the minimum detectable effect on the distance to the card is about 0.15, the size of the change-score effect. For amounts, Arm C ends where Arm B ends after the card (difference -0.001 in the distance to the card). For onset, Arm C’s expectations are 2.5 days later than Arm B’s after the card (standard error 1.19; Webb $p = 0.13$), so the earlier onset appears only when farmers revise an answer already given. Adding literacy, sex, schooling, and age, on which

Arm C is less well balanced, leaves the conclusions unchanged (distance to the card -0.083 , clustered $p = 0.13$; expected onset -0.01 days; confidence $+0.40$).

Table 28: Beliefs in the post-only arm (registered hypothesis H6).

	C – A, first subcounty FE	C – (A+B), first pair FE	C – B, second subcounty FE
Distance to card	+0.037 (0.053) [0.56]	-0.060 (0.055) [0.54]	-0.001 (0.048) [0.97]
RPS against 2025	+0.042 (0.057) [0.45]	+0.033 (0.056) [0.69]	+0.068 (0.063) [0.28]
Mean picture	-0.099 (0.090) [0.56]	+0.084 (0.085) [0.55]	+0.004 (0.099) [1.00]
Expected onset (days)	+1.82 (1.35) [0.53]	-0.04 (1.26) [0.98]	+2.49 (1.19) [0.16]
Confidence, amount (1–5)	+0.325 (0.123) [0.004]	+0.393 (0.103) [0.004]	+0.013 (0.079) [0.81]
Respondents / villages	393 / 34	686 / 60	412 / 36

Notes: Each cell reports the Arm C coefficient, its village-clustered standard error, and in brackets the randomization-inference p-value from reassigning Arm C within its ten donor pairs under the rerandomization rule (252 admissible assignments; 32 for the comparison with Arm B, which uses the six pairs with an Arm B partner). “First” is the first-stated belief, pre-card in Arms A and B; “second” is Arm B’s belief after the card. Source: authors’ computation.

H.5 Card content and experimenter demand

To separate the content of the card from its format, we compute for each respondent the distance of her expectations before and after the card to each of the eight distinct subcounty cards. Relative to Arm A, the distance to one’s own card falls by 0.134 and the average distance to the seven other cards by 0.091 (one-sided wild cluster bootstrap $p = 0.039$). The excess movement toward one’s own card is -0.043 (standard error 0.027; one-sided $p = 0.056$ by bootstrap and 0.064 by randomization inference), -0.044 with strata and -0.041 with pair fixed effects (one-sided p of 0.058 and 0.118), and there is no excess for the ranked probability score (-0.050 ; $p = 0.24$). About 70 percent of the fall in the distance to one’s own card is therefore also a fall in the distance to every card, consistent with beliefs moving toward the central, spread-out shape that all cards share (every card lies on pictures three to seven). With eight distinct cards across 11 subcounties, the test has little power to detect a content-specific response.

Several checks bear on experimenter demand. Seven enumerators each conducted between 33 and 49 interviews per arm in Arms A and B. With ANCOVA and strata fixed effects, the effects on the distance to the card, the ranked probability score, and confidence do not differ across enumerators (joint $p = 0.62$, 0.23, and 0.23; bootstrap $p = 0.69$, 0.30, and 0.30). The onset effect is borderline heterogeneous ($p = 0.069$; bootstrap $p = 0.099$): two enumerators record -6.9 and -5.3 days, the other five between -2.6 and -0.1 . Outcomes that a general wish to please would also move do not respond: expecting an extreme event (-0.011 ; $p = 0.72$), its expected severity (-0.10 ; $p = 0.67$), planned adaptation ($+0.045$; $p = 0.19$), and the mean picture, mass on the extremes, entropy, and sharpness of expectations. The 120 Arm B respondents who said they would not adjust their expectations moved at least as far toward the card as those who said they would (-0.181 against -0.127 relative to Arm A; this split conditions on a post-card answer and is descriptive). The card shifts revisions in

the distance to the card from upward to downward (34.1 against 24.8 percent downward, 25.6 against 32.8 percent upward) rather than raising the share who revise at all.

H.6 Expected onset

The rainfall rule behind the onset benchmark places onset on the first admissible day of its search window (10 February) in 48 of the 55 subcounty-seasons of 2021–2025 and within one day of it in all 55, before the first interval of the elicitation calendar. Field staff and local respondents reported that the main planting rains had not yet begun at the time of the interviews, and CHIRPS shows steady rainfall of 30 to 75 millimeters per week from February to April with no dry spell longer than six days, so the rule detects early rain rather than the planting rains farmers refer to; the same gap of 26 to 32 days separates recalled onset from the benchmark in every study season (Appendix B.3). Agronomic definitions that screen out false starts, of the type proposed by Sivakumar (1988) and Marteau et al. (2009), place onset between days 64 and 76, close to farmers’ recalled onsets, and against these records the card does not reduce the distance between beliefs and the record (difference-in-differences between -0.18 and $+0.36$ days, all $p \geq 0.61$). We therefore describe the treatment effect as a shift toward earlier expected onset. Within subcounties, pre-card onset beliefs do not respond to the rainfall received before the interview ($p = 0.22$).

Table 29 reports alternative estimators for the onset effect. The effect survives the wild cluster bootstrap ($p = 0.0055$), a two-limit Tobit that treats the calendar’s first and last intervals as censoring points (-3.17), and dropping the respondents whose initial belief sits at the calendar floor (-3.28). Added to the eleven-outcome updating family estimated with subcounty fixed effects, it has a Romano–Wolf q -value of 0.097. With interview-date fixed effects alone the change-score estimate is -1.77 (bootstrap $p = 0.078$), and with strata and interview-date fixed effects it is -1.95 (Table 27). Interview dates are nested in subcounties along the fieldwork route (the first week covered only three subcounties of Butaleja, and four dates contain a single arm), so the date fixed effects absorb mainly geography; by week the effect is -6.31 , -0.84 , and -0.95 days, and timing and place cannot be separated. The tercile pattern of Figure 7 holds in the Tobit specification, where the late tercile moves most (-5.36 days). A pure level shift is rejected against revision that grows with the initial belief (Arm B $\times$ initial belief -0.149 ; bootstrap $p = 0.047$), but because the benchmark lies below the calendar’s first interval, the data cannot tell revision toward the record from revision toward the earliest date the calendar allows.

H.7 Beliefs, memory, planned adaptation, and literacy

Table 30 adds richer pre-determined controls and fixed effects to the associations of Section 5.3.2. The associations with expecting an extreme event are unchanged by interview-date or village fixed effects but vanish with enumerator fixed effects: for planned spending the coefficient falls from 0.659 to -0.008 . Enumerator identity explains 28 percent of the within-subcounty variation in whether a farmer expects an extreme, and the share expecting one ranges from 19 to 96 percent across the seven enumerators. Because respondents were

Table 29: Robustness of the onset treatment effect.

Specification	Estimate	SE	<i>p</i>
Change score, no fixed effects (Section 5.4.1)	−2.59	0.86	0.0055 (bootstrap)
Change score, respondents at the calendar floor excluded ($N = 439$)	−3.28	1.01	0.0037 (bootstrap)
Two-limit Tobit ANCOVA, censored at days 49 and 111	−3.17	1.08	0.003 (clustered)
early / middle / late tercile of initial belief	−0.40 / −4.55 / −5.36	0.81 / 0.004 / 0.002	
ANCOVA, pair fixed effects	−2.79	0.65	0.002 (bootstrap)
Post-only, pair fixed effects	−2.53	0.77	0.018 (bootstrap)
Change score, controls for rainfall up to the interview	−2.59	0.84	0.003 (clustered)
Change score, interview-date fixed effects only	−1.77	0.78	0.078 (bootstrap)
ANCOVA, interview-date fixed effects only	−1.70	0.67	0.015 (clustered)

Notes: Arms A and B, 567 respondents in 50 villages unless noted. Outcome: expected onset in days on the fielded calendar. Bootstrap *p*-values are wild cluster bootstrap *p*-values with 9,999 replications; clustered *p*-values use village-clustered standard errors. Tercile standard errors are 1.67, 1.57, and 1.76. Source: authors' computation.

not randomly assigned to enumerators, enumerator fixed effects remove interviewer-induced co-movement between answers but also any genuine difference in beliefs and plans between the respondents each enumerator interviewed, so the specification is conservative. The association with having recalled a past extreme is robust: with the full set of controls and subcounty, enumerator, and interview-date fixed effects, farmers who recall an extreme still plan to spend 0.499 log points more ($p = 0.008$) and are 12.1 percentage points more likely to plan any adaptation ($p = 0.042$). The link from recalling an extreme to expecting one falls from 19.4 to 8.2 percentage points with enumerator fixed effects ($p = 0.061$) and is robust to an index of yes-saying (18.1 points). The token-based probability of an adverse season does not predict planned protection (for mass on picture three or drier, -0.009 log points of spending per token, $p = 0.80$).

Only eight of the 404 farmers who plan any adaptation plan to buy crop insurance, and seven of them were interviewed by the same enumerator. Among adapters in Arms A and B, none of the 160 who do not expect an extreme plans insurance, against 7 of the 167 who do; with enumerator fixed effects the coefficient is 0.020 ($p = 0.15$). The cells are too thin to support an association, and the main text does not report one.

The event-level recall tests in Appendix Figure 16 treat each named event as independent. Clustered at higher levels, recalled flood years remain wetter than usual (mean seasonal rainfall percentile 0.652; wild cluster bootstrap $p = 0.0001$ by village, 0.005 by subcounty, and 0.010 by year), and years recalled as droughts are also wetter, not drier, than usual (0.609; $p = 0.0001$, 0.0095, and 0.12). Recalled drought years are not marked by longer dry spells at any level (0.527; $p = 0.27$, 0.46, and 0.66). The flood and drought events fall in 63 and 95 distinct subcounty-years.

Literacy predicts where farmers start. Respondents who cannot read recall past seasons 0.33 standard deviations less accurately on the ranked probability score (wild cluster bootstrap $p = 0.0001$; Section 5.1.2), and schooling, recorded in four levels, adds nothing once literacy is held fixed: each additional level changes the score by -0.02 standard deviations,

Table 30: Beliefs, memory, and planned adaptation with enumerator effects.

	Log planned spending	Plans any adaptation	Uses fertilizer
<i>A. Expects an extreme event in 2026</i>			
Paper controls, subcounty FE	0.659 (0.0002)	0.144 (0.002)	0.125 (0.004)
+ interview-date FE	0.712 (0.0002)	0.141 (0.002)	0.128 (0.003)
+ village FE	0.739 (0.0001)	0.151 (0.001)	0.130 (0.002)
Rich pre-determined controls, subcounty FE	0.573 (0.005)	0.088 (0.060)	0.122 (0.007)
Paper controls + enumerator FE	−0.008 (0.96)	−0.024 (0.65)	0.026 (0.59)
Rich controls + enumerator FE	0.006 (0.98)	−0.029 (0.58)	0.035 (0.47)
Paper controls + enumerator and date FE	0.068 (0.74)	−0.024 (0.65)	0.031 (0.53)
<i>B. Recalls a past extreme event (Arms A and B)</i>			
Paper controls, subcounty FE (Section 5.3.2)	0.767 (0.0001)	0.155 (0.005)	−0.019 (0.71)
Rich controls + subcounty, enumerator, and date FE	0.499 (0.008)	0.121 (0.042)	−0.061 (0.33)
Expects an extreme event in 2026			
<i>C. Recalling a past extreme and expecting another</i>			
Paper controls, subcounty FE (Section 5.3.1)		0.194 (0.0002)	
+ yes-saying index		0.181 (0.0004)	
+ enumerator FE		0.082 (0.061)	
Rich controls + subcounty, enumerator, and date FE		0.071 (0.12)	

Notes: Coefficients with wild cluster bootstrap p-values in parentheses; 50 village clusters. Arms A and B. Planned spending is observed for the 323 respondents planning any adaptation; the other outcomes use 566 or 567 respondents. Paper controls are cultivated area, age, and household size. Rich controls add sex, literacy, education, land ownership, household headship, marital status, extension contact, access to weather information and forecasts, the number of seasons recalled, a yes-saying index built from four factual pre-card items, and crop indicators. Seven enumerator codes are recovered from the respondent identifier; the 18 respondents in Arms A and B whose identifiers do not follow the six-character format, including the two with malformed enumerator codes (Appendix A), form one further group. Source: authors' computation.

a negligible improvement in accuracy ($p = 0.75$). Before the card, their amount beliefs lie 0.28 standard deviations further from their card than those of literate farmers (with subcounty fixed effects; 0.32 without, $p = 0.0003$) and 0.36 standard deviations further from the realized 2025 season ($p = 0.0001$), whereas their expected onset does not differ ($p = 0.75$).

Table 31 reports the interaction of Arm B with literacy for the three headline outcomes. For the distance to the card, the card reduces the distance by 0.153 among literate farmers and by 0.104 among farmers who cannot read; the interaction is -0.049 (standard error 0.103; 95 percent confidence interval $[-0.26, 0.16]$; bootstrap $p = 0.63$). No difference is detected for any outcome, but the intervals are wide. For the distance to the card, the interaction ranges from about twice the average effect in favor of literate farmers to 1.24 times it in favor of farmers who cannot read; the minimum detectable difference between the two groups at 80 percent power is about 0.29, more than twice the average effect; and

equivalence within the size of the average effect is not established for any outcome. Under ANCOVA with subcounty fixed effects the point estimates tilt further toward literate farmers (for the distance to the card, -0.024 for farmers who cannot read against -0.110), while with enumerator fixed effects the difference is essentially unchanged (0.044 , $p = 0.67$). Splitting at any schooling instead of literacy gives the same conclusion (difference 0.065 , $p = 0.46$).

For the distance to the card, the effect does not detectably grow with a farmer’s own initial distance. By tercile of that distance the effects are -0.03 , -0.21 , and -0.17 (bootstrap $p = 0.45$, 0.017 , and 0.20), each netting out Arm A’s mechanical revision within the tercile, and neither a linear trend across terciles ($p = 0.26$), the continuous interaction with the initial distance (-0.12 per unit, $p = 0.32$), nor a test of equal effects ($p = 0.14$) rejects a constant effect. Because farmers who cannot read begin 0.20 units of earth-mover distance further from the card, this pooled gradient would predict that they respond 0.025 units more than literate farmers; the point estimate goes the other way, 0.049 units less. For expected onset the effect does grow with the initial belief (Appendix H.6), but initial onset beliefs do not differ by literacy, so the positional pattern does not produce a difference by literacy. The data therefore show that the card reaches farmers who cannot read without a detectable loss, not that it helps them more.

Table 31: Treatment effects by literacy.

	Average effect	Literate $\times$ B (SE) [p]	Effect: cannot read / can read	Interaction / average effect, 95% CI	TOST p ($\pm 100\%$)
Distance to card	-0.127	-0.049 (0.103) [0.63]	-0.104 / -0.153	$[-2.02, 1.24]$	0.23
RPS against 2025	-0.143	-0.034 (0.129) [0.79]	-0.130 / -0.164	$[-2.06, 1.58]$	0.20
Expected onset (days)	-2.65	-0.28 (1.88) [0.88]	-2.48 / -2.76	$[-1.53, 1.32]$	0.11

Notes: Change-score specification of Section 5.4.1, Arms A and B, 564 respondents, 50 village clusters. The survey records literacy for 565 of the 567 respondents in Arms A and B (684 of 686 overall, as in Table 1); one further respondent is missing from the merged analysis file. Bracketed p -values are wild cluster bootstrap p -values. TOST: two one-sided tests of equivalence with a margin equal to the average effect. Source: authors’ computation.

The signed test of Section 5.4.3 uses the change in planned adaptation. Conditional on plans stated before the card (ANCOVA), the Arm B $\times$ push interaction is $+0.058$ (standard error 0.036 ; $p = 0.13$), $+0.052$ with pair fixed effects ($p = 0.12$), and $+0.063$ with enumerator fixed effects ($p = 0.099$). Pre-card plans are balanced in level, but a placebo regression of pre-card plans on the push gives -0.059 ($p = 0.17$), 72 percent of the change-score estimate. Across the four definitions of the push the ANCOVA estimates range from $+0.051$ to $+0.072$; only one has $p < 0.05$ (0.031 ; Romano–Wolf $q = 0.083$ across the four). The sign is the predicted one in every specification, but the test is not significant once pre-card plans are held fixed.

H.8 Persistence of seasonal anomalies

Across the 11 subcounties and 25 seasons of 2001–2025, seasonal rainfall anomalies (normal scores relative to each subcounty’s 2001–2020 climate) show little persistence. The first-order

autocorrelation is 0.10 (standard error 0.06; Webb $p = 0.15$), and 0.01 once common year effects are removed; the lag-one autocorrelation averages 0.09 across subcounties, ranging from -0.24 to 0.38 . Past seasons therefore say little about the next season beyond the distribution from which seasons are drawn, which limits the predictive value of any record of past seasons. We accordingly read revisions toward the record as alignment of beliefs with local history, not as improvements in forecast accuracy.

I Validity of the visual instrument by literacy and enumerator

The token elicitation of Section 4.2 is designed to measure beliefs without relying on reading. This appendix examines whether it measures them equally well for farmers who can and cannot read, and how far its outputs depend on the enumerator who conducted the interview. The analyses were run after the first version of the paper and are exploratory; none belongs to a registered family. Literacy is the survey’s literacy indicator, and schooling distinguishes respondents with any formal education from those with none. The enumerator is identified by the first two digits of the respondent identifier, which yields seven codes; the 18 respondents in Arms A and B whose identifiers do not follow the six-character format form one further group, and dropping them leaves the results unchanged. Standard errors are clustered by village, and p-values are wild cluster bootstrap p-values with 9,999 Rademacher replications unless stated otherwise.

I.1 Test–retest reliability

Farmers in Arm A stated their expectation for 2026 twice, separated by the extreme-event, driver, and planned-adaptation modules, and received no information in between (274 respondents in 24 villages; 272 with a literacy code). Table 32 reports the stability of their allocations. For the rainfall pictures, the correlation of the mean picture across the two rounds is 0.42, the correlations of the dry- and wet-tail mass are 0.40 and 0.47, 40 percent of allocations are identical, and on average 3.9 of the ten tokens move. Reliability is therefore moderate, and it is the same for farmers who cannot read: the correlation of the mean picture is 0.44 for literate and 0.41 for non-literate farmers (difference 0.03; village bootstrap 95 percent confidence interval $[-0.30, 0.32]$), and the tail correlations are, if anything, higher among farmers who cannot read. Neither the earth-mover distance between the two allocations ($p = 0.74$), the share of identical allocations ($p = 0.58$), the absolute change in the mean picture ($p = 0.86$), nor the slope of the second-round mean on the first ($p = 0.67$) differs by literacy, and the distance still does not differ with enumerator fixed effects (clustered $p = 0.38$). Farmers without schooling have a lower point estimate for the correlation (0.31 against 0.51), but the difference is not statistically distinguishable (95 percent confidence interval $[-0.14, 0.47]$; $p = 0.33$).

Among the rainfall-picture measures, the one significant difference concerns the direction of drift, not its size. Between rounds, the mean picture of literate farmers moves 0.09 pictures wetter and that of non-literate farmers 0.19 pictures drier (difference 0.29, 95 percent confidence interval $[0.09, 0.50]$; $p = 0.005$). The onset calendar is more stable overall (correlation 0.70) but noisier for farmers who cannot read: the noise standard deviation of expected onset, $\text{sd}(\Delta)/\sqrt{2}$, is 8.6 days for non-literate and 6.5 days for literate farmers (difference 2.1 days, 95 percent confidence interval $[0.4, 3.7]$; $p = 0.02$). This is consistent with the calendar’s reliance on dated intervals; the pictures, not the calendar, are the literacy-free part of the instrument.

The second elicitation is not a clean repeat. It follows other modules, and the share of identical allocations ranges from 3 to 83 percent across enumerators, which suggests that some enumerators left the first allocation in place. The overall reliability therefore mixes instability of beliefs with differences in protocol; the comparison by literacy is unaffected by this, because it holds with enumerator fixed effects.

Table 32: Test–retest reliability in Arm A by literacy and schooling.

	All	Literate	Cannot read	Any schooling	No schooling	Literate – cannot read [<i>p</i>]
<i>A. Rainfall pictures</i>						
Correlation of the mean picture	0.42	0.44	0.41	0.51	0.31	0.03 [0.99]
Correlation of dry tail (pictures 1–3)	0.40	0.36	0.50	0.39	0.43	−0.14 [0.41]
Correlation of wet tail (pictures 7–9)	0.47	0.43	0.51	0.56	0.34	−0.08 [0.42]
Share of identical allocations	0.40	0.38	0.42	0.43	0.36	−0.04 [0.58]
Tokens moved (of 10)	3.85	3.80	3.89	3.57	4.31	−0.09 [0.91]
Earth-mover distance between rounds	0.64	0.62	0.66	0.60	0.71	−0.04 [0.74]
Change in the mean picture	−0.06	0.09	−0.19	0.02	−0.18	0.29 [0.005]
Noise SD of the mean picture	0.71	0.66	0.75	0.65	0.79	−0.09 [0.58]
<i>B. Onset calendar</i>						
Correlation of expected onset	0.70	0.75	0.65	0.69	0.70	0.10 [0.15]
Share of identical allocations	0.43	0.41	0.46	0.43	0.44	−0.05 [0.31]
Noise SD of expected onset (days)	7.6	6.5	8.6	7.4	7.8	−2.10 [0.020]
Respondents	274	137	135	169	105	

Notes: Arm A, 24 villages. Both elicitations concern the 2026 season; no information was given between them. The noise standard deviation is $\text{sd}(\Delta)/\sqrt{2}$, where Δ is the change between rounds. The earth-mover distance is measured in pictures. The last column reports the literate minus non-literate difference with a village cluster bootstrap *p*-value (9,999 replications) in brackets; wild cluster bootstrap *p*-values for the respondent-level quantities are 0.58 for identical allocations, 0.89 for tokens moved, and 0.74 for the earth-mover distance. Two respondents lack a literacy code. Source: authors’ computation.

I.2 Internal consistency of recall

The token recall can be checked against the independent free-recall module, in which farmers named past extreme events without being prompted by year. At the respondent-season level, with respondent and subcounty-by-season fixed effects so that the record is absorbed ($N = 2,559$), seasons a farmer freely named as a drought are recalled 0.36 pictures drier on the token scale (standard error 0.15; $p = 0.022$), with 8.2 percentage points more mass on pictures one to three ($p = 0.025$) and no change in the wet tail ($p = 0.59$). Seasons named as a flood are recalled 0.36 pictures wetter ($p = 0.077$), with 6.5 percentage points more mass on pictures seven to nine ($p = 0.14$). With respondent and year fixed effects, the two coefficients are -0.41 ($p = 0.010$) and 0.45 ($p = 0.045$). The interactions with literacy are not significant: -0.05 for the drought coefficient on the mean ($p = 0.87$), -0.07 on the dry-tail mass ($p = 0.42$), and an imprecise -0.46 for the flood coefficient on the mean ($p = 0.31$). The two recall modules agree with each other, and about equally well for farmers who can and cannot read. The test concerns the consistency of two memory measures, not

their accuracy: the years farmers name as droughts are not unusually dry in the record (Appendix Figure 16).

I.3 Response style by literacy

Table 33 compares how farmers who can and cannot read allocate their tokens. Farmers who cannot read place all ten tokens on a single picture more often, in 55 against 43 percent of forward allocations (9 percentage points with subcounty and enumerator fixed effects), and in 39 against 30 percent of recalled seasons. They do not default to the central picture: they are 10 percentage points less likely to place any token on picture five, place no more mass on it, and are no more likely to put all their tokens there. Coverage does not differ. The number of seasons recalled and the share completing all five are the same, and in no season does the share of respondents who recall it differ by literacy (all $p \geq 0.10$). Farmers without schooling complete all five seasons more often (0.67 against 0.53; $p = 0.0002$), but the difference disappears with enumerator fixed effects (-0.013 ; $p = 0.59$) and reflects which enumerators interviewed whom (Section I.4).

Recall sensitivity is not flatter for farmers who cannot read. The slope of the recalled mean picture on the record is 0.077 for literate farmers ($p = 0.049$) and 0.114 for farmers who cannot read ($p = 0.026$); the interaction is 0.037 (standard error 0.054; $p = 0.49$), and 0.041 with year fixed effects ($p = 0.45$). The 0.33 standard deviation gap in recall accuracy (Section 5.1.2) has a different source. At the respondent-season level with subcounty-by-season fixed effects, the ranked probability score of farmers who cannot read is 0.217 higher ($p = 0.0001$). They have larger absolute errors (0.18 pictures; $p = 0.0001$) and more concentrated allocations (10 percentage points more point masses, $p = 0.0005$; lower entropy, $p = 0.0006$), but no level shift (signed error -0.10 ; $p = 0.11$). Farmers who cannot read thus track differences between seasons as well as literate farmers do, but place more concentrated allocations further from the record, which the ranked probability score penalizes heavily.

I.4 Enumerator dependence

Each enumerator worked in 56 to 59 of the 60 villages (Appendix A; 52 to 59 with the nonstandard identifiers pooled), and each village was surveyed on a single day. Within a village, differences between enumerators therefore compare respondents interviewed in the same place on the same day. Table 34 reports, for the 567 respondents in Arms A and B before the card (50 villages), the share of within-village variance explained by enumerator identity, net of its mean under 1,999 permutations of enumerator labels within village (about 1.3 percent), with 95 percent confidence intervals from 1,999 village bootstrap draws.

The location of the elicited distributions is almost free of interviewer effects. Enumerator identity explains 0.5 percent of the within-village variance of the expected mean picture, 0.7 percent of the expected wet-tail mass, and 0.9 percent of the recalled mean picture, none of them significant (permutation $p \geq 0.12$), and 4.2 percent of the expected dry-tail mass. Verbal belief questions are strongly interviewer-dependent: enumerator identity explains 28.0

Table 33: Response style by literacy.

	Literate	Cannot read	Difference (p)	With subcounty and enumerator FE (p)
<i>A. Expectations for 2026, before the card (Arms A and B)</i>				
All tokens on one picture	0.43	0.55	0.127 (0.005)	0.091 (0.036)
All tokens on picture 5	0.26	0.30	0.036 (0.38)	0.007 (0.86)
Any token on picture 5	0.68	0.58	−0.101 (0.033)	−0.111 (0.010)
Mass on picture 5	0.47	0.43	−0.037 (0.37)	−0.057 (0.12)
Pictures used	1.83	1.66	−0.172 (0.022)	−0.131 (0.031)
Normalized entropy	0.21	0.17	−0.043 (0.015)	−0.031 (0.042)
Onset: all tokens on one interval	0.56	0.62	0.061 (0.17)	0.028 (0.47)
<i>B. Recall of 2021–2025 (all arms)</i>				
Share of seasons with all tokens on one picture	0.30	0.39	0.086 (0.002)	0.078 (0.005)
Mass on picture 5	0.23	0.18	−0.047 (0.004)	−0.059 (0.0001)
Seasons recalled (0–5)	3.91	3.79	−0.116 (0.33)	−0.095 (0.19)
All five seasons recalled	0.59	0.58	−0.009 (0.82)	−0.003 (0.91)

Notes: Panel A: 564 respondents in Arms A and B with a literacy code (295 literate, 269 cannot read). Panel B: 683 respondents in all arms (350 and 333); recall precedes any card in every arm. Differences are cannot read minus literate, with wild cluster bootstrap p -values in parentheses; 50 (Panel A) and 60 (Panel B) village clusters. Normalized entropy is the entropy of the allocation divided by its maximum. Source: authors' computation.

percent of the variance in whether a farmer expects an extreme event, whose share ranges from 19 to 96 percent across enumerators, 26.2 percent in whether memory is named as a driver of expectations, and 22.4 percent in expected severity; for planned adaptation spending the share is 20.2 percent. Averaged over measures, the seven token belief and recall measures (the four in Panel A, the entropy of expectations, expected onset, and recall accuracy) are less enumerator-dependent than the eleven verbal belief and plan items (difference -0.049 , 95 percent confidence interval $[-0.077, -0.021]$; $p < 0.001$), and verbal belief items are more enumerator-dependent than verbal factual items (0.062 , $[0.031, 0.090]$). Recall sensitivity does not differ across the seven enumerators: their slopes lie between -0.016 and 0.294 , and equality is not rejected ($\chi^2(6) = 4.65$; $p = 0.59$).

Other outputs of the instrument depend substantially on the interviewer. Enumerator identity explains 35 to 41 percent of the within-village variance in the entropy of expectations and the number of pictures used, 25 percent in whether all tokens are placed on one picture, 24 percent in expected onset, and 10 percent in recall accuracy. It explains 61 percent in whether a respondent completed all five recalled seasons, a share that ranges from 3 to 97 percent across enumerators. How widely tokens are spread, when the rains are expected, and which seasons enter the recall panel are therefore enforced by interviewer protocol to a substantial degree, and the schooling gap in completing all five seasons disappears with enumerator fixed effects (Section I.3). Adding the five response-style measures (pictures used, point masses in expectations and recall, and mass on picture five in each), the twelve token measures as a whole are not less enumerator-dependent than the verbal items (difference -0.015 , $[-0.045, 0.014]$; $p = 0.29$).

Table 34: Enumerator dependence of token and verbal measures.

	Share (%)	95% CI	Perm. p	Range (SD)
<i>A. Location of token distributions</i>				
Expected mean picture	0.5	[−0.2, 4.2]	0.22	0.36
Expected wet-tail mass (pictures 7–9)	0.7	[−0.2, 5.0]	0.16	0.38
Recalled mean picture	0.9	[−0.2, 6.2]	0.12	0.40
Expected dry-tail mass (pictures 1–3)	4.2	[2.0, 9.2]	0.002	0.68
<i>B. Verbal beliefs and plans</i>				
Expects an extreme event	28.0	[22.5, 35.4]	< 0.001	1.55
Memory named as driver	26.2	[20.2, 34.8]	< 0.001	1.48
Expected severity	22.4	[16.7, 30.3]	< 0.001	1.12
Weather concern	15.7	[11.3, 22.3]	< 0.001	1.35
Recalls a past extreme	13.6	[9.2, 21.9]	< 0.001	1.03
Confidence, amount	1.7	[0.3, 7.3]	0.032	0.51
Planned adaptation spending (log)	20.2	[16.1, 26.7]	< 0.001	1.42
Plans any adaptation	15.1	[11.5, 21.5]	< 0.001	1.29
<i>C. Verbal factual items</i>				
Received a forecast	18.8	[12.9, 27.6]	< 0.001	1.21
Uses irrigation	9.1	[5.5, 16.0]	< 0.001	0.99
Uses fertilizer	6.8	[3.8, 13.1]	< 0.001	1.05
<i>D. Token dispersion, onset, and recall</i>				
Pictures used	40.5	[34.4, 48.0]	< 0.001	2.00
Entropy of expectations	35.4	[30.3, 42.3]	< 0.001	1.88
All tokens on one picture	24.6	[19.9, 32.0]	< 0.001	1.51
Recall: seasons with all tokens on one picture	22.6	[18.4, 29.2]	< 0.001	1.24
Expected onset	23.9	[16.7, 33.7]	< 0.001	1.49
Recall accuracy (ranked probability score)	10.2	[6.9, 16.4]	< 0.001	0.97
Seasons recalled (0–5)	61.5	[55.4, 68.5]	< 0.001	1.94
All five seasons recalled	61.4	[54.1, 70.0]	< 0.001	1.87

Notes: Arms A and B before the card: 567 respondents, 50 villages, eight enumerator groups (seven codes and one group for the 18 identifiers that do not follow the six-character format, as in Appendix H.7). Share: partial R^2 of enumerator indicators within village minus its mean under 1,999 permutations of enumerator labels within village, in percent; confidence intervals are from 1,999 village bootstrap draws. The permutation p-value is bounded below by 1/2,000. Range: range of enumerator means in standard deviations of the measure. Panels B and C show a selection of the eleven verbal belief and plan items and the six verbal factual items that enter the class averages in the text. Dropping the unmatched identifiers gives 0.2 percent for the expected mean picture and 28.1 percent for expecting an extreme event. Source: authors' computation.

The main estimates rest on the location of the distributions. Recall sensitivity β and the pass-through λ are estimated on mean pictures, β does not differ across enumerators, and specifications with respondent fixed effects absorb enumerator level effects. The treatment effects on the distance-based belief measures compare two allocations elicited by the same enumerator in one interview and do not differ across enumerators (Appendix H.5). Results based on dispersion or onset, including the onset treatment effect, should be read with the interviewer component in mind; Appendix H.5 reports the treatment effects by enumerator, where the onset effect is borderline heterogeneous, and Appendix H.7 reports the associations with planned adaptation and the literacy comparisons with enumerator fixed effects.

Enumerators were not randomly assigned within villages, so these shares also contain any sorting of respondents to enumerators; the share of respondents who cannot read ranges from 41 to 62 percent across enumerators, and the literacy comparisons above hold with enumerator fixed effects.

For farmers who cannot read, the picture instrument is as reliable on repetition, as consistent with independent verbal recall, and as sensitive to differences between seasons as for literate farmers, and the location of the distributions they report is almost free of interviewer effects, unlike answers to verbal belief questions. What differs by literacy is response style, with more concentrated allocations and a noisier onset calendar, while the spread of every respondent's allocation, expected onset, and recall completeness depend substantially on the interviewer, which argues for tighter standardization of these steps of the protocol in future rounds.

Online Appendix. The accompanying Online Appendix describes the study site and contains the survey instrument, the elicitation protocol, the enumerator manual, the consent form, the fieldwork procedures, the pilot, and computational details of the inference procedures.

Online Appendix

The Online Appendix collects the documentation of the study: the study site in detail, the survey instrument item by item, the enumerator script for the two token elicitation, the complete enumerator manual as fielded, the consent form, the fieldwork protocol, the pilot and what it changed, and the implementation detail behind the inference procedures. The material a reader needs to evaluate the paper’s claims is in Appendices A–I above.

OA-1 Study setting

The study covers 60 villages in 42 parishes and 11 subcounties across Bugiri, Butaleja, and Mayuge districts in eastern Uganda. The study subcounties are Buwunga, Kapyanga, Muterere, and Nankoma in Bugiri; Budumba, Kachonga, and Mazimasa in Butaleja; and Bukatube, Kigandalo, Kityerera, and Mpungwe in Mayuge. As Appendix A of the paper explains, exact village coordinates were not collected, so villages are represented geographically by their parish centroids. These centroids span 0.332 to 0.993 degrees north and 33.363 to 34.081 degrees east. Mean elevation within a 500-meter buffer ranges from 1,076 to 1,248 meters, with a mean of 1,147 meters, placing the study area on the Lake Victoria basin plateau rather than on the slopes of Mount Elgon. This distinction is relevant because the 2025 pilot was conducted in Bulambuli district on the Elgon slopes; Section OA-7 describes the pilot and the changes made before the main study.

Rainfall is bimodal. The survey focuses on the first agricultural season, described in the instrument as running approximately from February through July. The objective-weather construction described in Appendix B of the paper allows the first-season growing window to extend through 30 September depending on detected rainfall onset. The second agricultural season, which runs approximately from August through December, is not elicited. Over the 25 seasons from 2001 through 2025, seasonal rainfall averages computed from CHIRPS (Funk et al., 2015) range from 697.8 to 730.3 millimeters across the 11 subcounties, with within-subcounty coefficients of variation between 0.113 and 0.173. Across the five seasons displayed in the information treatment, subcounty rainfall totals range from 610.6 to 885.3 millimeters. Long-run mean rainfall differs by only 32.5 millimeters across the 11 subcounties, consistent with the relatively limited spatial variation in rainfall documented in the paper.

Agriculture in the study area is predominantly smallholder and rainfed. All 686 respondents identify farming as their occupation. Among crops planned for the upcoming season, 95.5 percent of respondents name maize, 77.3 percent beans, 52.0 percent cassava, 45.5 percent groundnuts, 15.3 percent soybeans, 13.0 percent rice, and 10.8 percent sweet potatoes; no other crop is named by more than seven percent of respondents. Mean respondent age is 41.4 years, 44.3 percent of respondents are women, mean household size is 7.75, and 350 of 684 respondents with nonmissing literacy responses report being able to read and write. Mean cultivated area is 2.79 acres and the median is two acres. Of the 686 respondents, 269 report no formal education, 281 primary education, 119 secondary education, 13 tertiary education, and four vocational training.

District-level statistics describe a similarly agriculture-dependent setting. In the 2014 National Population and Housing Census, 88.8 percent of households in Bugiri, 95.3 percent in Butaleja, and 83.7 percent in Mayuge report growing crops. Subsistence farming is reported as the main livelihood by 80.8, 89.8, and 74.2 percent of households in the three districts, respectively, compared with 69 percent nationally (Uganda Bureau of Statistics, 2017). The Annual Agricultural Survey reports that the average agricultural household nationally operates 1.3 hectares over 2.1 parcels, while mean parcel size is 0.47 hectares in Busoga, which includes Bugiri and Mayuge, and 0.37 hectares in Bukedi, which includes Butaleja (Uganda Bureau of Statistics, 2022). The bimodal rainfall regime and irrigated rice production in the wider region are documented by FEWS NET (FEWS NET, 2010). The Mayuge District Development Plan reports annual rainfall ranging from approximately 900 millimeters in the north to 1,200 millimeters in the south (Mayuge District Local Government, 2015).

OA-2 Survey instrument

This section reproduces the pre-season survey instrument in field order. The instrument contains 328 numbered entries, comprising 311 respondent-facing questions, 15 stored calculations, and two enumerator-facing notes. The entries are organized into 19 modules following the structure of the fielded questionnaire. For each item, we report the fielded variable name, question wording, response type, answer options, routing conditions, and relevant validation constraints. Question wording and answer options are reproduced as fielded rather than edited for presentation.

Language. The machine-readable questionnaire contains a single English-language label for each item and no stored written translations. Interviews were administered face to face. Where respondents did not use English, enumerators translated orally during administration. Because no written translation or back-translation is preserved in the study records, we reproduce only the English fielded instrument.

Interview timing. The fielded questionnaire does not retain interview start and end timestamps, so realized interview duration cannot be recovered from the available survey data.

Experimental routing. Experimental arm is assigned at the village level and fixed by the village selected at the beginning of the questionnaire. Arms A and B complete the pre-information forward-belief modules, while Arm C is routed directly to the information treatment and subsequent forward elicitation. All three arms complete the final forward elicitation. Arm A therefore repeats the same elicitation without receiving information in between, providing the repeated-elicitation benchmark used in the experimental analysis.

Token elicitations. Two visual elicitation instruments recur throughout the questionnaire. The seasonal-weather exercise asks respondents to allocate ten physical tokens across nine ordered pictures ranging from extreme drought to severe flooding. The planting-onset exercise asks respondents to allocate ten tokens across ten dated intervals. Section OA-3 reproduces the standardized instructions and enumerator protocol for both exercises.

Reading the arm gating. Assignment is at the village level and is computed on the device by the `calculate` field `arm` (Module OA-2.6), which maps each of the 60 sampled villages to one of three arms: 24 villages to Arm A, 26 to Arm B, and 10 to Arm C. Two features of the ordering carry most of the identification weight in this paper, and both are properties of the form itself rather than of the analysis.

First, the pre-card adaptation block `adapt_group` carries the relevance condition `arm = 'A' or arm = 'B' or arm = 'C'`, which is exhaustive over the three arms, and it sits at form position 324–329, ahead of the information block `info_treatment_group` at position 336. Every respondent in every arm therefore answers the pre-card adaptation questions before any information is shown, and `adapt_any_pre` is defined for all three arms.

Second, the pre-card belief blocks `belief_amount_future` and `onset_future_group` carry the relevance condition `arm = 'A' or arm = 'B'`. Arm C respondents are

routed past them entirely, so Arm C has no pre-card belief elicitation at all and `fut_pre_total_tokens` is zero for every one of the 119 Arm C respondents. Arm C’s single forward elicitation is the `post_amount` and `post_onset_group` block, which is administered after the card.

The second forward elicitation itself, `post_amount` and `post_onset_group`, carries *no* relevance condition. All three arms answer it. For Arm A it is a second reading of an unchanged instrument with nothing in between, which is what makes Arm A a within-respondent placebo for the elicitation itself rather than a pure no-contact control.

Table OA1: The `village_list` choice list and the village-to-arm assignment encoded in the `arm` calculate field

Option code	Option label (subcounty, district)	Arm
bubalule_b	Bubalule B (Kityerera, Mayuge)	C
bubinge	Bubinge (Kityerera, Mayuge)	A
bufuta_a	Bufuta A (Bukatube, Mayuge)	A
bugalo	Bugalo (Mazimasa, Butaleja)	B
bugodo_b	Bugodo B (Kapyanga, Bugiri)	C
bugolola_a	Bugolola A (Budumba, Butaleja)	B
buguluma	Buguluma (Bukatube, Mayuge)	A
buhwaya	Buhwaya (Budumba, Butaleja)	A
bukerewe	Bukerewe (Buwunga, Bugiri)	A
bulomi	Bulomi (Mazimasa, Butaleja)	B
bulumba	Bulumba (Kachonga, Butaleja)	B
bunghanga	Bunghanga (Budumba, Butaleja)	A
busimo	Busimo (Kityerera, Mayuge)	B
busolo	Busolo (Kapyanga, Bugiri)	B
buswikira	Buswikira (Mpungwe, Mayuge)	B
butegwa_b	Butegwa B (Nankoma, Bugiri)	B
buwembula	Buwembula (Muterere, Bugiri)	A
buwuge	Buwuge (Buwunga, Bugiri)	B
bwalula_a	Bwalula A (Nankoma, Bugiri)	A
dumbu	Dumbu (Budumba, Butaleja)	C
kasokwe	Kasokwe (Mpungwe, Mayuge)	A
kazinga	Kazinga (Kigandalo, Mayuge)	B
kigulamo	Kigulamo (Kigandalo, Mayuge)	A
kikandwa	Kikandwa (Kityerera, Mayuge)	A
kiteigalwa	Kiteigalwa (Buwunga, Bugiri)	A
kyaluya	Kyaluya (Buwunga, Bugiri)	C
kyando	Kyando (Bukatube, Mayuge)	B
kyete	Kyete (Mpungwe, Mayuge)	B
lubanga	Lubanga (Kachonga, Butaleja)	B

Table OA1, continued

Option code	Option label (subcounty, district)	Arm
lubanyi	Lubanyi (Muterere, Bugiri)	A
lubembe	Lubembe (Mazimasa, Butaleja)	A
lukunhu	Lukunhu (Kigandalo, Mayuge)	A
lwaba	Lwaba (Kapyanga, Bugiri)	A
maleka	Maleka (Kigandalo, Mayuge)	B
matiki	Matiki (Nankoma, Bugiri)	B
mauta	Mauta (Bukatube, Mayuge)	B
mawanga	Mawanga (Buwunga, Bugiri)	B
mitimito	Mitimito (Kityerera, Mayuge)	B
mpumu	Mpumu (Mpungwe, Mayuge)	C
muhula_kagondo	Muhula Kagondo (Kachonga, Butaleja)	A
musima	Musima (Mpungwe, Mayuge)	B
mwezi	Mwezi (Mpungwe, Mayuge)	A
nabuyanja	Nabuyanja (Budumba, Butaleja)	A
nahanja	Nahanja (Mazimasa, Butaleja)	A
nahigogola	Nahigogola (Kachonga, Butaleja)	B
nakabale	Nakabale (Mazimasa, Butaleja)	A
nakalanga	Nakalanga (Bukatube, Mayuge)	C
nakavule	Nakavule (Kapyanga, Bugiri)	B
namalenha	Namalenha (Buwunga, Bugiri)	A
namatooke	Namatooke (Mpungwe, Mayuge)	C
namawa_a	Namawa A (Kachonga, Butaleja)	C
namujukume	Namujukume (Kachonga, Butaleja)	A
namunasa	Namunasa (Mazimasa, Butaleja)	B
napindo	Napindo (Budumba, Butaleja)	B
ndhokero_b	Ndhokero B (Bukatube, Mayuge)	C
ngunga	Ngunga (Muterere, Bugiri)	B
nongo	Nongo (Muterere, Bugiri)	B
nsono	Nsono (Nankoma, Bugiri)	C
nvunwa	Nvunwa (Kigandalo, Mayuge)	A
wandegeire	Wandegeire (Buwunga, Bugiri)	B

Notes: This is the complete `village_list` choice list of the XLSForm, given verbatim: the option code as stored, and the option label as it appeared on the enumerator's screen, with subcounty and district in parentheses. The Arm column is not part of the choice list; it is the value the `calculate` field `arm` assigns to a respondent who selects that village. Arm A: 24 villages. Arm B: 26 villages. Arm C: 10 villages. The mapping is hard-coded in the form and is not re-drawn on the device, so assignment is fixed before the enumerator opens the questionnaire.

The two token instruments. Two elicitation devices recur throughout the instrument and are described once here. The *rainfall amount* exercise asks the respondent to place ten physical tokens across nine ordered pictures of a field, running from image 1 (extreme drought) through image 5 (optimal) to image 9 (severe flood); the nine labels are given verbatim as the question text of the nine token cells in Module OA-2.7. The *onset timing* exercise asks for ten tokens across ten dated intervals of the planting window. The interval edges are printed on the enumerator’s calendar card and are not stored in the XLSForm, which labels the cells only as “onset interval 1” through “onset interval 10”; they are reproduced in Table OA2.

Table OA2: Onset intervals shown on the enumerator’s calendar card

Interval	Date range	Text printed alongside
1	15–21 Feb	Very dry; land still hard; few early land preparers start ploughing.
2	22–28 Feb	First scattered showers in some low-lying areas; usually still unreliable.
3	1–6 Mar	Land preparation intensifies; some early planters risk maize if showers continue.
4	7–13 Mar	Transitional period; first rains often occur; seedbed preparation peaks.
5	14–20 Mar	Some areas record 1–2 good rains; not yet sustained in the highlands.
6	21–27 Mar	Typical onset for lower slopes; first main planting begins.
7	28 Mar–3 Apr	Onset consolidates; continuous rain spells; widespread maize and bean planting.
8	4–10 Apr	Average onset period for mid-slopes; moisture reliable.
9	11–17 Apr	Late planters begin; upper slopes catch up.
10	18–24 Apr	Late onset years often start here; second wave of planters.

Notes: Reproduced from the enumerator calendar card “FARMER SURVEY CALENDAR: Start of Rains”, whose instruction line reads: “Allocate 10 tokens across the intervals to indicate when you remember / expect the seasonal rains started.” These edges are not stored in the XLSForm and must be read from the card to interpret `y2025_int1–y2021_int10`, `fut_pre_int1–fut_pre_int10`, and `fut_post_int1–fut_post_int10`. The intervals are not uniform: nine of the ten are exactly seven days wide and the third, 1–6 March, is six, so the calendar spans 15 February to 24 April and nothing outside it, with midpoints at days 49, 56, 62.5, 69, 76, 83, 90, 97, 104 and 111 of the year. The physical card is reproduced as the calendar card reproduced in the paper’s design section. Every day-scale magnitude in this draft is computed against those midpoints; Appendix B of the paper documents the construction.

OA-2.1 Identification and location

Arms: All arms. Administered before randomization is revealed to the enumerator.

No.	Variable	Question, type, options, and routing
1	resp_id → respondent_id text	Respondent ID <i>Required.</i>
2	Date → interview_date datetime	Date
3	village_name → village_name_label select_one village_list	Village name (select from list) <i>Required.</i> <i>Options:</i> the 60 study villages. The list is too long to sit in this cell; all 60 option codes and labels, with the arm each village is assigned to, are given in Table OA1. <i>Appearance:</i> minimal
4	village_confirm → (echo column, dropped) select_one yn	<i>Dynamic text:</i> concat("Is the village you selected: ", jr:choice-name(\${village_name}, "\${village_name}"), "?") <i>Required.</i> <i>Options</i> (code — label as read out), yn: yes — Yes no — No <i>Constraint:</i> . = 'yes' <i>Message:</i> "Please go back and reselect the correct village before continuing"
5	Village_ID → village_id integer	Village ID
6	district text	District or region

OA-2.2 Demographic information

XLSForm group(s): demo_group.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block demo_group — “Demographic information”.		
<i>Block relevance:</i> none; every respondent who reaches this point answers it.		

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No.	Variable	Question, type, options, and routing
7	age <i>integer</i>	Age in years <i>Required.</i> <i>Constraint:</i> . >= 10 and . <= 99 <i>Message:</i> "Please enter an age between 10 and 99"
8	gender <i>select_one gender</i>	Gender <i>Required.</i> <i>Options</i> (code — label as read out), gender : male — Male female — Female other — Other
9	marital_status <i>select_one marital</i>	Marital status <i>Options</i> (code — label as read out), marital : single — Single married — Married widowed — Widowed divorced — Divorced
10	hh_head <i>select_one yn</i>	Are you the head of the household <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
11	hh_size <i>integer</i>	Household size (including Children) <i>Required.</i> <i>Constraint:</i> . > 0 <i>Message:</i> "Household size must be greater than zero"
12	literate → literacy <i>select_one yn</i>	Can you read and write <i>Options:</i> the 2-item yn list, given in full at Q4.
13	educ_level → education <i>select_one educ</i>	Highest level of education attended <i>Options</i> (code — label as read out), educ : none — No formal education primary — Primary education secondary — Secondary education vocational — Vocational training tertiary — Tertiary education

OA-2.3 Farming background, land tenure, and decision rights

XLSForm group(s): Farming_Background.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block Farming_Background — “ <i>Farming Background</i> ”.		
<i>Block relevance:</i> none; every respondent who reaches this point answers it.		
14	farmer_status → is_farmer <i>select_one yn</i>	Are you a farmer <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
15	land_ownership → tenure_own , tenure_rent , tenure_sharecrop , tenure_communal , tenure_other <i>select_multiple landown</i>	Which types of land tenure apply to the land you farm (tick all that apply) <i>Required.</i> <i>Options</i> (code — label as read out), landown : own — Own rent — Rent sharecrop — Sharecrop communal — Communal other — Other
16	land_own_acres → acres_owned <i>decimal</i>	How many acres are owned by your household or close family (Grand-Parents, Parents, Children or Siblings) <i>Relevance:</i> Asked only if the option own was ticked at land_ownership . selected (\${land_ownership}, 'own') <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”
17	land_rent_acres → acres_rented <i>decimal</i>	How many acres are rented <i>Relevance:</i> Asked only if the option rent was ticked at land_ownership . selected (\${land_ownership}, 'rent') <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”
18	land_sharecrop_acres → acres_sharecrop <i>decimal</i>	How many acres are farmed under sharecropping arrangements <i>Relevance:</i> Asked only if the option sharecrop was ticked at land_ownership . selected (\${land_ownership}, 'sharecrop') <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”
19	land_communal_acres → acres_communal <i>decimal</i>	How many acres are communal land that you farm <i>Relevance:</i> Asked only if the option communal was ticked at land_ownership . selected (\${land_ownership}, 'communal') <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”

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No.	Variable	Question, type, options, and routing
20	land_other_acres → acres_other <i>decimal</i>	How many acres are under other land tenure arrangements <i>Relevance:</i> Asked only if the option other was ticked at land_ownership . selected (\${land_ownership}, 'other') <i>Constraint:</i> . >= 0 <i>Message:</i> "Value must be zero or positive"
21	decision_role <i>select_one yn</i>	Are you involved in farm related decision making <i>Options:</i> the 2-item yn list, given in full at Q4.
22	top_decision_1 <i>select_one dec_parties</i>	Most important decision maker for crop and planting decisions <i>Required.</i> <i>Options</i> (code — label as read out), dec_parties : Individual — Survey Taker aid — Aid organizations (Individual) hh_head — Household head (if other than individual) spouse — Spouse elders — Elders other_dm — Other Decision Maker parents — Parents na_none — NA / No one neighbors — Neighbors other_dm_2 — Other Decision Maker chief — Village official or chief other_dm_3 — Other Decision Maker official_info — Official information source na_none_2 — NA / No one <i>Displayed subset:</i> the nine substantive parties plus other_dm ; the "NA / No one" codes are hidden here.
23	top_decision_1_other → other_decision_1 <i>text</i>	Specify other decision maker (first) <i>Relevance:</i> Asked only if "Other Decision Maker" (other_dm) was chosen at top_decision_1 . \${top_decision_1} = 'other_dm'
24	top_decision_2 <i>select_one dec_parties</i>	Second most important decision maker <i>Required.</i> <i>Options:</i> the 14-item dec_parties list, given in full at Q22. <i>Displayed subset:</i> the nine substantive parties plus na_none and other_dm_2 . <i>Constraint:</i> . != \${top_decision_1} <i>Message:</i> "Please select a different decision maker from the first choice"

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No.	Variable	Question, type, options, and routing
25	<code>top_decision_2_</code> <code>other</code> → <code>other_decision_2</code> <i>text</i>	Specify other decision maker (second) <i>Relevance:</i> Asked only if “Other Decision Maker” (<code>other_dm_2</code>) was chosen at <code>top_decision_2</code> . <code>\${top_decision_2} = 'other_dm_2'</code>
26	<code>top_decision_3</code> <i>select_one dec_parties</i>	Third most important decision maker <i>Required.</i> <i>Options:</i> the 14-item <code>dec_parties</code> list, given in full at Q22. <i>Displayed subset:</i> the nine substantive parties plus <code>other_dm_3</code> and <code>na_none_2</code> . <i>Constraint:</i> . != <code>\${top_decision_1}</code> and . != <code>\${top_decision_2}</code> <i>Message:</i> “Please select a different decision maker from the first two choices”
27	<code>top_decision_3_</code> <code>other</code> → <code>other_decision_3</code> <i>text</i>	Specify other decision maker (third) <i>Relevance:</i> Asked only if “Other Decision Maker” (<code>other_dm_3</code>) was chosen at <code>top_decision_3</code> . <code>\${top_decision_3} = 'other_dm_3'</code>
28	<code>parents_farmers</code> <i>select_one yn</i>	Were your parents farmers <i>Options:</i> the 2-item <code>yn</code> list, given in full at Q4.
29	<code>grandparents_farmers</code> <i>select_one yn</i>	Were your grandparents farmers <i>Options:</i> the 2-item <code>yn</code> list, given in full at Q4.
30	<code>farm_size_acres</code> → <code>total_cultivated_acres</code> <i>decimal</i>	How much land does your household cultivate in total in acres <i>Required.</i> <i>Constraint:</i> . > 0 <i>Message:</i> “Total cultivated area must be greater than zero”
31	<code>owned_area_acres</code> → <code>owned_cultivated_acres</code> <i>decimal</i>	How much of this cultivated land is owned by your household or direct relatives <i>Constraint:</i> . >= 0 and . <= <code>\${farm_size_acres}</code> <i>Message:</i> “Value must be between 0 and total cultivated acres”
32	<code>irrigation_access</code> <i>select_one yn</i>	Is there a water (irrigation) source (rivers, wells, pond, canals or similar sources) near or at your property <i>Options:</i> the 2-item <code>yn</code> list, given in full at Q4.
33	<code>irrigation_use</code> <i>select_one yn</i>	Do you use irrigation <i>Options:</i> the 2-item <code>yn</code> list, given in full at Q4. <i>Relevance:</i> Asked only if <code>irrigation_access</code> was answered Yes. <code>\${irrigation_access} = 'yes'</code>

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No.	Variable	Question, type, options, and routing
34	irrigated_area_acres → irrigated_acres decimal	How many acres of your cultivated land are irrigated <i>Relevance:</i> Asked only if irrigation_use was answered Yes. $\text{\$}\{\text{irrigation_use}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . $\leq \text{\$}\{\text{farm_size_acres}\}$ <i>Message:</i> “Irrigated acres must not exceed total cultivated acres”

OA-2.4 Crop production and agricultural practices

XLSForm group(s): crop_practices_group.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block crop_practices_group — “Crop production and agricultural practices”. <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
35	planned_crops → crop_maize ... crop_ other (22 dummies) select_multiple crop_list	Which crops do you plan to grow this season <i>Required.</i> <i>Options</i> (code — label as read out), crop_list: maize — Maize beans — Beans cassava — Cassava sweet_potatoes — Sweet potatoes groundnuts — Groundnuts sorghum — Sorghum millet — Millet rice — Rice bananas — Bananas or plantains simsim — Sesame Simsim cowpeas — Cowpeas soybeans — Soybeans sunflower — Sunflower coffee — Coffee tomatoes — Tomatoes onions — Onions cabbages — Cabbages pumpkin — Pumpkin vegetables — Mixed vegetables sugar_cane — Sugar Cane coconut — Coconut other — Other
36	planned_crops_other → crop_other_specify text	Please specify the other crops <i>Relevance:</i> Asked only if the option other was ticked at planned_crops. $\text{selected}(\text{\$}\{\text{planned_crops}\}, \text{'other'})$

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No.	Variable	Question, type, options, and routing
37	top_crop1 <i>select_one crop_list</i>	Biggest crop by area <i>Required.</i> <i>Options:</i> the 22-item crop_list list, given in full at Q35. <i>Displayed subset:</i> restricted to the crops ticked at planned_crops .
38	top_crop2 <i>select_one crop_list</i>	Second biggest crop by area <i>Required.</i> <i>Options:</i> the 22-item crop_list list, given in full at Q35. <i>Displayed subset:</i> restricted to the crops ticked at planned_crops . <i>Constraint:</i> . != \${top_crop1} <i>Message:</i> “Please select a different crop from the first choice”
39	top_crop3 <i>select_one crop_list</i>	Third biggest crop by area <i>Required.</i> <i>Options:</i> the 22-item crop_list list, given in full at Q35. <i>Displayed subset:</i> restricted to the crops ticked at planned_crops . <i>Constraint:</i> \${top_crop1} != \${top_crop2} and \${top_crop1} != \${top_crop3} and \${top_crop2} != \${top_crop3} <i>Message:</i> “Please select three different crops”
Block crop1_details — “Details for your biggest crop”.		
<i>Block relevance:</i> Asked only if a crop was named at top_crop1 . \${top_crop1} != ”		
40	crop1_area <i>decimal</i>	How many acres will you plant with \${top_crop1}? <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= \${farm_size_acres} <i>Message:</i> “Must not exceed total cultivated acres”
41	crop1_plant_weeks → (not renamed; 11 week dummies in the export) <i>select_multiple plant_weeks</i>	In which weeks between 1 February and 15 April do you plan to plant \${top_crop1}? (Select at most two.) <i>Options</i> (code — label as read out), plant_weeks : w1 — 1 to 7 February w7 — 15 to 21 March w2 — 8 to 14 February w8 — 22 to 28 March w3 — 15 to 21 February w9 — 29 March to 4 April w4 — 22 to 28 February w10 — 5 to 11 April w5 — 1 to 7 March w11 — 12 to 18 April w6 — 8 to 14 March <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> “Please select one or two weeks”

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No.	Variable	Question, type, options, and routing
42	crop1_harvest_ weeks → crop1_harvest_weeks_ label select_multiple harvest_ weeks	In which weeks between 1 June and 31 August do you expect to harvest $\{\text{top_crop1}\}$? (Select at most two.) Options (code — label as read out), harvest_weeks: h1 — 1 to 7 June h7 — 15 to 21 July h2 — 8 to 14 June h8 — 22 to 31 July h3 — 15 to 21 June h9 — 1 to 7 August h4 — 22 to 30 June h10 — 8 to 14 August h5 — 1 to 7 July h11 — 15 to 21 August h6 — 8 to 14 July h12 — 22 to 31 August Constraint: count-selected(.) >= 1 and count-selected(.) <= 2 Message: “Please select one or two weeks”
43	crop1_critical_ months → crop1_critical_months_ label select_multiple month_list	Months when rainfall is most critical for $\{\text{top_crop1}\}$ (Select at most two.) Options (code — label as read out), month_list: jan — January jul — July feb — February aug — August mar — March sep — September apr — April oct — October may — May nov — November jun — June dec — December Constraint: count-selected(.) >= 1 and count-selected(.) <= 2 Message: “Please select one or two weeks”
44	irrigated_area_ acres_crop1 → crop1_irrigated_acres decimal	How many acres of your cultivated land are irrigated for $\{\text{top_crop1}\}$? Relevance: Asked only if irrigation_use was answered Yes. $\{\text{irrigation_use}\} = \text{'yes'}$ Constraint: . >= 0 and . <= $\{\text{crop1_area}\}$ Message: “Irrigated acres must not exceed cultivated acres for this crop”

Block crop2_details — “Details for your second biggest crop”.

Block relevance: Asked only if a crop was named at top_crop2. \${top_crop2} != ''

45	crop2_area decimal	How many acres will you plant with \${top_crop2}? <i>Required.</i> <i>Constraint:</i> . >= 0 and . + \${crop1_area} <= \${farm_size_acres} <i>Message:</i> “Sum of 1st and 2nd crop areas must not exceed total cultivated acres”
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No.	Variable	Question, type, options, and routing
46	crop2_plant_weeks → (not renamed) <i>select_multiple plant_weeks</i>	In which weeks between 1 February and 15 April do you plan to plant \${top_crop2}? (Select at most two.) <i>Options:</i> the 11-item plant_weeks list, given in full at Q41. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> “Please select one or two weeks”
47	crop2_harvest_weeks → (not renamed) <i>select_multiple harvest_weeks</i>	In which weeks between 1 June and 31 August do you expect to harvest \${top_crop2}? (Select at most two.) <i>Options:</i> the 12-item harvest_weeks list, given in full at Q42. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> “Please select one or two weeks”
48	crop2_critical_months → (not renamed) <i>select_multiple month_list</i>	Months when rainfall is most critical for \${top_crop2} (Select at most two.) <i>Options:</i> the 12-item month_list list, given in full at Q43. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> “Please select one or two weeks”
49	irrigated_area_acres_crop2 → crop2_irrigated_acres <i>decimal</i>	How many acres of your cultivated land are irrigated for \${top_crop2}? <i>Relevance:</i> Asked only if irrigation_use was answered Yes. \${irrigation_use} = 'yes' <i>Constraint:</i> . >= 0 and . <= \${crop2_area} <i>Message:</i> “Irrigated acres must not exceed cultivated acres for this crop”
Block crop3_details — “Details for your third biggest crop”.		
<i>Block relevance:</i> Asked only if a crop was named at top_crop3 . \${top_crop3} != ''		
50	crop3_area <i>decimal</i>	How many acres will you plant with \${top_crop3}? <i>Required.</i> <i>Constraint:</i> . >= 0 and . + \${crop1_area} + \${crop2_area} <= \${farm_size_acres} <i>Message:</i> “Sum of all three crop areas must not exceed total cultivated acres”
51	crop3_plant_weeks → (not renamed) <i>select_multiple plant_weeks</i>	In which weeks between 1 February and 15 April do you plan to plant \${top_crop3}? (Select at most two.) <i>Options:</i> the 11-item plant_weeks list, given in full at Q41. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> “Please select one or two weeks”

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No.	Variable	Question, type, options, and routing
52	crop3_harvest_weeks → (not renamed) select_multiple harvest_weeks	In which weeks between 1 June and 31 August do you expect to harvest \${top_crop3}? (Select at most two.) <i>Options:</i> the 12-item harvest_weeks list, given in full at Q42. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> "Please select one or two weeks"
53	crop3_critical_months → (not renamed) select_multiple month_list	Months when rainfall is most critical for \${top_crop3} (Select at most two.) <i>Options:</i> the 12-item month_list list, given in full at Q43. <i>Constraint:</i> count-selected(.) >= 1 and count-selected(.) <= 2 <i>Message:</i> "Please select one or two weeks"
54	irrigated_area_acres_crop3 → crop3_irrigated_acres decimal	How many acres of your cultivated land are irrigated for \${top_crop3}? <i>Relevance:</i> Asked only if irrigation_use was answered Yes. \${irrigation_use} = 'yes' <i>Constraint:</i> . >= 0 and . <= \${crop3_area} <i>Message:</i> "Irrigated acres must not exceed cultivated acres for this crop"
55	farm_practices_used → prac_mechanized, prac_fertilizer, prac_pest_control, prac_improved_seeds, prac_crop_rotation, prac_other select_multiple farm_practices	Does your household use any of the following for farming <i>Options</i> (code — label as read out), farm_practices: mechanized — Mechanized equipment electric or petrol powered improved_seeds — Use of improved seeds fertilizer — Use of fertilizers rotation — Crop rotation pest_control — Pest control methods other — Other methods
56	farm_practices_other → prac_other_specify text	Please specify other farming practices <i>Relevance:</i> Asked only if the option other was ticked at farm_practices_used . selected(\${farm_practices_used}, 'other')

OA-2.5 Access to agricultural information and forecasts

XLSForm group(s): info_access_group.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block info_access_group — “Access to agricultural information”.		
<i>Block relevance:</i> none; every respondent who reaches this point answers it.		
57	ext_access → extension_access <i>select_one yn</i>	Do you have access to agricultural extension services <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
58	weather_info_any <i>select_one yn</i>	Do you receive any information on past present or future weather <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
59	weather_info_level <i>select_one info_level</i>	At what level is this weather information provided (If you receive information at multiple levels, select the most detailed one, e.g. select "District" rather than "National" if you receive both.) <i>Options</i> (code — label as read out), info_level : national — National regional — Regional district — District village — Village other — Other <i>Relevance:</i> Asked only if weather_info_any was answered Yes. \${weather_info_any} = 'yes'
60	weather_info_level_other <i>text</i>	Please specify the other level <i>Relevance:</i> Asked only if “Other” was chosen at weather_info_level . \${weather_info_level} = 'other'
61	weather_info_specific → weather_info_subcounty_specific <i>select_one yn</i>	Is this weather information specific to your location (sub county level or below) <i>Options:</i> the 2-item yn list, given in full at Q4. <i>Relevance:</i> Asked only if weather_info_any was answered Yes. \${weather_info_any} = 'yes'
62	weather_info_past <i>select_one yn</i>	Is this information about past weather <i>Options:</i> the 2-item yn list, given in full at Q4. <i>Relevance:</i> Asked only if weather_info_any was answered Yes. \${weather_info_any} = 'yes'

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No.	Variable	Question, type, options, and routing
63	weather_info_past_seasons → weather_info_n_seasons integer	How many past seasons are reported <i>Relevance:</i> Asked only if weather_info_past was answered Yes. <code>\${weather_info_past} = 'yes'</code> <i>Constraint:</i> . >= 0 <i>Message:</i> "Value must be zero or positive"
64	weather_info_content → info_content_rain, info_content_temp, info_content_soil, info_content_storms, info_content_other select_multiple info_content	What information is reported <i>Options</i> (code — label as read out), info_content : rain — Rainfall temp — Temperature soil — Soil moisture storms — Storms other — Other <i>Relevance:</i> Asked only if weather_info_any was answered Yes. <code>\${weather_info_any} = 'yes'</code>
65	weather_info_content_other → info_content_other_specify text	Please specify the other reported information <i>Relevance:</i> Asked only if the option other was ticked at weather_info_content . selected (<code>\${weather_info_content}</code> , 'other')
66	weather_info_sources → src_past_radio, src_past_tv, src_past_mobile, src_past_meetings, src_past_extension, src_past_neighbors, src_past_internet, src_past_other select_multiple info_source	What are your main sources of weather information <i>Options</i> (code — label as read out), info_source : radio — Radio extension — Agricultural extension agents tv — Television neighbors — Neighbors or other farmers phone — Mobile phone alerts internet — Internet via smartphone or device meetings — Community meetings other — Other <i>Relevance:</i> Asked only if weather_info_any was answered Yes. <code>\${weather_info_any} = 'yes'</code>
67	weather_info_sources_other → src_past_other_specify text	Please specify the other weather information source <i>Relevance:</i> Asked only if the option other was ticked at weather_info_sources . selected (<code>\${weather_info_sources}</code> , 'other')
68	forecast_any select_one yn	Do you receive weather forecasts <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.

continued from previous page

No.	Variable	Question, type, options, and routing
69	forecast_level <i>select_one info_level</i>	At what level is the weather forecast provided (If you receive information at multiple levels, select the most detailed one, e.g. select "District" rather than "National" if you receive both.) <i>Options:</i> the 5-item info_level list, given in full at Q59. <i>Relevance:</i> Asked only if forecast_any was answered Yes. \${forecast_any} = 'yes'
70	forecast_level_ other <i>text</i>	Please specify the other level <i>Relevance:</i> Asked only if "Other" was chosen at forecast_level . \${forecast_level} = 'other'
71	forecast_specific <i>select_one yn</i>	Is this forecast specific to your location (sub county level or below) <i>Options:</i> the 2-item yn list, given in full at Q4. <i>Relevance:</i> Asked only if forecast_any was answered Yes. \${forecast_any} = 'yes'
72	forecast_content → forecast_content_rain , forecast_content_temp , forecast_content_soil , forecast_content_storms , forecast_content_other <i>select_multiple info_content</i>	What is forecasted <i>Options:</i> the 5-item info_content list, given in full at Q64. <i>Relevance:</i> Asked only if forecast_any was answered Yes. \${forecast_any} = 'yes'
73	forecast_content_ other → forecast_content_ other_specify <i>text</i>	Please specify the other forecasted information <i>Relevance:</i> Asked only if the option other was ticked at forecast_content . selected(\${forecast_content}, 'other')
74	forecast_sources → src_fcst_radio , src_fcst_tv , src_fcst_mobile , src_fcst_meetings , src_fcst_extension , src_fcst_neighbors , src_fcst_internet , src_fcst_other <i>select_multiple info_source</i>	What are your main sources of weather forecasts <i>Options:</i> the 8-item info_source list, given in full at Q66. <i>Relevance:</i> Asked only if forecast_any was answered Yes. \${forecast_any} = 'yes'

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No.	Variable	Question, type, options, and routing
75	<code>forecast_sources_</code> <code>other</code> → <code>src_fcst_other_specify</code> <i>text</i>	Please specify the other weather forecast source <i>Relevance:</i> Asked only if the option <code>other</code> was ticked at <code>forecast_sources</code> . <code>selected(\${forecast_sources}, 'other')</code>

OA-2.6 Experimental assignment

XLSForm group(s): `arm_group`.

Arms: All arms. The arm is computed from the village, not drawn on the device.

No.	Variable	Question, type, options, and routing
Block <code>arm_group</code> — “ <i>Experimental assignment</i> ”.		
<i>Block relevance:</i> none; every respondent who reaches this point answers it.		
76	<code>arm</code> <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Calculation:</i> a 60-branch nested <code>if()</code> on <code>village_name</code> ; the full village-to-arm mapping it encodes is reproduced in Table OA1.
77	<code>arm_note</code> → (<i>echo column, dropped</i>) <i>note</i>	<i>Dynamic text:</i> <code>concat("✓ This village is assigned to Treatment Arm ", \${arm})</code>

OA-2.7 Recalled rainfall amount, 2021–2025

XLSForm group(s): `belief_amount_2025` ... `belief_amount_2021`.

Arms: All arms. This is the memory instrument and it runs before any treatment.

No.	Variable	Question, type, options, and routing
Block <code>belief_amount_2025</code> — “ <i>Memory of rainfall amount in the first agricultural season (February-July) 2025</i> ”.		
<i>Block relevance:</i> none; every respondent who reaches this point answers it.		
78	<code>rem_2025</code> <i>select_one yn</i>	Do you remember the rainfall conditions in the first agricultural season (February-July) 2025 <i>Required.</i> <i>Options:</i> the 2-item <code>yn</code> list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
79	y2025_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
80	y2025_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
81	y2025_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
82	y2025_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
83	y2025_img5 <i>integer</i>	Tokens on image 5 optimal <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
84	y2025_img6 <i>integer</i>	Tokens on image 6 slightly wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
85	y2025_img7 <i>integer</i>	Tokens on image 7 very wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
86	y2025_img8 <i>integer</i>	Tokens on image 8 flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
87	y2025_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2025_img1}+\${y2025_img2}+\${y2025_img3}+\${y2025_img4}+\${y2025_img5}+\${y2025_img6}+\${y2025_img7}+\${y2025_img8}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
88	y2025_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes' <i>Calculation:</i> \${y2025_img1}+\${y2025_img2}+\${y2025_img3}+\${y2025_img4}+\${y2025_img5}+\${y2025_img6}+\${y2025_img7}+\${y2025_img8}+\${y2025_img9}
89	comment_belief_amount_2025 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_2025 was answered Yes. \${rem_2025} = 'yes'
Block belief_amount_2024 — "Memory of rainfall amount in the first agricultural season (February-July) 2024". <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
90	rem_2024 <i>select_one yn</i>	Do you remember the rainfall conditions in the first agricultural season (February-July) 2024 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
91	y2024_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
92	y2024_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
93	y2024_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
94	y2024_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
95	y2024_img5 <i>integer</i>	Tokens on image 5 optimal <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
96	y2024_img6 <i>integer</i>	Tokens on image 6 slightly wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
97	y2024_img7 <i>integer</i>	Tokens on image 7 very wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
98	y2024_img8 <i>integer</i>	Tokens on image 8 flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
99	y2024_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2024_img1}+\${y2024_img2}+\${y2024_img3}+\${y2024_img4}+\${y2024_img5}+\${y2024_img6}+\${y2024_img7}+\${y2024_img8}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
100	y2024_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes' <i>Calculation:</i> \${y2024_img1}+\${y2024_img2}+\${y2024_img3}+\${y2024_img4}+\${y2024_img5}+\${y2024_img6}+\${y2024_img7}+\${y2024_img8}+\${y2024_img9}
101	comment_belief_amount_2024 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_2024 was answered Yes. \${rem_2024} = 'yes'
Block belief_amount_2023 — "Memory of rainfall amount in the first agricultural season (February-July) 2023". <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
102	rem_2023 <i>select_one yn</i>	Do you remember the rainfall conditions in the first agricultural season (February-July) 2023 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
103	y2023_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
104	y2023_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
105	y2023_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
106	y2023_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
107	y2023_img5 <i>integer</i>	Tokens on image 5 optimal <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
108	y2023_img6 <i>integer</i>	Tokens on image 6 slightly wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
109	y2023_img7 <i>integer</i>	Tokens on image 7 very wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
110	y2023_img8 <i>integer</i>	Tokens on image 8 flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
111	y2023_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2023_img1}+\${y2023_img2}+\${y2023_img3}+\${y2023_img4}+\${y2023_img5}+\${y2023_img6}+\${y2023_img7}+\${y2023_img8}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
112	y2023_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes' <i>Calculation:</i> \${y2023_img1}+\${y2023_img2}+\${y2023_img3}+\${y2023_img4}+\${y2023_img5}+\${y2023_img6}+\${y2023_img7}+\${y2023_img8}+\${y2023_img9}
113	comment_belief_amount_2023 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_2023 was answered Yes. \${rem_2023} = 'yes'
Block belief_amount_2022 — "Memory of rainfall amount in the first agricultural season (February-July) 2022". <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
114	rem_2022 <i>select_one yn</i>	Do you remember the rainfall conditions in the first agricultural season (February-July) 2022 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
115	y2022_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
116	y2022_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
117	y2022_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
118	y2022_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
119	y2022_img5 <i>integer</i>	Tokens on image 5 optimal <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
120	y2022_img6 <i>integer</i>	Tokens on image 6 slightly wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
121	y2022_img7 <i>integer</i>	Tokens on image 7 very wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
122	y2022_img8 <i>integer</i>	Tokens on image 8 flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
123	y2022_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2022_img1}+\${y2022_img2}+\${y2022_img3}+\${y2022_img4}+\${y2022_img5}+\${y2022_img6}+\${y2022_img7}+\${y2022_img8}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
124	y2022_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes' <i>Calculation:</i> \${y2022_img1}+\${y2022_img2}+\${y2022_img3}+\${y2022_img4}+\${y2022_img5}+\${y2022_img6}+\${y2022_img7}+\${y2022_img8}+\${y2022_img9}
125	comment_belief_amount_2022 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_2022 was answered Yes. \${rem_2022} = 'yes'
Block belief_amount_2021 — "Memory of rainfall amount in the first agricultural season (February-July) 2021". <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
126	rem_2021 <i>select_one yn</i>	Do you remember the rainfall conditions in the first agricultural season (February-July) 2021 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
127	y2021_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
128	y2021_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
129	y2021_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
130	y2021_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
131	y2021_img5 <i>integer</i>	Tokens on image 5 optimal <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
132	y2021_img6 <i>integer</i>	Tokens on image 6 slightly wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
133	y2021_img7 <i>integer</i>	Tokens on image 7 very wet <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
134	y2021_img8 <i>integer</i>	Tokens on image 8 flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
135	y2021_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2021_img1}+\${y2021_img2}+\${y2021_img3}+\${y2021_img4}+\${y2021_img5}+\${y2021_img6}+\${y2021_img7}+\${y2021_img8}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
136	y2021_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes' <i>Calculation:</i> \${y2021_img1}+\${y2021_img2}+\${y2021_img3}+\${y2021_img4}+\${y2021_img5}+\${y2021_img6}+\${y2021_img7}+\${y2021_img8}+\${y2021_img9}
137	comment_belief_amount_2021 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_2021 was answered Yes. \${rem_2021} = 'yes'

OA-2.8 Recalled onset timing, 2021–2025

XLSForm group(s): onset_2025 ... onset_2021.

Arms: All arms. Runs before any treatment.

No.	Variable	Question, type, options, and routing
Block onset_2025 — “Memory of onset timing in the first agricultural season (February-July) 2025”. <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
138	rem_onset_2025 <i>select_one yn</i>	Do you remember the onset timing in the first agricultural season (February-July) 2025 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
139	y2025_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”
140	y2025_int2 <i>integer</i>	Tokens for onset interval 2 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”
141	y2025_int3 <i>integer</i>	Tokens for onset interval 3 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”
142	y2025_int4 <i>integer</i>	Tokens for onset interval 4 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”
143	y2025_int5 <i>integer</i>	Tokens for onset interval 5 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”
144	y2025_int6 <i>integer</i>	Tokens for onset interval 6 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. $\text{\$}\{\text{rem_onset_2025}\} = \text{'yes'}$ <i>Constraint:</i> . ≥ 0 and . ≤ 10 <i>Message:</i> “Enter a number between 0 and 10”

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No.	Variable	Question, type, options, and routing
145	y2025_int7 <i>integer</i>	Tokens for onset interval 7 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
146	y2025_int8 <i>integer</i>	Tokens for onset interval 8 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
147	y2025_int9 <i>integer</i>	Tokens for onset interval 9 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
148	y2025_int10 <i>integer</i>	Tokens for onset interval 10 <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2025_int1}+\${y2025_int2}+\${y2025_int3}+\${y2025_int4}+\${y2025_int5}+\${y2025_int6}+\${y2025_int7}+\${y2025_int8}+\${y2025_int9}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
149	y2025_onset_total <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes' <i>Calculation:</i> \${y2025_int1}+\${y2025_int2}+\${y2025_int3}+\${y2025_int4}+\${y2025_int5}+\${y2025_int6}+\${y2025_int7}+\${y2025_int8}+\${y2025_int9}+\${y2025_int10}
150	comment_onset_2025 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_onset_2025 was answered Yes. \${rem_onset_2025} = 'yes'

Block onset_2024 — "Memory of onset timing in the first agricultural season (February-July) 2024".

Block relevance: none; every respondent who reaches this point answers it.

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No.	Variable	Question, type, options, and routing
151	rem_onset_2024 <i>select_one yn</i>	Do you remember the onset timing in the first agricultural season (February-July) 2024 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
152	y2024_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
153	y2024_int2 <i>integer</i>	Tokens for onset interval 2 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
154	y2024_int3 <i>integer</i>	Tokens for onset interval 3 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
155	y2024_int4 <i>integer</i>	Tokens for onset interval 4 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
156	y2024_int5 <i>integer</i>	Tokens for onset interval 5 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
157	y2024_int6 <i>integer</i>	Tokens for onset interval 6 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
158	y2024_int7 <i>integer</i>	Tokens for onset interval 7 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
159	y2024_int8 <i>integer</i>	Tokens for onset interval 8 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
160	y2024_int9 <i>integer</i>	Tokens for onset interval 9 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
161	y2024_int10 <i>integer</i>	Tokens for onset interval 10 <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2024_int1}+\${y2024_int2}+\${y2024_int3}+\${y2024_int4}+\${y2024_int5}+\${y2024_int6}+\${y2024_int7}+\${y2024_int8}+\${y2024_int9}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
162	y2024_onset_total <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes' <i>Calculation:</i> \${y2024_int1}+\${y2024_int2}+\${y2024_int3}+\${y2024_int4}+\${y2024_int5}+\${y2024_int6}+\${y2024_int7}+\${y2024_int8}+\${y2024_int9}+\${y2024_int10}
163	comment_onset_2024 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_onset_2024 was answered Yes. \${rem_onset_2024} = 'yes'

Block onset_2023 — "Memory of onset timing in the first agricultural season (February-July) 2023".

Block relevance: none; every respondent who reaches this point answers it.

164	rem_onset_2023 <i>select_one yn</i>	Do you remember the onset timing in the first agricultural season (February-July) 2023 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
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No.	Variable	Question, type, options, and routing
165	y2023_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
166	y2023_int2 <i>integer</i>	Tokens for onset interval 2 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
167	y2023_int3 <i>integer</i>	Tokens for onset interval 3 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
168	y2023_int4 <i>integer</i>	Tokens for onset interval 4 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
169	y2023_int5 <i>integer</i>	Tokens for onset interval 5 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
170	y2023_int6 <i>integer</i>	Tokens for onset interval 6 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
171	y2023_int7 <i>integer</i>	Tokens for onset interval 7 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
172	y2023_int8 <i>integer</i>	Tokens for onset interval 8 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
173	y2023_int9 <i>integer</i>	Tokens for onset interval 9 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
174	y2023_int10 <i>integer</i>	Tokens for onset interval 10 <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2023_int1}+\${y2023_int2}+\${y2023_int3}+\${y2023_int4}+\${y2023_int5}+\${y2023_int6}+\${y2023_int7}+\${y2023_int8}+\${y2023_int9}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
175	y2023_onset_total <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes' <i>Calculation:</i> \${y2023_int1}+\${y2023_int2}+\${y2023_int3}+\${y2023_int4}+\${y2023_int5}+\${y2023_int6}+\${y2023_int7}+\${y2023_int8}+\${y2023_int9}+\${y2023_int10}
176	comment_onset_2023 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_onset_2023 was answered Yes. \${rem_onset_2023} = 'yes'

Block onset_2022 — "Memory of onset timing in the first agricultural season (February-July) 2022".

Block relevance: none; every respondent who reaches this point answers it.

177	rem_onset_2022 <i>select_one yn</i>	Do you remember the onset timing in the first agricultural season (February-July) 2022 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
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No.	Variable	Question, type, options, and routing
178	y2022_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
179	y2022_int2 <i>integer</i>	Tokens for onset interval 2 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
180	y2022_int3 <i>integer</i>	Tokens for onset interval 3 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
181	y2022_int4 <i>integer</i>	Tokens for onset interval 4 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
182	y2022_int5 <i>integer</i>	Tokens for onset interval 5 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
183	y2022_int6 <i>integer</i>	Tokens for onset interval 6 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
184	y2022_int7 <i>integer</i>	Tokens for onset interval 7 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
185	y2022_int8 <i>integer</i>	Tokens for onset interval 8 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
186	y2022_int9 <i>integer</i>	Tokens for onset interval 9 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
187	y2022_int10 <i>integer</i>	Tokens for onset interval 10 <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2022_int1}+\${y2022_int2}+\${y2022_int3}+\${y2022_int4}+\${y2022_int5}+\${y2022_int6}+\${y2022_int7}+\${y2022_int8}+\${y2022_int9}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
188	y2022_onset_total <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes' <i>Calculation:</i> \${y2022_int1}+\${y2022_int2}+\${y2022_int3}+\${y2022_int4}+\${y2022_int5}+\${y2022_int6}+\${y2022_int7}+\${y2022_int8}+\${y2022_int9}+\${y2022_int10}
189	comment_onset_2022 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_onset_2022 was answered Yes. \${rem_onset_2022} = 'yes'

Block onset_2021 — "Memory of onset timing in the first agricultural season (February-July) 2021".

Block relevance: none; every respondent who reaches this point answers it.

190	rem_onset_2021 <i>select_one yn</i>	Do you remember the onset timing in the first agricultural season (February-July) 2021 <i>Required.</i> <i>Options:</i> the 2-item yn list, given in full at Q4.
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No.	Variable	Question, type, options, and routing
191	y2021_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
192	y2021_int2 <i>integer</i>	Tokens for onset interval 2 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
193	y2021_int3 <i>integer</i>	Tokens for onset interval 3 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
194	y2021_int4 <i>integer</i>	Tokens for onset interval 4 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
195	y2021_int5 <i>integer</i>	Tokens for onset interval 5 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
196	y2021_int6 <i>integer</i>	Tokens for onset interval 6 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
197	y2021_int7 <i>integer</i>	Tokens for onset interval 7 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"

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No.	Variable	Question, type, options, and routing
198	y2021_int8 <i>integer</i>	Tokens for onset interval 8 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
199	y2021_int9 <i>integer</i>	Tokens for onset interval 9 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> "Enter a number between 0 and 10"
200	y2021_int10 <i>integer</i>	Tokens for onset interval 10 <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${y2021_int1}+\${y2021_int2}+\${y2021_int3}+\${y2021_int4}+\${y2021_int5}+\${y2021_int6}+\${y2021_int7}+\${y2021_int8}+\${y2021_int9}) = 10 <i>Message:</i> "Total number of tokens for this year must be 10"
201	y2021_onset_total <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes' <i>Calculation:</i> \${y2021_int1}+\${y2021_int2}+\${y2021_int3}+\${y2021_int4}+\${y2021_int5}+\${y2021_int6}+\${y2021_int7}+\${y2021_int8}+\${y2021_int9}+\${y2021_int10}
202	comment_onset_2021 <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section <i>Relevance:</i> Asked only if rem_onset_2021 was answered Yes. \${rem_onset_2021} = 'yes'

OA-2.9 Pre-card forward expectations and confidence

XLSForm group(s): belief_amount_future, onset_future_group, prediction_confidence.

Arms: **Arms A and B only.** Arm C has no pre-card belief elicitation.

No.	Variable	Question, type, options, and routing
Block belief_amount_future — “ <i>Expectations about rainfall amount upcoming season</i> ”. <i>Block relevance:</i> Arms A and B only; Arm C respondents skip this item. $\text{\$}\{\text{arm}\} = \text{'A'}$ or $\text{\$}\{\text{arm}\} = \text{'B'}$		
203	fut_img1 → fut_pre_img1 integer	Tokens on image 1 extreme drought <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
204	fut_img2 → fut_pre_img2 integer	Tokens on image 2 drought <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
205	fut_img3 → fut_pre_img3 integer	Tokens on image 3 very dry <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
206	fut_img4 → fut_pre_img4 integer	Tokens on image 4 slightly dry <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
207	fut_img5 → fut_pre_img5 integer	Tokens on image 5 optimal <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
208	fut_img6 → fut_pre_img6 integer	Tokens on image 6 slightly wet <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
209	fut_img7 → fut_pre_img7 integer	Tokens on image 7 very wet <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
210	fut_img8 → fut_pre_img8 integer	Tokens on image 8 flood <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”

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No.	Variable	Question, type, options, and routing
211	fut_img9 → fut_pre_img9 <i>integer</i>	Tokens on image 9 severe flood <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${fut_img1}+\${fut_img2}+\${fut_img3}+\${fut_img4}+\${fut_img5}+\${fut_img6}+\${fut_img7}+\${fut_img8}) = 10 <i>Message:</i> “Total number of tokens must be 10”
212	fut_total_tokens → fut_pre_total_tokens <i>calculate</i>	<i>Not asked. Stored calculation.</i> <i>Calculation:</i> \${fut_img1}+\${fut_img2}+\${fut_img3}+\${fut_img4}+\${fut_img5}+\${fut_img6}+\${fut_img7}+\${fut_img8}+\${fut_img9}
213	comment_belief_amount_future → comment_fut_pre_amount <i>text</i>	Enumerator: note any comments or remarks made by the participant during this section
Block onset_future_group — “ <i>Expectations about onset timing upcoming season</i> ”. <i>Block relevance:</i> Arms A and B only; Arm C respondents skip this item. \${arm} = 'A' or \${arm} = 'B'		
214	fut_int1 → fut_pre_int1 <i>integer</i>	Tokens for onset interval 1 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
215	fut_int2 → fut_pre_int2 <i>integer</i>	Tokens for onset interval 2 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
216	fut_int3 → fut_pre_int3 <i>integer</i>	Tokens for onset interval 3 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
217	fut_int4 → fut_pre_int4 <i>integer</i>	Tokens for onset interval 4 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
218	fut_int5 → fut_pre_int5 <i>integer</i>	Tokens for onset interval 5 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”

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No.	Variable	Question, type, options, and routing
219	fut_int6 → fut_pre_int6 integer	Tokens for onset interval 6 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
220	fut_int7 → fut_pre_int7 integer	Tokens for onset interval 7 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
221	fut_int8 → fut_pre_int8 integer	Tokens for onset interval 8 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
222	fut_int9 → fut_pre_int9 integer	Tokens for onset interval 9 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
223	fut_int10 → fut_pre_int10 integer	Tokens for onset interval 10 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${fut_int1}+\${fut_int2}+\${fut_int3}+\${fut_int4}+\${fut_int5}+\${fut_int6}+\${fut_int7}+\${fut_int8}+\${fut_int9}) = 10 <i>Message:</i> “Total number of tokens must be 10”
224	fut_onset_total → fut_pre_onset_total calculate	<i>Not asked. Stored calculation.</i> <i>Calculation:</i> \${fut_int1}+\${fut_int2}+\${fut_int3}+\${fut_int4}+\${fut_int5}+\${fut_int6}+\${fut_int7}+\${fut_int8}+\${fut_int9}+\${fut_int10}
225	comment_onset_future_group → comment_fut_pre_onset text	Enumerator: note any comments or remarks made by the participant during this section

Block prediction_confidence — “*Prediction Confidence*”.

Block relevance: Arms A and B only; Arm C respondents skip this item. \${arm} = ‘A’ or \${arm} = ‘B’

continued from previous page

No.	Variable	Question, type, options, and routing
226	<code>conf_amount</code> → <code>conf_rain_pre</code> <i>select_one conf_scale</i>	Confidence in rainfall amount expectations <i>Required.</i> <i>Options</i> (code — label as read out), <code>conf_scale</code> : 5 — 5 Very High Confidence 4 — 4 High Confidence 3 — 3 Moderate Confidence 2 — 2 Low Confidence 1 — 1 Very Low Confidence
227	<code>conf_onset</code> → <code>conf_onset_pre</code> <i>select_one conf_scale</i>	Confidence in onset expectations <i>Required.</i> <i>Options</i> : the 5-item <code>conf_scale</code> list, given in full at Q226.

OA-2.10 Memory of extreme events

XLSForm group(s): `extreme_memory_grp`.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block <code>extreme_memory_grp</code> — “ <i>Memory of extreme events</i> ”.		
<i>Block relevance</i> : none; every respondent who reaches this point answers it.		
228	<code>extreme_past_any</code> <i>select_one yn</i>	Do you remember any years with major weather events (e.g. drought, flood, hailstorm, storm, etc.) <i>Options</i> : the 2-item <code>yn</code> list, given in full at Q4.
229	<code>extreme_past_n</code> <i>integer</i>	How many major weather events do you remember (0 to 5) <i>Required.</i> <i>Relevance</i> : Asked only if <code>extreme_past_any</code> was answered Yes. <code>\${extreme_past_any} = 'yes'</code> <i>Constraint</i> : <code>. >= 0 and . <= 5</code> <i>Message</i> : “Enter a number between 0 and 5”

Block `extreme_event1` — “*Details of event 1*”.

Block relevance: Asked only if the respondent said they remember at least 1 major weather event(s). `${extreme_past_n} >= 1`

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No.	Variable	Question, type, options, and routing
230	extreme1_type <i>select_one extreme_type</i>	Type of event <i>Options</i> (code — label as read out), extreme_type : drought — Drought flood — Flood hailstorm — Hailstorm storm — Storm other — Other <i>Relevance</i> : Asked only if the respondent said they remember at least 1 major weather event(s). \${extreme_past_n} >= 1
231	extreme1_type_ other <i>text</i>	If other, specify the type of event <i>Relevance</i> : Asked only if “Other” was chosen at extreme1_type . \${extreme1_type} = 'other'
232	extreme1_severity <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Required</i> . <i>Relevance</i> : Asked only if the respondent said they remember at least 1 major weather event(s). \${extreme_past_n} >= 1 <i>Constraint</i> : . >= 1 and . <= 10 <i>Message</i> : “Enter a value from 1 to 10”
233	extreme1_year <i>integer</i>	In which year did this event happen <i>Required</i> . <i>Relevance</i> : Asked only if the respondent said they remember at least 1 major weather event(s). \${extreme_past_n} >= 1 <i>Constraint</i> : . >= 1900 and . <= 2100 <i>Message</i> : “Enter a valid year”
Block extreme_event2 — “Details of event 2”.		
<i>Block relevance</i> : Asked only if the respondent said they remember at least 2 major weather event(s). \${extreme_past_n} >= 2		
234	extreme2_type <i>select_one extreme_type</i>	Type of event <i>Options</i> : the 5-item extreme_type list, given in full at Q230. <i>Relevance</i> : Asked only if the respondent said they remember at least 2 major weather event(s). \${extreme_past_n} >= 2
235	extreme2_type_ other <i>text</i>	If other, specify the type of event <i>Relevance</i> : Asked only if “Other” was chosen at extreme2_type . \${extreme2_type} = 'other'

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No.	Variable	Question, type, options, and routing
236	extreme2_severity <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 2 major weather event(s). \${extreme_past_n} >= 2 <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> "Enter a value from 1 to 10"
237	extreme2_year <i>integer</i>	In which year did this event happen <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 2 major weather event(s). \${extreme_past_n} >= 2 <i>Constraint:</i> . >= 1900 and . <= 2100 <i>Message:</i> "Enter a valid year"
Block extreme_event3 — "Details of event 3". <i>Block relevance:</i> Asked only if the respondent said they remember at least 3 major weather event(s). \${extreme_past_n} >= 3		
238	extreme3_type <i>select_one extreme_type</i>	Type of event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if the respondent said they remember at least 3 major weather event(s). \${extreme_past_n} >= 3
239	extreme3_type_ other <i>text</i>	If other, specify the type of event <i>Relevance:</i> Asked only if "Other" was chosen at extreme3_type . \${extreme3_type} = 'other'
240	extreme3_severity <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 3 major weather event(s). \${extreme_past_n} >= 3 <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> "Enter a value from 1 to 10"

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No.	Variable	Question, type, options, and routing
241	extreme3_year <i>integer</i>	In which year did this event happen <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 3 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 3$ <i>Constraint:</i> . ≥ 1900 and . ≤ 2100 <i>Message:</i> “Enter a valid year”
Block extreme_event4 — “Details of event 4”.		
<i>Block relevance:</i> Asked only if the respondent said they remember at least 4 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 4$		
242	extreme4_type <i>select_one extreme_type</i>	Type of event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if the respondent said they remember at least 4 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 4$
243	extreme4_type_ other <i>text</i>	If other, specify the type of event <i>Relevance:</i> Asked only if “Other” was chosen at extreme4_type . $\text{\$}\{\text{extreme4_type}\} = \text{'other'}$
244	extreme4_severity <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 4 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 4$ <i>Constraint:</i> . ≥ 1 and . ≤ 10 <i>Message:</i> “Enter a value from 1 to 10”
245	extreme4_year <i>integer</i>	In which year did this event happen <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 4 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 4$ <i>Constraint:</i> . ≥ 1900 and . ≤ 2100 <i>Message:</i> “Enter a valid year”

Block extreme_event5 — “Details of event 5”.

Block relevance: Asked only if the respondent said they remember at least 5 major weather event(s). $\text{\$}\{\text{extreme_past_n}\} \geq 5$

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No.	Variable	Question, type, options, and routing
246	extreme5_type <i>select_one extreme_type</i>	Type of event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if the respondent said they remember at least 5 major weather event(s). \${extreme_past_n} >= 5
247	extreme5_type_other <i>text</i>	If other, specify the type of event <i>Relevance:</i> Asked only if “Other” was chosen at extreme5_type . \${extreme5_type} = 'other'
248	extreme5_severity <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 5 major weather event(s). \${extreme_past_n} >= 5 <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> “Enter a value from 1 to 10”
249	extreme5_year <i>integer</i>	In which year did this event happen <i>Required.</i> <i>Relevance:</i> Asked only if the respondent said they remember at least 5 major weather event(s). \${extreme_past_n} >= 5 <i>Constraint:</i> . >= 1900 and . <= 2100 <i>Message:</i> “Enter a valid year”

OA-2.11 Pre-card expectations of extreme events

XLSForm group(s): extreme_expect_group.

Arms: Arms A and B only.

No.	Variable	Question, type, options, and routing
Block extreme_expect_group — “ <i>Expectations of extreme events</i> ”.		
<i>Block relevance:</i> Arms A and B only; Arm C respondents skip this item. \${arm} = 'A' or \${arm} = 'B'		
250	extreme_future_any → extreme_future_any_pre <i>select_one yn</i>	Do you expect a major weather event in the upcoming season <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
251	extreme_future_type → extreme_future_type_pre <i>select_one extreme_type</i>	Type of expected event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if extreme_future_any was answered Yes. \${extreme_future_any} = 'yes'
252	extreme_future_type_other → extreme_future_type_pre_other <i>text</i>	If other, specify <i>Relevance:</i> Asked only if “Other” was chosen at extreme_future_type . \${extreme_future_type} = 'other'
253	extreme_future_severity → extreme_future_severity_pre <i>integer</i>	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Relevance:</i> Asked only if extreme_future_any was answered Yes. \${extreme_future_any} = 'yes' <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> “Value must be between 1 and 10”

OA-2.12 Drivers of expectations and social learning

XLSForm group(s): factors_group, social_group.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block factors_group — “Factors influencing expectations”. <i>Block relevance:</i> All three arms (the condition is exhaustive and gates nothing out). \${arm} = 'A' or \${arm} = 'B' or \${arm} = 'C'		

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No.	Variable	Question, type, options, and routing
254	factors_considered → driver_memory, driver_farmers, driver_leaders, driver_govt, driver_tradition, driver_religion, driver_env, driver_media, driver_ext, driver_other <i>select_multiple factors</i>	What factors do you consider when forming rainfall expectations <i>Options</i> (code — label as read out), factors : memory — Personal memory of past seasons religious — Religious or spiritual guidance discuss — Discussions with other farmers env — Environmental indicators leaders — Information from community leaders media — Media sources gov — Government weather forecasts extension — Agricultural extension services trad — Traditional knowledge or beliefs other — Other
255	factors_considered_other → driver_other_specify <i>text</i>	If other, specify <i>Relevance</i> : Asked only if the option other was ticked at factors_considered . <code>selected(\${factors_considered}, 'other')</code>
Block social_group — “Social interactions and information sharing”. <i>Block relevance</i> : All three arms (the condition is exhaustive and gates nothing out). <code>\${arm} = 'A' or \${arm} = 'B' or \${arm} = 'C'</code>		
256	discuss_expectations <i>select_one yn</i>	Do you discuss your rainfall expectations with others <i>Options</i> : the 2-item yn list, given in full at Q4.
257	discuss_with → discuss_family, discuss_neighbors, discuss_leaders, discuss_extension, discuss_friends, discuss_religious, discuss_others <i>select_multiple discuss_who</i>	With whom do you usually discuss <i>Options</i> (code — label as read out), discuss_who : family — Family members friends — Friends neighbors — Neighboring farmers religious — Religious leaders leaders — Community leaders extension — Agricultural extension agents other — Others <i>Relevance</i> : Asked only if discuss_expectations was answered Yes. <code>\${discuss_expectations} = 'yes'</code>
258	discuss_with_other → discuss_other_specify <i>text</i>	If other, specify <i>Relevance</i> : Asked only if the option other was ticked at discuss_with . <code>selected(\${discuss_with}, 'other')</code>

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No.	Variable	Question, type, options, and routing
259	discuss_freq <i>select_one freq</i>	How often do you have these discussions <i>Options</i> (code — label as read out), freq : daily — Daily weekly — Weekly monthly — Monthly occas — Occasionally <i>Relevance:</i> Asked only if discuss_expectations was answered Yes. \${discuss_expectations} = 'yes'
260	trusted_opinions → trust_experienced_farmers , trust_elders , trust_experts , trust_religious , trust_media , trust_govt , trust_others <i>select_multiple trust_who</i>	Whose opinions do you trust most <i>Options</i> (code — label as read out), trust_who : farmers — Experienced farmers elders — Elders in the community experts — Agricultural experts religious — Religious leaders media — Media sources gov — Government officials other — Others
261	trusted_opinions_other → trust_other_specify <i>text</i>	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at trusted_opinions . selected(\${trusted_opinions}, 'other')
262	trust_reasons → trust_quality_experience , trust_quality_accuracy , trust_quality_education , trust_quality_age , trust_quality_wealth , trust_quality_family , trust_quality_status , trust_quality_other <i>select_multiple trust_why</i>	What qualities make you trust someones knowledge <i>Options</i> (code — label as read out), trust_why : exper — Farming experience accuracy — Past accuracy in predictions educ — Education level age — Age wealth — Wealth family — Family relation status — Social status other — Other
263	trust_reasons_other → trust_quality_other_specify <i>text</i>	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at trust_reasons . selected(\${trust_reasons}, 'other')

OA-2.13 Pre-card planned adaptation

XLSForm group(s): adapt_group.

Arms: **All three arms.** This is the block that yields adapt_any_pre for every respondent.

No.	Variable	Question, type, options, and routing
Block adapt_group — “Planned adaptation strategies based on expectations”. <i>Block relevance:</i> All three arms (the condition is exhaustive and gates nothing out). $\text{\$}\{\text{arm}\} = \text{'A'}$ or $\text{\$}\{\text{arm}\} = \text{'B'}$ or $\text{\$}\{\text{arm}\} = \text{'C'}$		
264	adapt_any → adapt_any_pre <i>select_one yn</i>	Are you planning to change your farming practices based on your rainfall expectations <i>Options:</i> the 2-item yn list, given in full at Q4.
265	adapt_types → adapt_pre_variety, adapt_pre_planting_date, adapt_pre_irrigation, adapt_pre_fertilizer, adapt_pre_diversify, adapt_pre_insurance, adapt_pre_off_farm, adapt_pre_credit, adapt_pre_training, adapt_pre_other <i>select_multiple adapt_ops</i>	What changes are you planning to implement <i>Options</i> (code — label as read out), adapt_ops : variety — Changing crop varieties dates — Adjusting planting dates irrigation — Investing in irrigation fert — Modifying fertiliser use diversify — Diversifying crops insurance — Purchasing insurance offfarm — Seeking off farm employment credit — Accessing credit or loans training — Attending training or workshops other — Other <i>Relevance:</i> Asked only if adapt_any was answered Yes. $\text{\$}\{\text{adapt_any}\} = \text{'yes'}$
266	adapt_types_other → adapt_pre_other_specify <i>text</i>	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at adapt_types. <code>selected(\text{\\$}\{\text{adapt_types}\}, \text{'other'})</code>
267	adapt_cost → adapt_pre_cost_ugx <i>decimal</i>	Estimated total cost of these adaptations (in Ugandan Shillings) <i>Relevance:</i> Asked only if adapt_any was answered Yes. $\text{\$}\{\text{adapt_any}\} = \text{'yes'}$ <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”

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No.	Variable	Question, type, options, and routing
268	adapt_finance → adapt_pre_fin_savings, adapt_pre_fin_family_ loans, adapt_pre_fin_ bank, adapt_pre_fin_ microfinance, adapt_pre_ fin_govt, adapt_pre_fin_ assets, adapt_pre_fin_ community, adapt_pre_ fin_other select_multiple_finance	How do you plan to finance these adaptations <i>Options</i> (code — label as read out), finance : savings — Own savings gov — Government subsidies or grants family — Loans from family assets — Sale of assets or friends bank — Bank loans group — Community savings groups micro — Microfinance other — Other institutions <i>Relevance</i> : Asked only if adapt_any was answered Yes. \${adapt_any} = 'yes'
269	adapt_finance_ other → adapt_pre_fin_other_ specify text	If other, specify <i>Relevance</i> : Asked only if the option other was ticked at adapt_finance. selected(\${adapt_finance}, 'other')

OA-2.14 Risk perceptions and attitudes

XLSForm group(s): risk_group.

Arms: All arms.

No.	Variable	Question, type, options, and routing
Block risk_group — “ <i>Risk perceptions and attitudes</i> ”. <i>Block relevance</i> : none; every respondent who reaches this point answers it.		
270	risk_willingness select_one conf_scale3	How willing are you to take risks in farming <i>Required</i> . <i>Options</i> (code — label as read out), conf_scale3 : 5 — 5 Extremely Willing 4 — 4 Very Willing 3 — 3 Somewhat Willing 2 — 2 A Little Willing 1 — 1 Not Willing At All

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No.	Variable	Question, type, options, and routing
271	weather_concern <i>select_one conf_scale4</i>	How concerned are you about weather impact on your income <i>Required.</i> <i>Options</i> (code — label as read out), conf_scale4 : 5 — 5 Extremely Concerned 4 — 4 Very Concerned 3 — 3 Somewhat Concerned 2 — 2 A Little Concerned 1 — 1 Not Concerned At All

OA-2.15 Historical rainfall information card and immediate reaction

XLSForm group(s): info_treatment_group.

Arms: **Arms B and C only.** Arm A never sees the card.

No.	Variable	Question, type, options, and routing
Block info_treatment_group — “ <i>Historical rainfall information block</i> ”. <i>Block relevance:</i> Arms B and C only; Arm A never sees the information card. \${arm} = 'B' or \${arm} = 'C'		
272	info_given_note → info_card_shown_note <i>note</i>	Show the respondent the local historical rainfall charts now
273	info_differs <i>select_one yn</i>	Does this information differ from what you remembered about past rainfall <i>Options:</i> the 2-item yn list, given in full at Q4.
274	info_surprise <i>select_one conf_scale2</i>	How surprised are you by this information <i>Options</i> (code — label as read out), conf_scale2 : 5 — 5 Extremely Surprised 4 — 4 Very Surprised 3 — 3 Somewhat Surprised 2 — 2 A Little Surprised 1 — 1 Not Surprised At All
275	adjust_expectations <i>select_one yn</i>	After seeing this information would you like to adjust your expectations <i>Options:</i> the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
276	conf_updated <i>select_one conf_scale</i>	Confidence in updated expectations <i>Options:</i> the 5-item conf_scale list, given in full at Q226. <i>Relevance:</i> Asked only if adjust_expectations was answered Yes. \${adjust_expectations} = 'yes'
277	info_useful <i>select_one yn</i>	Is historical rainfall information helpful for planning <i>Options:</i> the 2-item yn list, given in full at Q4.
278	info_future <i>select_one yn</i>	Would you like to receive similar information in the future <i>Options:</i> the 2-item yn list, given in full at Q4.

OA-2.16 Second forward elicitation

XLSForm group(s): **post_amount**, **post_onset_group**.

Arms: **All three arms.** Neither group carries a relevance condition.

No.	Variable	Question, type, options, and routing
Block post_amount — “ <i>Expectations about rainfall amount upcoming season</i> ”. <i>Block relevance:</i> none; every respondent who reaches this point answers it.		
279	post_img1 → fut_post_img1 <i>integer</i>	Tokens on image 1 extreme drought <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
280	post_img2 → fut_post_img2 <i>integer</i>	Tokens on image 2 drought <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
281	post_img3 → fut_post_img3 <i>integer</i>	Tokens on image 3 very dry <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
282	post_img4 → fut_post_img4 <i>integer</i>	Tokens on image 4 slightly dry <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”

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No.	Variable	Question, type, options, and routing
283	post_img5 → fut_post_img5 integer	Tokens on image 5 optimal <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
284	post_img6 → fut_post_img6 integer	Tokens on image 6 slightly wet <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
285	post_img7 → fut_post_img7 integer	Tokens on image 7 very wet <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
286	post_img8 → fut_post_img8 integer	Tokens on image 8 flood <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
287	post_img9 → fut_post_img9 integer	Tokens on image 9 severe flood <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${post_img1}+\${post_img2}+\${post_img3}+\${post_img4}+\${post_img5}+\${post_img6}+\${post_img7}+\${post_img8}) = 10 <i>Message:</i> “Total number of tokens must be 10”
288	post_total_tokens → fut_post_total_tokens calculate	<i>Not asked. Stored calculation.</i> <i>Calculation:</i> \${post_img1}+\${post_img2}+\${post_img3}+\${post_img4}+\${post_img5}+\${post_img6}+\${post_img7}+\${post_img8}+\${post_img9}
289	comment_post_amount → comment_fut_post_amount text	Enumerator: note any comments or remarks made by the participant during this section

Block post_onset_group — “Expectations about onset timing upcoming season”.

Block relevance: none; every respondent who reaches this point answers it.

290	post_int1 → fut_post_int1 integer	Tokens for onset interval 1 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
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No.	Variable	Question, type, options, and routing
291	post_int2 → fut_post_int2 integer	Tokens for onset interval 2 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
292	post_int3 → fut_post_int3 integer	Tokens for onset interval 3 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
293	post_int4 → fut_post_int4 integer	Tokens for onset interval 4 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
294	post_int5 → fut_post_int5 integer	Tokens for onset interval 5 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
295	post_int6 → fut_post_int6 integer	Tokens for onset interval 6 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
296	post_int7 → fut_post_int7 integer	Tokens for onset interval 7 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
297	post_int8 → fut_post_int8 integer	Tokens for onset interval 8 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
298	post_int9 → fut_post_int9 integer	Tokens for onset interval 9 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 <i>Message:</i> “Enter a number between 0 and 10”
299	post_int10 → fut_post_int10 integer	Tokens for onset interval 10 <i>Required.</i> <i>Constraint:</i> . >= 0 and . <= 10 and (. + \${post_int1}+\${post_int2}+\${post_int3}+\${post_int4}+\${post_int5}+\${post_int6}+\${post_int7}+\${post_int8}+\${post_int9}) = 10 <i>Message:</i> “Total number of tokens must be 10”

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No.	Variable	Question, type, options, and routing
300	post_onset_total → fut_post_onset_total calculate	Not asked. Stored calculation. Calculation: \${post_int1}+\${post_int2}+\${post_int3}+\${post_int4}+\${post_int5}+\${post_int6}+\${post_int7}+\${post_int8}+\${post_int9}+\${post_int10}
301	comment_post_onset_group → comment_fut_post_onset text	Enumerator: note any comments or remarks made by the participant during this section

OA-2.17 Post-card confidence, extremes, and adaptation (Arms A and B)

XLSForm group(s): prediction_confidence2, extreme_expect_group2, adapt_group2.

Arms: Arms A and B only.

No.	Variable	Question, type, options, and routing
Block prediction_confidence2 — “ <i>Prediction Confidence</i> ”.		
<i>Block relevance</i> : Arms A and B only; Arm C respondents skip this item. \${arm} = 'A' or \${arm} = 'B'		
302	conf_amount2 → conf_rain_post select_one conf_scale	Confidence in updated rainfall amount expectations <i>Required.</i> <i>Options</i> : the 5-item conf_scale list, given in full at Q226.
303	conf_onset2 → conf_onset_post select_one conf_scale	Confidence in updated onset expectations <i>Required.</i> <i>Options</i> : the 5-item conf_scale list, given in full at Q226.
Block extreme_expect_group2 — “ <i>Expectations of extreme events</i> ”.		
<i>Block relevance</i> : Arms A and B only; Arm C respondents skip this item. \${arm} = 'A' or \${arm} = 'B'		
304	extreme_future_any2 → extreme_future_any_post select_one yn	Do you expect a major weather event in the upcoming season <i>Options</i> : the 2-item yn list, given in full at Q4.

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No.	Variable	Question, type, options, and routing
305	extreme_future_type2 → extreme_future_type_post select_one extreme_type	Type of expected event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if extreme_future_any2 was answered Yes. \${extreme_future_any2} = 'yes'
306	extreme_future_type2_other → extreme_future_type_post_other text	If other, specify <i>Relevance:</i> Asked only if “Other” was chosen at extreme_future_type2 . \${extreme_future_type2} = 'other'
307	extreme_future_severity2 → extreme_future_severity_post integer	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Relevance:</i> Asked only if extreme_future_any2 was answered Yes. \${extreme_future_any2} = 'yes' <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> “Value must be between 1 and 10”

Block adapt_group2 — “Planned adaptation strategies based on expectations”.

Block relevance: Arms A and B only; Arm C respondents skip this item. **\${arm} = 'A'** or **\${arm} = 'B'**

308	adapt_any2 → adapt_any_post select_one yn	Are you planning to change your farming practices based on your updated rainfall expectations <i>Options:</i> the 2-item yn list, given in full at Q4.
309	adapt_types2 → adapt_post_variety ... adapt_post_other (10 dummies) select_multiple adapt_ops	What changes are you planning to implement <i>Options:</i> the 10-item adapt_ops list, given in full at Q265. <i>Relevance:</i> Asked only if adapt_any2 was answered Yes. \${adapt_any2} = 'yes'
310	adapt_types2_other → adapt_post_other_specify text	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at adapt_types2 . selected(\${adapt_types2}, 'other')
311	adapt_cost2 → adapt_post_cost_ugx decimal	Estimated total cost of these adaptations (in Ugandan Shillings) <i>Relevance:</i> Asked only if adapt_any2 was answered Yes. \${adapt_any2} = 'yes' <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”

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No.	Variable	Question, type, options, and routing
312	adapt_finance2 → adapt_post_fin_savings ... adapt_post_fin_ other (8 dummies) select_multiple finance	How do you plan to finance these adaptations <i>Options:</i> the 8-item finance list, given in full at Q268. <i>Relevance:</i> Asked only if adapt_any2 was answered Yes. \${adapt_any2} = 'yes'
313	adapt_finance2_ other → adapt_post_fin_other_ specify text	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at adapt_finance2. selected(\${adapt_finance2}, 'other')

OA-2.18 Confidence, extremes, and adaptation (Arm C)

XLSForm group(s): prediction_confidence3, extreme_expect_group3, adapt_group3.

Arms: **Arm C only.**

No.	Variable	Question, type, options, and routing
Block prediction_confidence3 — “ <i>Prediction Confidence</i> ”.		
<i>Block relevance:</i> Arm C only. \${arm} = 'C'		
314	conf_amount3 → conf_rain_C select_one conf_scale	Confidence in rainfall amount expectations <i>Required.</i> <i>Options:</i> the 5-item conf_scale list, given in full at Q226.
315	conf_onset3 → conf_onset_C select_one conf_scale	Confidence in onset expectations <i>Required.</i> <i>Options:</i> the 5-item conf_scale list, given in full at Q226.
Block extreme_expect_group3 — “ <i>Expectations of extreme events</i> ”.		
<i>Block relevance:</i> Arm C only. \${arm} = 'C'		
316	extreme_future_ any3 → extreme_future_any_C select_one yn	Do you expect a major weather event in the upcoming season <i>Options:</i> the 2-item yn list, given in full at Q4.
317	extreme_future_ type3 → extreme_future_type_C select_one extreme_type	Type of expected event <i>Options:</i> the 5-item extreme_type list, given in full at Q230. <i>Relevance:</i> Asked only if extreme_future_any3 was answered Yes. \${extreme_future_any3} = 'yes'

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No.	Variable	Question, type, options, and routing
318	extreme_future_type3_other → extreme_future_type_C_other text	If other, specify <i>Relevance:</i> Asked only if “Other” was chosen at extreme_future_type3. \${extreme_future_type3} = 'other'
319	extreme_future_severity3 → extreme_future_severity_C integer	Severity of this event on a scale from 1 to 10 (1= Not Severe at All, 5= Moderately Severe, 10 = Extremely Severe) <i>Relevance:</i> Asked only if extreme_future_any3 was answered Yes. \${extreme_future_any3} = 'yes' <i>Constraint:</i> . >= 1 and . <= 10 <i>Message:</i> “Value must be between 1 and 10”
Block adapt_group3 — “Planned adaptation strategies based on expectations”. <i>Block relevance:</i> Arm C only. \${arm} = 'C'		
320	adapt_any3 → adapt_any_C select_one yn	Are you planning to change your farming practices based on your rainfall expectations <i>Options:</i> the 2-item yn list, given in full at Q4.
321	adapt_types3 → adapt_C_variety ... adapt_C_other (10 dummies) select_multiple adapt_ops	What changes are you planning to implement <i>Options:</i> the 10-item adapt_ops list, given in full at Q265. <i>Relevance:</i> Asked only if adapt_any3 was answered Yes. \${adapt_any3} = 'yes'
322	adapt_types3_other → adapt_C_other_specify text	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at adapt_types3. selected(\${adapt_types3}, 'other')
323	adapt_cost3 → adapt_C_cost_ugx decimal	Estimated total cost of these adaptations (in Ugandan Shillings) <i>Relevance:</i> Asked only if adapt_any3 was answered Yes. \${adapt_any3} = 'yes' <i>Constraint:</i> . >= 0 <i>Message:</i> “Value must be zero or positive”
324	adapt_finance3 → adapt_C_fin_savings ... adapt_C_fin_other (8 dummies) select_multiple finance	How do you plan to finance these adaptations <i>Options:</i> the 8-item finance list, given in full at Q268. <i>Relevance:</i> Asked only if adapt_any3 was answered Yes. \${adapt_any3} = 'yes'

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No.	Variable	Question, type, options, and routing
325	adapt_finance3_ other → adapt_C_fin_other_ specify text	If other, specify <i>Relevance:</i> Asked only if the option other was ticked at adapt_finance3. selected(\${adapt_finance3}, 'other')

OA-2.19 Plot location

Arms: All arms.

No.	Variable	Question, type, options, and routing
326	latitude_farm_ manual text	Latitude coordinate of interviewees farm (typical range for eastern Uganda: 0.5 to 2.5) <i>Required.</i> <i>Appearance:</i> numbers
327	longitude_farm_ manua → longitude_farm_manual text	Longitude coordinate of interviewees farm (typical range for eastern Uganda: 33 to 35) <i>Required.</i> <i>Appearance:</i> numbers
328	gps_main_plot → gps_main_plot_label, gps_lat, gps_lon, gps_ alt, gps_precision geopoint	GPS of main farming plot <i>Required.</i>

The enumerator manual instructs enumerators to start from the interview location and mark the respondent's main plot on the tablet map (item 328). The recorded points are nonetheless clustered at interview sites rather than spread across plots; Appendix D of the paper documents their geometry. The two manual coordinate fields (items 326 and 327) were added after the pilot as a fallback and are not used in the analysis.

OA-2.20 Items in the form that we could not fully resolve

Six features of the XLSForm are worth flagging for anyone replicating from it, none of which changes what was asked but all of which will trip up a naive read of the workbook.

First, the **choices** sheet contains five blank spacer rows with an empty **list_name** and an empty **name**: four of them separate the four five-point scales **conf_scale**, **conf_scale2**,

`conf_scale3`, and `conf_scale4` from one another and from `extreme_type`, and one sits between `finance` and `village_list`. They define no options and are inert, but a script that groups the sheet by `list_name` will report a phantom, unnamed, five-item choice list.

Second, the choice list `arm`, with the three labels “Group A”, “Group B”, and “Group C”, is declared in the `choices` sheet but is never referenced by any question. Arm membership is stored by the `calculate` field `arm`, which writes the bare letters A, B, and C, so those labels never reach the data.

Third, the variable `longitude_farm_manua` is missing its final letter in the form. The analysis dictionary repairs it to `longitude_farm_manual`, and because that dictionary addresses columns by position rather than by header text the repair is silent; the two names must be matched by column position.

Fourth, the plant-week, harvest-week, and critical-month multi-selects for the second and third crops are collected by the form but are assigned no analysis name in `survey_dictionary.R`, so they are present in the raw export and absent from the analysis frame. For the first crop, only the harvest-week and critical-month selections are named; the plant-week selection is not.

Fifth, the constraint message on all nine of the “at most two” crop-calendar multi-selects reads “Please select one or two weeks”, including on the three *months* questions, where the unit is a month and not a week. This is a copy-and-paste artifact in the message string only. The constraint itself, `count-selected(.) >= 1 and count-selected(.) <= 2`, is correct in every case.

Sixth, two option labels contain spelling slips that we reproduce verbatim above because they are what the enumerator read from the screen: “5 Extremely Surpised” in the surprise scale `conf_scale2`, and “Sesame Simsim” in the crop list, where the intended word is the Ugandan term *simsim* for sesame.

OA-3 Token elicitation protocol

Both belief elicitations followed standardized enumerator instructions. Respondents were given exactly ten physical tokens, counted in front of them before each exercise. Enumerators were instructed not to suggest an answer, characterize any outcome as normal or desirable, or allocate tokens on the respondent's behalf. Respondents could move tokens until satisfied with their allocation, after which the enumerator confirmed that exactly ten tokens had been used before recording the response. The survey form independently enforced the ten-token total.

Seasonal-weather exercise. Before respondents allocated tokens, enumerators described the nine pictures in order. Picture 1 represented extreme drought, picture 5 optimal rainfall and growing conditions, and picture 9 severe flooding, with intermediate pictures representing progressively wetter conditions. The standardized descriptions were:

Picture 1: "An extreme drought. The soil is cracked and plants are dry or dead and almost nothing can grow."

Picture 2: "A drought. There is very little rain and plants are badly stressed."

Picture 3: "Very dry conditions. Plants are weak and growing slowly."

Picture 4: "Slightly dry conditions. Plants grow but they are smaller and the soil is drying."

Picture 5: "Optimal rainfall. The soil is moist and plants are healthy."

Picture 6: "Slightly wet conditions. There is more water than usual but plants still grow normally."

Picture 7: "Very wet conditions. Water stays for some days and plants begin to weaken."

Picture 8: "A flood. Water stands on the field and the crop is damaged."

Picture 9: "A severe flood. The whole field is under water and the crop is destroyed."

Enumerators then explained the token allocation using the following standardized instruction:

"You have ten tokens. Each token is like a vote. You can put your tokens on any of the pictures. If you think one picture shows the season very well you can put all ten tokens on that picture. If you think the season was between two pictures you can divide the tokens between them. You must use all ten tokens."

The same picture card was used in all study villages. The pictures are generic representations of the ordered weather scale rather than photographs of the respondent's own village. Treatment variation therefore does not arise from the elicitation images themselves.

Order and examples. The seasonal-weather exercise was introduced before the planting-onset exercise. Respondents completed it for each remembered season from 2021 through 2025 and for the upcoming 2026 season, with an additional forward elicitation where required by experimental assignment. Worked examples were framed around farmers in other locations and were explicitly identified as examples. Their purpose was to demonstrate that respondents could either concentrate all tokens on one outcome or distribute them across outcomes.

Planting-onset exercise. The onset exercise used the same ten-token procedure over ten dated intervals from 15 February through 24 April. Enumerators introduced the task by explaining that the intervals represented possible dates at which the main rains could begin. Respondents could concentrate or spread their tokens in the same manner as in the seasonal-weather exercise. The fielded calendar is reproduced in the paper and its interval construction is documented in Appendix B.

Respondents unable to recall a season. Enumerators could use up to two neutral probes before recording that a respondent could not remember a past season. Respondents who continued to report no memory were not required to allocate tokens. The resulting recall indicator therefore measures whether a season could be elicited under this protocol, not whether the underlying memory objectively existed.

Standardization rules. Respondents who requested more or fewer than ten tokens were reminded that ten tokens were used for every participant. Respondents reluctant to use extreme pictures were reminded that any picture was permissible. These instructions standardized the elicitation without directing respondents toward the center or tails of the scale.

OA-4 Enumerator manual

Enumerators worked from a 59-page field manual dated 19 December 2025. The manual presents the questionnaire screen by screen and distinguishes respondent-facing text from enumerator instructions. It documents the permitted probes, routing rules, consistency checks, and common administration errors. The manual was finalized before the main field-work and was used throughout the March 2026 survey.

For completeness, the fielded manual is provided as a separate supplementary file, *Enumerator Manual: Expectations, Shocks, and Climate Adaptation*, version of 19 December 2025. It is reproduced without editorial revision.

OA-5 Consent and ethics documentation

Written informed consent was obtained from all respondents before the survey began. Consent was recorded on paper using a standardized information-and-consent form and retained separately from the tablet questionnaire. The form identifies Erasmus University Rotterdam and the Erasmus Research Institute of Management (ERIM) as the sources of research support and records ethics approval under reference ETH2425-0832. The approval was granted by the Internal Review Board of the Erasmus School of Economics on 20 May 2025 and was valid for three years.

The consent form explains the purpose of the study, the voluntary nature of participation, the types of personal and research data collected, confidentiality and data-retention procedures, possible discomfort associated with recalling adverse weather events, and respondents' right to stop the interview or decline individual questions. It also states that anonymized research data may be retained and reused for research and educational purposes.

Consent was recorded on paper rather than through the electronic questionnaire. The fielded survey form contains no digital consent item, while the study documentation records signed and scanned consent forms among the retained project materials. The fielded consent form was written in English. No written translated version or back-translation is preserved in the study records.

The original fielded consent form is provided as a separate supplementary file, *Consent Form – Fielded Version*. We reproduce the original document rather than an edited transcription so that handwritten amendments, strike-throughs, formatting, and unresolved template language remain visible exactly as they appeared in the field.

OA-6 Fieldwork procedures

Fieldwork took place from 10 through 31 March 2026 across the 60 study villages. Seven enumerators administered the survey face to face using the fielded electronic questionnaire, printed elicitation materials, and the enumerator manual described in Section OA-4.

Interview administration. The manual instructed enumerators to conduct interviews in a private setting where possible, record only answers given by the respondent, follow the questionnaire routing as programmed, and consult the field team rather than improvise when an item was unclear. Respondent-facing wording was distinguished from enumerator instructions in the field materials. The two token exercises followed the standardized protocol described in Section OA-3. Enumerators also received a one-page field aid summarizing the two token tasks.

Built-in data checks. The electronic questionnaire imposed a series of validation rules during administration. Token allocations were required to sum to ten, individual token entries were restricted to the feasible range, acreage and household variables were constrained to nonnegative values where appropriate, and event severity and event year were restricted to their permitted ranges. Villages were selected from the fixed study list and subsequently confirmed, while experimental arm was assigned automatically from village identity rather than entered by the enumerator. Free-text comment fields allowed enumerators to record unusual circumstances during the belief modules.

The district variable remained a free-text field and so contains multiple spellings of the three study districts. All geographic assignment in the analysis therefore uses the standardized village-to-district mapping rather than this free-text field.

Consent. Written consent was obtained before interviewing using the form reproduced in Section OA-5. Consent was recorded on paper and retained separately from the tablet questionnaire.

Language. The electronic questionnaire and enumerator manual preserve the field instrument in English and contain no stored written translations. Interviews were conducted face to face, with oral translation by enumerators where needed. Because the study records do not preserve a written translation protocol or identify the language used in each interview, we do not assign a specific language of administration at the respondent level.

Training and monitoring documentation. The surviving study records do not contain a sufficiently detailed training log to reconstruct the duration or content of enumerator training beyond the field manuals and pilot materials described here. They likewise do not contain a separate back-check dataset or systematic back-check log. We therefore restrict claims about field quality control to the procedures documented in the instrument and field materials.

OA-7 Pilot and changes to the field protocol

The pilot was conducted on 24 and 29 December 2025 in Namikelo, Sango, and Burukuru villages in Bulambuli district on the slopes of Mount Elgon. Thirty interviews were completed, ten in each village, by five enumerators. All pilot respondents followed the same experimental sequence. No pilot observation is included in the analysis reported in the paper.

Comparing the pilot questionnaire and materials with those fielded in March 2026 identifies eleven changes.

1. *Recall years were changed to a consecutive five-season window.* The pilot elicited 2025, 2024, 2023, 2021, and 2019. The final instrument instead elicited each season from 2021 through 2025. This aligned the recall window with the five seasons displayed in the information treatment.
2. *Village identification was converted from free text to a closed list.* The final questionnaire required selection from the 60 study villages followed by explicit confirmation, eliminating spelling variation in the village identifier.
3. *Experimental assignment was automated.* In the pilot, enumerators selected the assigned arm from the allocation sheet. In the final instrument, experimental arm was determined automatically from village identity. Enumerators therefore had no discretion over assignment once the village was selected correctly.
4. *Enumerator comment fields were added.* The final questionnaire introduced comment fields after the recall and forward-belief modules so that unusual responses or administration circumstances could be recorded.
5. *Interview date and village identifier fields were added.* The final questionnaire records an interview date and numeric village identifier alongside the respondent identifier.
6. *A manual coordinate fallback was added.* Numeric latitude and longitude fields were introduced as a fallback when the primary map-based location capture could not be used.
7. *Separate “other” fields were added to decision-maker questions.* The final instrument records a separate text response for each ranked decision-maker item when “other” is selected.
8. *A crop-ranking item was removed.* A separate rank-order question used in the pilot was dropped because the final questionnaire already collected information on the respondent’s three principal crops through dedicated crop modules.
9. *The post-information reaction block was shortened.* Three reaction items used in the pilot were removed before the main fieldwork. The final instrument retained questions about whether the information differed from memory, surprise, willingness to revise expectations, confidence after revision, usefulness of historical information, and demand

for similar information in the future. As discussed in the paper, the reaction questions occur between the information treatment and the second forward elicitation in Arm B, before the single elicitation in Arm C, and not at all in Arm A.

10. *The planting-onset calendar was corrected.* The pilot materials attached year labels to the calendar and included an erroneous February date in the 2026 calendar. The final field card removed year labels so that the same instrument could be used for both recalled and expected onset and correctly defined the third interval as 1–6 March.
11. *Questionnaire metadata were standardized.* Form labels and version information were updated for the final field instrument.

Elements retained from the pilot. The nine-picture weather scale, ten-token format, ten planting-onset intervals, experimental-arm structure, and core read-aloud instructions were retained for the main study. One site-specific feature of the onset calendar also remained. Several interval descriptions refer to highlands and slopes, reflecting the Elgon pilot location, although the main study was conducted on the lower-elevation plateau. We therefore do not interpret absolute cross-sectional differences between reported and satellite-based onset as measurement of forecast accuracy. As Appendix B of the paper explains, onset is used only for within-person and randomized comparisons.

The available study records do not contain a separate written pilot debrief. We therefore restrict this account to changes that can be established by comparing the pilot and fielded instruments and materials.

OA-8 Computational details for statistical inference

Appendix C of the paper describes the inferential procedures and reports the resulting estimates. This section provides additional computational detail on the accelerated wild cluster bootstrap, the Romano–Wolf stepdown procedure, and the double-selection specifications. File-level replication instructions are provided separately in the replication package.

OA-8.1 Wild cluster bootstrap implementation

The restricted wild cluster bootstrap is computationally demanding because each hypothesis requires repeated estimation of the model and its cluster-robust variance. We therefore use a matrix implementation that partials out fixed effects before the bootstrap replications, pre-computes the relevant cross-products, and subsequently updates only the bootstrap outcome and studentized statistic. For specifications in which villages are nested within a fixed-effect unit, the weighted restricted residual is itself residualized within each bootstrap replication before the cluster-robust variance is computed. This preserves the residual transformation that would result from re-estimating the complete fixed-effect model.

The accelerated implementation was validated against direct repeated estimation on smaller bootstrap samples. Coefficients and test statistics coincide numerically, while differences in simulated p-values at full replication counts are within the Monte Carlo variation implied by the bootstrap.

OA-8.2 Romano–Wolf stepdown

The stepdown procedure follows Romano & Wolf (2005) and the resampling implementation described by Clarke et al. (2020). A single set of cluster-level bootstrap weights is shared across all hypotheses in a family. This is important because individual hypotheses can be estimated on different subsets of respondents while sharing villages. Whenever a village enters more than one hypothesis, it therefore receives the same bootstrap weight in the same replication, preserving dependence among the test statistics.

For each hypothesis, unrestricted and null-restricted specifications are estimated as described in Appendix C. Bootstrap outcomes are generated under the null and studentized statistics are calculated for each replication. Statistics are transformed according to the direction of the registered test, using the signed statistic for one-sided hypotheses and its absolute value for two-sided hypotheses. Hypotheses are ordered by their observed transformed statistic. At each step, the bootstrap reference statistic is the maximum over the hypotheses remaining in the family. Adjusted p-values are calculated from the corresponding exceedance frequency and made weakly increasing along the stepdown sequence. A bootstrap replication is retained for the family only when every hypothesis produces a finite statistic, preserving the common-draw structure.

Accelerated implementations are used for families sharing a common regression structure. These implementations were validated against the general procedure on smaller bootstrap samples before being used for the full set of replications.

OA-8.3 Double selection

Double selection is used in the heterogeneity analyses to reduce discretion over which correlated pre-treatment characteristics enter as controls (Belloni et al., 2014). For each target predictor, selection is conducted twice: once with the outcome as the dependent variable and once with the target predictor as the dependent variable. The final control set is the union of variables selected in these two equations. The reported coefficient is then obtained by re-estimating the outcome equation on the target predictor and the selected controls, with standard errors clustered at the village level.

Updating screen. The candidate set contains 22 pre-treatment characteristics. Continuous characteristics are standardized over the pooled Arms A and B sample so that coefficients are comparable across arms. Binary characteristics enter without standardization. Penalization uses the theoretically motivated heteroskedasticity-robust penalty with post-selection refitting rather than cross-validation. The final regressions use village-clustered inference.

Memory-quality screen. The same procedure is applied to the candidate predictors of memory accuracy, concentration, signed bias, and related memory outcomes. Subcounty indicators and experimental-arm assignment are always retained rather than subjected to selection. They are partialled out of the outcome and candidate predictors before the selection step. Candidate controls that are perfectly collinear after this transformation are excluded. Specifications with insufficient cluster-level degrees of freedom relative to the number of free parameters are classified as not estimable rather than retained with unstable cluster-robust inference.

The replication package reports the complete candidate sets, penalty parameters, estimability rules, random seeds, and numerical validation checks.